

The Compass



Volume 3-1, January 2026

Community Mathematics Centre

Schools of the Krishnamurti Foundation India
(KFI)

Statement of intent

The magazine is published by the Community Mathematics Centre. It is a collaborative initiative of the schools of the Krishnamurti Foundation India (KFI).

Its intention is:

- To provide material for mathematics and science teachers who teach at the high school or middle school levels, and for students who study at these levels.
- To serve as a platform where teachers and students can share their findings and discoveries with others who share their love for the subject.

In addition to mathematical content, in each issue we shall share a few ‘snippets’ from the world of science — short extracts abstracted from articles in online magazines, selected on the basis of intrinsic interest.

The Compass is published as an online magazine free of cost. Articles may be downloaded and used freely by teachers, but with proper attribution.

Edited by Shailesh Shirali & Padmapriya Shirali & Shashidhar Jagadeeshan on behalf of schools of the KFI

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Articles in this issue

I Letters to the Editor	1
Letter from Prof. Jyotirmoy Sarkar	2
II Miscellaneous Articles – 1	5
2026	6
A Beginner's Guide to QR Codes	10
Gems From Geometry – 5	17
Solutions to the Geometry Problems	22
III News From Around the World	31
Science Keeps Changing. So Why Should We Trust It?	32
Jane Goodall	36
Madhav Gadgil	47

Putting Nature on the Balance Sheet	51
Gladys Mae West	54
Human on a Bicycle	58
Income Inequality Around the World	62
The Paris Climate Agreement At 10	65
The Power of Self-Correcting AI Models	73
New Math AI Startup Flexes Its Muscles	76
Mathematicians Find One Pi Formula to Rule Them All	79
OpenClaw AI Chatbots Are Running Amok	85
How Much Energy Is Released When Supermassive Black Holes Collide?	88
Mathematicians Solve ‘Moving Sofa Problem’	92
Mathematicians Make a Breakthrough on 2,000-Year-Old Problem of Curves	98
How Cricket Balls Move	101
Finger Counting and Math Skills	104
The Untold Story of Our Remarkable Hands	107

IV Contributed Articles	113
Don't Rely on Analogy!	114
Divisibility property of sum of digits of n and $n+1$	121
Fermat's 'Little' Theorem	126
Solving a Set of Simultaneous Equations by Long Division	129
Primorial Numbers	135
Blanket-the-Big-Triangle Puzzle	139
V Miscellaneous Articles – 2	149
Miscellaneous Problems in Number Theory – 1	150
Rational Multiples of π with Rational Cosine	159
Infinitely Many Primes – 1	166
Fermat Primes and their Cousins	174
Grandma's Identities	186
Call for Articles	189

Colour coding

The magazine has articles addressed to teachers who teach at different levels: Lower Primary level; Upper Primary and High School levels; and Senior Secondary and Junior College levels. There is also a Problem Corner. In addition we have the Snippets.

For the convenience of the reader, we use the following colour coding to distinguish between the various levels. Of course, such a division can never be exact. So, our apologies in advance for errors of judgement!

- Primary & Upper Primary (Classes 4, 5, 6, 7)
- Upper Primary & High School (Classes 6, 7, 8, 9, 10)
- Senior Secondary & Junior College (Classes 10, 11, 12)
- Problem Corner
- Snippets

Part I

Letters to the Editor

Letter from Prof. Jyotirmoy Sarkar

Dear Editors,

I read with interest the article [1] in this magazine. I commend the authors for adopting multiple approaches to solving a problem, which is a productive intellectual pursuit. In that spirit, may I present a third solution, nay, a more general result?

[Editor's note: The problem studied in [1] is from Round 2 of the Hungarian Mathematical Olympiad 2001/02.]

Aspect Ratio and Angles in a Rectangle

Proposition 1: Let $ABCD$ be a rectangle with $AB = x$, $AD = 1$. On diagonal BD , drop a perpendicular from vertex A , meeting BD at E . Let $\theta = \angle ECB$. Then $\tan \theta = x^3$.

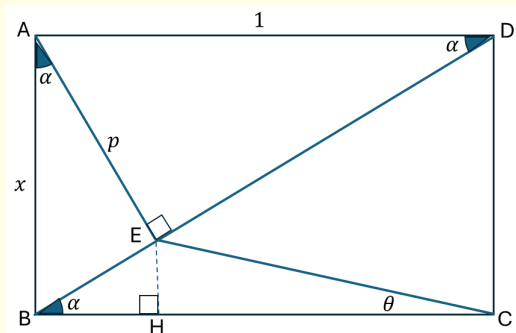


Figure 1: Drop a perpendicular from one vertex to a diagonal of a rectangle with aspect ratio $1 : x$. Join the foot to the opposite vertex. What are the two angles into which the opposite corner is split?

Proof: Refer to Figure 1. Since $\angle BAD = 90^\circ$, by the Pythagorean theorem, $BD = \sqrt{1 + x^2}$. Let $p = AE$. Twice the area of right- $\triangle BAD$ can be written in two ways: (i) $AD \cdot AB = 1 \cdot x$, and (ii) $BD \cdot AE = \sqrt{1 + x^2} \cdot p$. Hence, $p = x / \sqrt{1 + x^2}$.

On BC drop a perpendicular from E , meeting BC at H . Each of the right triangles AED , BEA and EHB , has an angle $\alpha = \arctan(x)$. Hence, they are similar triangles. Comparing the small, medium, and large sides of these three right triangles, we have

$$AE : ED : DA = BE : EA : AB = EH : HB : BE,$$

$$\text{or } p : ED : 1 = BE : p : x = EH : HB : BE.$$

Hence, sequentially, we derive $BE = px$, $HB = p^2$, and $EH = p^2x$.

Finally, from the right- $\triangle EHC$, we have

$$\tan \theta = \frac{EH}{HC} = \frac{EH}{BC - BH} = \frac{xp^2}{1 - p^2} = \frac{x^3/(1 + x^2)}{1 - x^2/(1 + x^2)} = x^3.$$

Thus, we have proved $x = \sqrt[3]{\tan \theta}$, and equivalently, $\theta = \arctan(x^3)$. ■

Remark: In [1], $\theta = 30^\circ$. Hence, $x^3 = \tan 30^\circ = 1/\sqrt{3}$, and so $x = 1/3^{1/6}$. Diligent readers can show that

$$ED = \frac{1}{\sqrt{1 + x^2}} = \sqrt{1 - p^2},$$

$$EC = \frac{1 - x^2 + x^4}{1 + x^2} = \frac{1}{1 - p^2} - 3p^2.$$

Further reading

[1] Anudeep Prashanth and Dhruv Perody. “Two solutions to a geometry problem.” *The Compass*, 1 67–72, July 2024. See <https://tinyurl.com/TwoSolutions>

About the author

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Part II

Miscellaneous Articles – 1

2026 and its many properties

As we move from 2025 to 2026, it is nice to list as many properties as we can of the number 2026. Here are some that I could assemble.

1. Prime factorization:

$$2026 = 2 \times 1013.$$

Please verify for yourself that 1013 is a prime number.

So the divisors of 2026 are 1, 2, 1013, and 2026 itself.

Numbers with exactly two prime divisors (not necessarily distinct) are known as *semiprimes*, so 2026 is a semiprime.

The first few semiprimes are 4, 6, 9, 10, 14, 15, 21, 22, 25, 26,

After 2026 the next nine semiprimes are

$2031 = 3 \times 677,$	$2033 = 19 \times 107,$	$2038 = 2 \times 1019,$
$2041 = 13 \times 157,$	$2042 = 2 \times 1021,$	$2045 = 5 \times 409,$
$2047 = 23 \times 89,$	$2049 = 3 \times 683,$	$2051 = 7 \times 293.$

2. 2026 is a **deficient number**, as the sum of its proper divisors is less than the number itself: $1 + 2 + 1013 = 1016 < 2026$. (This is the property that defines a deficient number: *the sum of its proper divisors is less than the number itself*. If the reverse inequality holds, i.e., *the sum of its proper divisors is greater than the number itself*, the number is called **abundant**.)

If we consider the first few multiples of 2026 we find that the following are deficient: 2026, 4052, 8104. On the other hand, 6078 is abundant. So if we wait for another 4052 years, we will encounter a multiple of 2026 which is abundant!

3. 2026 can be written as *the sum of two primes* in many different ways. The representation with the largest difference between the two prime numbers is $2026 = 2003 + 23$.

The two prime numbers less than and closest to 2026 are 2017 and 2011, but $2026 - 2017$ and $2026 - 2011$ are not prime. So if we want the representation with the largest difference between the two prime numbers, we arrive at $2026 = 2003 + 23$, as stated.

You must be aware that there is a conjecture (Goldbach's conjecture) that every even number larger than 2 can be written as the sum of two prime numbers. This remains an open statement as of end-2025.

4. There is precisely one way of writing 2026 as *a sum of two or more consecutive positive integers*:

$$2026 = 505 + 506 + 507 + 508.$$

In general, if p is an odd prime, then $2p$ can be written as a sum of

two or more consecutive non-negative integers in just one way,

$$2p = \frac{p-3}{2} + \frac{p-1}{2} + \frac{p+1}{2} + \frac{p+3}{2},$$

and the expression will always have 4 summands (except for $p = 3$ where we have 3 summands, $6 = 1 + 2 + 3$). Please prove this claim for yourself.

5. We *cannot* write 2026 as a *difference of two squares*.

In general, a number of the form $2 \pmod{4}$ cannot be written as a difference of two squares. Please prove this claim for yourself.

6. We can write 2026 as a *sum of two squares*. Indeed we can do this in just one way:

$$2026 = 45^2 + 1^2.$$

In general, if p is a prime number of the form $1 \pmod{4}$, then p can be written as a sum of two squares in just one way, and so too for $2p$. Indeed, if $p = a^2 + b^2$, where $a > b$, then $2p = (a+b)^2 + (a-b)^2$. Here we have $1013 = 22^2 + 23^2$.

7. 2026 is a *happy number*. This means that under the iteration

$$n \mapsto \text{sum of the squares of the base-ten digits of } n,$$

we eventually reach 1.

Under the iteration, $2026 \mapsto 2^2 + 0^2 + 2^2 + 6^2 = 44$. Continuing, we have,

$$2026 \mapsto 44 \mapsto 32 \mapsto 13 \mapsto 10 \mapsto 1 \mapsto 1 \mapsto 1 \mapsto \dots,$$

and we have landed at 1.

8. Here are a few miscellaneous expressions for 2026 in terms of repeated digits (please check all the computations):

- $2026 = (1 + 1)^{11} - 11 - 11$
- $2026 = 2^{22/2} - 22$
- $2026 = 3^3 + (333 \times (3 + 3)) + \frac{3}{3}$
- $2026 = ((4 + 4)^4 - 44) \times \frac{4}{4 + 4}$
- $2026 = 5 \times 5 + \frac{(5 + 5)^{(5 - 5/5)} + 5}{5}$
- $2026 = 6 + \left(6 - \frac{6}{6}\right) \times \left(\frac{6 + 6}{6} + (6 \times 66) + 6\right)$
- $2026 = \left(\frac{7 + 7 + 7}{7}\right)^7 - 77 - 77 - 7$
- $2026 = \frac{8 + 8}{8} + \left(88 \times \left(8 + 8 + 8 - \frac{8}{8}\right)\right)$
- $2026 = (9 + 9) \times (99 + 9) + 9 \times 9 + \frac{9}{9}$

Now go ahead and find more properties of 2026 for yourself!

A Beginner's Guide to QR Codes

We see them everywhere: in little tea stalls, in autorickshaws, on menu cards, on restaurant tables, on online shopping websites, on posters for concerts, on product packaging, and even in TV commercials. They are those little black-and-white squares that look like a scrambled crossword puzzle. These are **QR codes**, and they are a powerful bridge between our physical world and the digital universe. There must be very few places where their reach has not been felt. But what exactly are they, and how does your phone magically understand them? Let's dive in and demystify the QR code.

From Barcodes to QR Codes: A Quick History

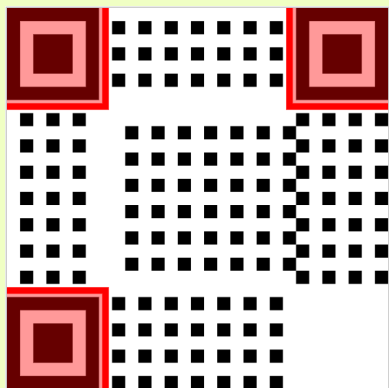
Before the QR code, there was the **barcode**: those vertical black and white stripes that could be seen on every product in the supermarket and the grocery store. Invented in the 1950s, these one-dimensional (1-D) barcodes

were revolutionary for inventory and checkout. However, they have a big limitation: being one-dimensional, they can only hold a small amount of information (usually just a string of numbers), and they can only be read horizontally.

In 1994, a Japanese company called Denso Wave, a subsidiary of Toyota, needed a better way to track car parts during manufacturing. They needed a code that could hold more information and be read faster. The result was the “**Quick Response**” code, or QR code. Unlike its 1-D predecessor, the QR code is two-dimensional (2-D), meaning that it stores information both horizontally and vertically. This simple change allowed it to hold hundreds of times more data than a traditional barcode, opening up a whole new world of possibilities.

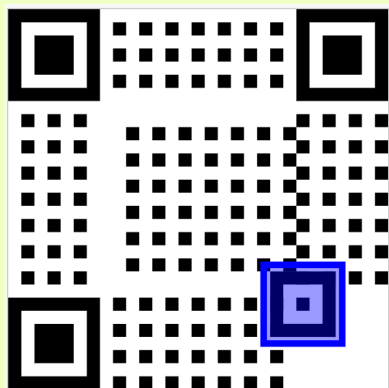
How Do QR Codes Work? The Magic Behind the Squares

In essence, a QR code is simply a visual way of storing information. Think of it like a special alphabet where letters are represented by black and white squares (called “modules”). When you point your phone’s camera at it, an app or the built-in camera software acts as a translator, reading the pattern and converting it back into usable data, e.g., a website link, or text. To do this accurately every time, the code has a few key components.



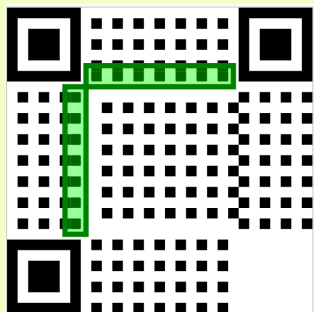
Finder Patterns: The three large squares in the corners tell the scanner where the QR code is and which way is up. This allows you to scan a code from any angle, even upside down!

Figure 2



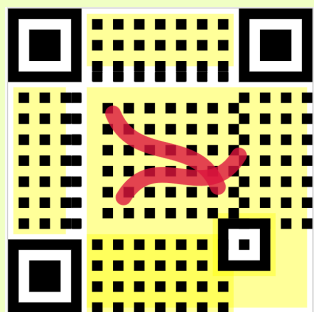
Alignment Pattern: The smaller square helps the scanner read the code even if it's distorted, for example, if you are scanning a code on a curved bottle or at a sharp angle.

Figure 3



Timing Patterns: These alternating black and white lines run between the finder patterns. They act like a ruler, defining the size of the grid and helping the scanner determine how large each individual module is.

Figure 4



Data and Error Correction (Highlighted Yellow): All the remaining modules contain the actual information. The most brilliant part is the built-in error correction. The data is stored with redundancy, much like how a RAID array in a computer stores extra data to prevent loss. This means that even if the QR code is partially damaged, dirty, or has a logo in the middle of it, your phone can often still read it by filling in the missing pieces. Up to 30% of the code can be damaged, and it will still work!

Figure 5

What Can You Store in a QR Code?

While linking to a website is the most common use, QR codes are incredibly versatile. You can create a QR code that, when scanned, does any of the following:

- Opens a website URL.
- Shows plain text (like a secret message or instructions or a recipe).
- Connects a phone to a Wi-Fi network without needing to type the password.
- Creates a new contact in your phone (a vCard).
- Adds an event to your calendar.
- Opens a specific location in a map application.
- Initiates a phone call or sends an SMS/email.
- Links to an app in the Google Play Store or Apple App Store.
- Facilitates a digital payment (this has become one of the commonest uses of QR codes).

Staying Safe with QR Codes

Alas, with great convenience comes some responsibility! A QR code is simply a gateway to information, and that information isn't always safe. Malicious actors can use QR codes to send you to phishing websites (fake sites designed to steal your passwords) or to download malware. This is sometimes called "Qishing" (QR code phishing).

Here are a few simple tips to stay safe

Trust the Source: Only scan codes from sources you trust, like a sealed product or a well-known company's official store.

Check for Tampering: Be cautious of QR codes in public places. Look closely to see if a sticker with a malicious code has been placed over the original one.

Preview the Link: Most modern smartphones will show you a preview of the URL before you open it. If the link looks suspicious or doesn't match what you expect, don't tap it!

Conclusion

From tracking auto parts to providing touchless menus in a pandemic, the humble QR code has become an indispensable part of our daily lives. It's a simple, elegant, and powerful technology that seamlessly connects the physical objects around us to the vast world of digital information. The next time you scan one, you'll know exactly what's happening behind that magical translation from a black-and-white square to an action on your phone.

Further reading

- [1] DENSO WAVE INCORPORATED. (n.d.). QR Code.com. Retrieved February 25, 2026, from <https://www.qrcode.com/en/history/>

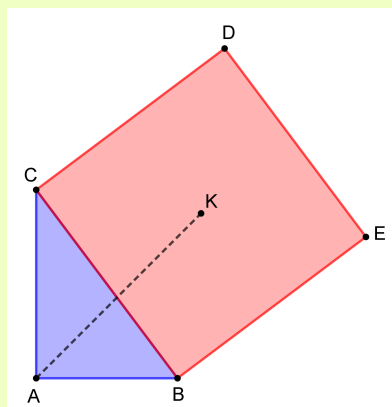
- [2] International Organization for Standardization. (2015). ISO/IEC 18004:2015: Information technology — Automatic identification and data capture techniques — QR Code bar code symbology specification. Retrieved from <https://www.iso.org/standard/62021.html>
- [3] BBC News. (2020, September 11). How the pandemic supercharged the QR code. Retrieved from <https://www.bbc.com/news/business-54090152>

Acknowledgement: This article has been prepared taking assistance from Gemini 2.5 Pro.

Gems From Geometry – 5

In this article we study some geometrical phenomena involving squares. More can be said about squares than you would expect! Many problems we study are from [1]. We give the problems first and the solutions later.

Problem 1



The figure shows $\triangle ABC$, with $\angle A = 90^\circ$, and a square $BCDE$ on side BC , outside the triangle.

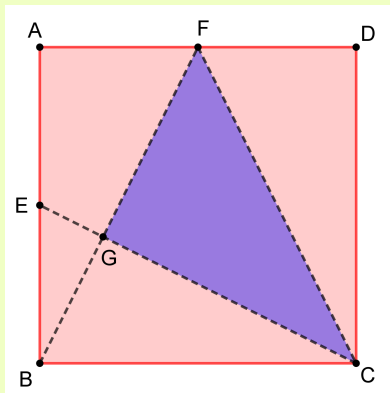
Segment AK joins vertex A to the centre K of the square. (So, K is the point of intersection of BD and CE .)

Problem: Show that AK bisects $\angle CAB$.

Figure 6

Question for exploration

What happens to the above conclusion if $\angle A \neq 90^\circ$?

Problem 2

The figure shows a square $ABCD$ in which E and F are the midpoints of sides AB and AD respectively.

Segments BF , CF and CE are drawn; CE and BF intersect at G .

Problem: Show that triangle FGC is right-angled with sides in the ratio $3 : 4 : 5$.

Figure 7

Question for exploration

Explore what happens happens to the above conclusion if E and F are located such that

$$AE = AF = k \cdot AB,$$

where k is some given number, $0 < k < 1$; in the new configuration, CE and CF are joined as earlier, but now G is the foot of the perpendicular from F to CE . What other ratios of sides do we obtain for different k ?

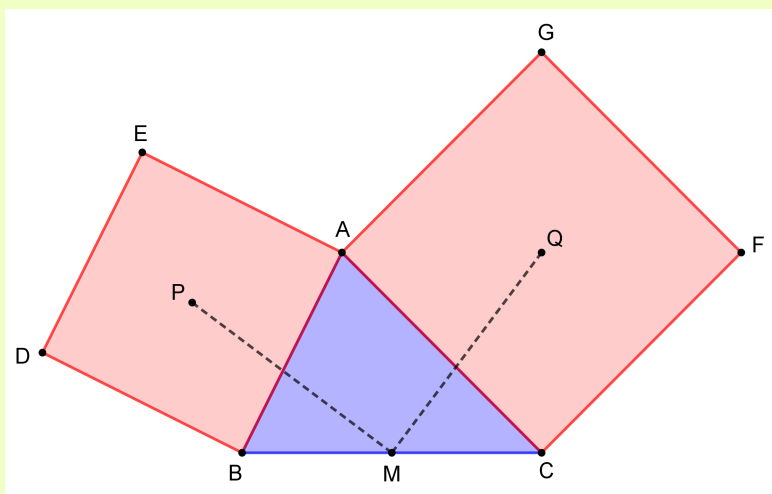
Problem 3

Figure 8: Von Aubel's theorem

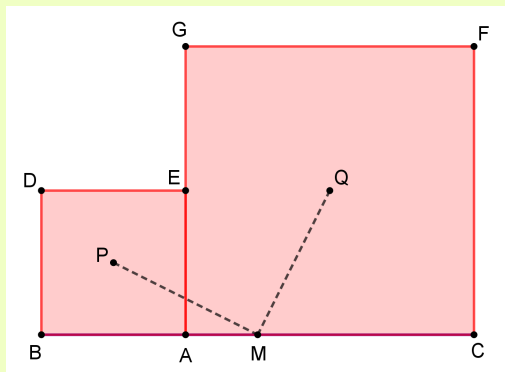
Given an arbitrary triangle ABC , draw squares $ABDE$ and $ACFG$ on sides AB and AC respectively, *outside* the triangle. Let P and Q be the centres of $ABDE$ and $ACFG$ respectively; let M be the midpoint of BC . Prove that $PM = QM$ and $PM \perp QM$.

This result is known as ***Von Aubel's theorem***. It can be proved in many different ways. Try to find some proofs by yourself.

It is always of interest to check out degenerate cases of theorems. That is, examine cases where two or more vertices coincide with one another, or the measure of some angle becomes 0° or 180° ; etc.

In the case of Von Aubel's theorem, let us study the degenerate triangle

where $\angle A = 180^\circ$. The resulting figure is shown below. Observe that the triangle has ‘collapsed’ into a line segment with vertex A on side BC .



Construct the squares $ABDE$ and $ACFG$ in the prescribed manner, locate their centres P and Q and also locate the midpoint M of BC . We must now prove that $PM = QM$ and $PM \perp QM$, as earlier.

Figure 9: A degenerate case of Von Aubel's theorem

Problem 4

This is a problem involving three squares and multiple triangles.

We are given an arbitrary triangle ABC . Squares $BCDE$, $ACFG$ and $ABHI$ are drawn on its sides BC , CA and AB , outside the triangle, as shown.

Line segments DF , GI and HE are now drawn, forming triangles CDF , AGI and BEH .

Prove that these three triangles have the same area as the original triangle ABC . (Note that these four triangles in general are not congruent to each other.) Try to look for a “pure geometry” proof.

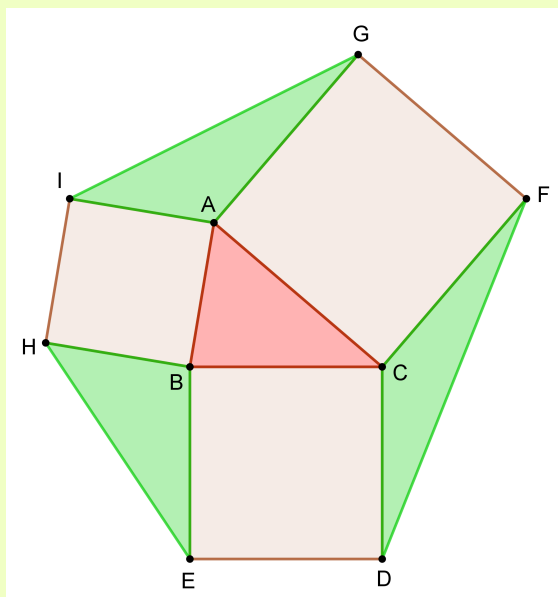


Figure 10: Four triangles with equal area

For the solutions, please see page 22.

Further reading

- [1] Duane Detemple & Sonia Harold, "A Round-Up of Square Problems," *Mathematics Magazine*, Vol. 69, No. 1, February 1996, from <https://www.jstor.org/stable/2691390>

Article contributed by Shailesh Shirali

Solutions to the Geometry Problems (on page 17)

Solution to Problem 1

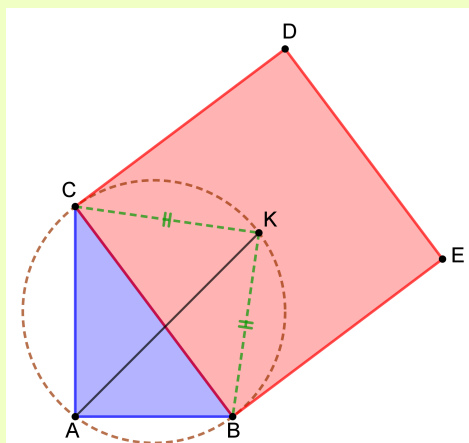


Figure 11

Draw the circumcircle of $\triangle ABC$. It passes through K since $\angle BKC$ and $\angle BAC$ are right angles.

Now $KB = KC$ (each length is half the diagonal of the square).

Hence $\angle KAB = \angle KAC$. So, AK bisects $\angle CAB$.

What happens to the above conclusion if $\angle A \neq 90^\circ$?

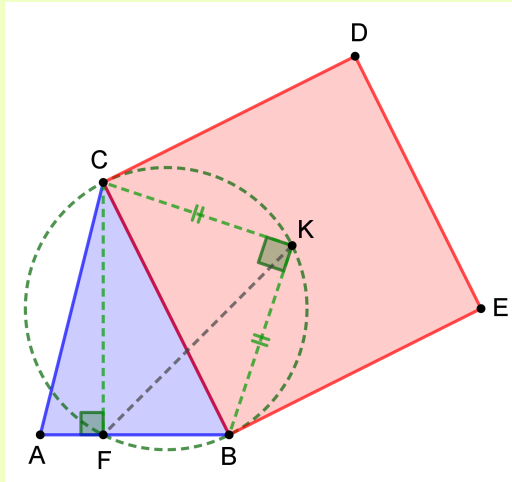


Figure 12: Generalization of the original problem

The circle will not pass through A since $\angle BAC \neq 90^\circ$. But it will pass through the foot F of the perpendicular from C to AB .

So, the conclusion for this situation is that FK bisects $\angle CFB$.

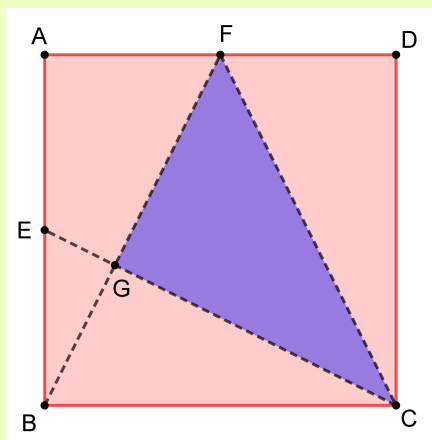
Solution of Problem 2

Figure 13

Observe that $\triangle FAB \cong \triangle EBC$, and

$$\angle GBC + \angle GCB = \angle GBC + \angle GBA = 90^\circ,$$

so, $\triangle FGC$ is right-angled.

Next, $EC = FC$, and

$$EG : GC = EB^2 : BC^2 = 1 : 4,$$

hence $GC : EC = 4 : 5$, therefore $GC : FC = 4 : 5$.

This implies that $FG : GC : FC = 3 : 4 : 5$, as claimed.

Question for exploration

What happens if E and F are located such that $AE = AF = k \cdot AB$, where $0 < k < 1$, and G is the foot of the perpendicular from F to CE ? What ratios $(FG : GC : FC)$ do we obtain for different values of k ?

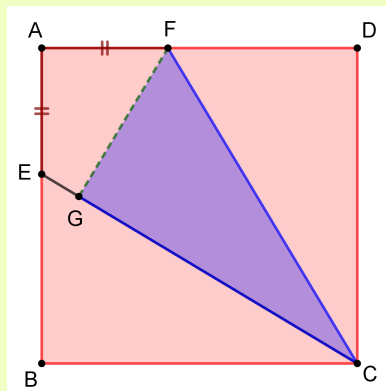


Figure 14

The configuration does not admit of a simple geometric solution, so we take recourse to coordinates. Assign coordinates: $A(0,1)$, $B(0,0)$, $C(1,0)$, $D(1,1)$, $E(0,1-k)$, $F(k,1)$. Then the equations of CE and FG are

$$\text{Equation of } CE: y = (1-k)(1-x),$$

$$\text{Equation of } FG: (y-1)(1-k) = x-k.$$

Solving for x, y , we find the coordinates of G :

$$G \left(\frac{k^2}{k^2 - 2k + 2}, \frac{2(1-k)^2}{k^2 - 2k + 2} \right).$$

This enables to find expressions for the sides of $\triangle FGC$, via the distance formula. After some straightforward (but tedious) algebra we get,

$$FG : GC : FC = k(2 - k) : 2(1 - k) : k^2 - 2k + 2.$$

By using the above expression we can generate PPTs (as many as we wish) by assigning different rational values to k . (We have thus hit upon a new way of generating PPTs!)

Examples

- If $k = \frac{1}{2}$ we get

$$FG : GC : FC = \frac{3}{4} : 1 : \frac{5}{4},$$

giving the PPT (3,4,5).

- If $k = \frac{1}{3}$ we get

$$FG : GC : FC = \frac{5}{9} : \frac{4}{3} : \frac{13}{9},$$

giving the PPT (5,12,13).

- If $k = \frac{1}{4}$ we get

$$FG : GC : FC = \frac{7}{16} : \frac{3}{2} : \frac{25}{16},$$

giving the PPT (7,24,25).

- If $k = \frac{3}{7}$ we get $[33/49, 8/7, 65/49]$

$$FG : GC : FC = \frac{33}{49} : \frac{8}{7} : \frac{65}{49},$$

giving the PPT (33,56,65).

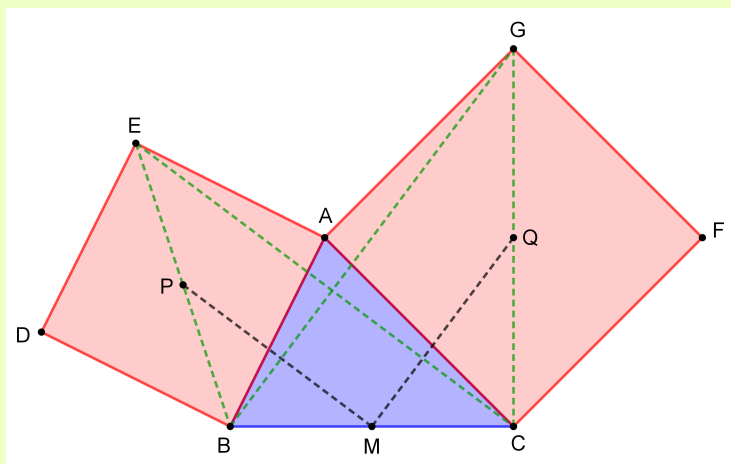
Solution to Problem 3: Proof of Von Aubel's theorem

Figure 15: Proof of Von Aubel's theorem

The most straightforward approach is through the use of the midpoint theorem. Draw additional lines as shown in the figure (above): BE , CG , CE , BG . By the midpoint theorem we have,

$$MP = \frac{1}{2}CE, \quad MP \parallel CE,$$

$$MQ = \frac{1}{2}BG, \quad MQ \parallel BG.$$

Next, consider $\triangle CEA$ and $\triangle GBA$. They are congruent to each other (side-angle-side or SAS congruence), hence $CE = GB$. And since $CA \perp GA$ (i.e., a pair of corresponding sides are perpendicular to each other), it follows that $CE \perp GB$. Therefore, $MP = MQ$ and $MP \perp MQ$. ■

An elegant and compact proof may be devised using the language of transformations. Consider a quarter-turn centred at P , followed by a quarter-turn centred at Q . Observe that the composite mapping takes B to C (via A). The composition of two like quarter-turns is a half-turn, and since it maps B to C , the half-turn must be centred at M . From the algebra of rotations we know that this implies that $MP = MQ$ and $MP \perp MQ$. (For those of you who are not familiar with the algebra of rotations, we will devote an article to this topic in a later issue.)

Another elegant proof may be devised using complex numbers. This draws on the fact that multiplication by the imaginary unit $i = \sqrt{-1}$ achieves a quarter-turn in the complex plane. Embed the entire configuration on the complex plane so that M lies at the origin and B and C are on the real axis. Let the affixes of A, B, C, D, P, G, Q be a, b, c, d, p, g, q respectively; then $c = -b$. We now have,

$$d - b = i(a - b), \quad \therefore d = ia + (1 - i)b, \quad \therefore 2p = (1 + i)a + (1 - i)b.$$

Similarly,

$$g - a = i(c - a), \quad \therefore q = ic + (1 - i)a, \quad \therefore 2q = (1 + i)c + (1 - i)a.$$

Hence we have,

$$2p = (1 + i)a + (1 - i)b, \quad 2q = -(1 + i)b + (1 - i)a.$$

It is easy to verify now that

$$i \cdot 2q = 2p,$$

i.e., $p = iq$. From this it follows that $MP = MQ$ and $MP \perp MQ$. ■

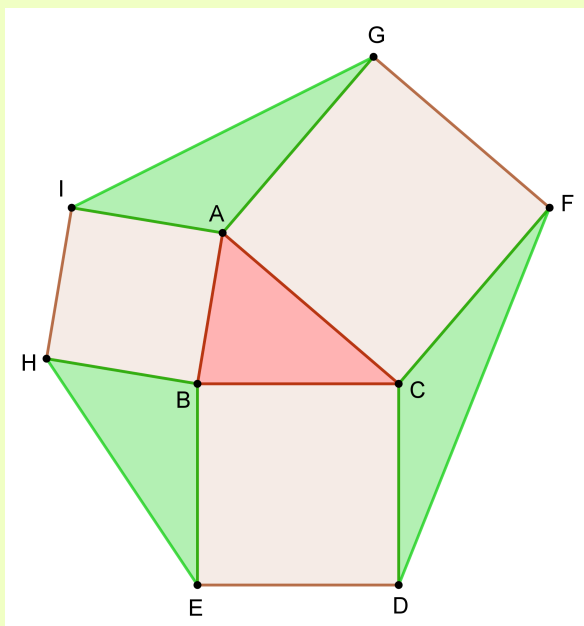
Problem 4

Figure 16: Four triangles with equal area

We must prove that triangles CDF , AGI and BEH have equal area. The simplest proof uses the trigonometric formula for area of a triangle. E.g., consider $\triangle AGI$. Let a, b, c denote the sides of $\triangle ABC$. Then we have, since $\angle GAI = 180^\circ - A$,

$$2 \cdot \text{Area of } \triangle AGI = AI \cdot AG \cdot \sin \angle GAI = bc \sin A = 2 \cdot \text{Area of } \triangle ABC.$$

Similarly for $\triangle CDF$ and $\triangle BEH$. This proves the claim. ■

While the above proof is very convincing, it is not visual. We are not able to see *why* these four triangles have equal area. But by drawing the completed parallelograms and looking at the “other halves” of these parallelograms (namely, $\triangle AJI$, $\triangle BHK$, $\triangle CLF$), the required visual element emerges (the idea is from the original reference). A close look reveals that the four shaded triangles are congruent to each other! (Try working out why!)

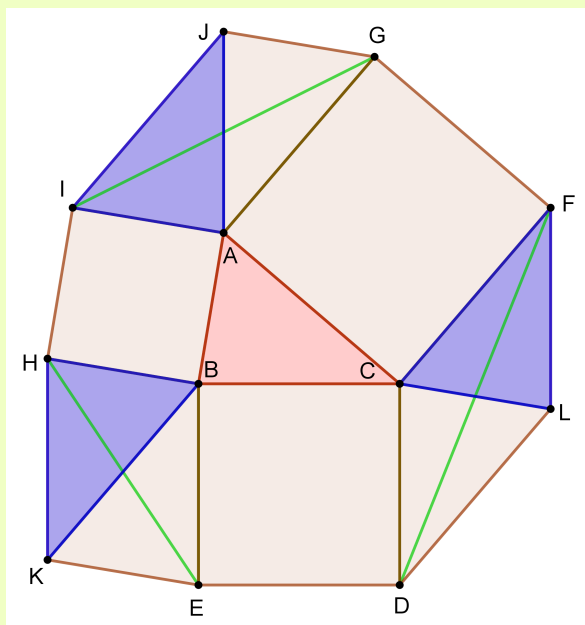


Figure 17: Four congruent triangles!

Article contributed by Shailesh Shirali

Part III

News From Around the World

Science Keeps Changing. So Why Should We Trust It?

Adapted from: "Science Keeps Changing. So Why Should We Trust It?"
<https://www.nytimes.com/2026/01/05/opinion/science-trust.html>? (New York Times). Author: Elay Shech

Elay Shech, a philosopher of science, opens by addressing a growing skeptical argument: since many well-established scientific theories have been overturned in the past, science cannot be trusted because it will inevitably change again. He cites real-world examples of this reasoning — Robert F. Kennedy Jr. used it to justify halting mRNA vaccine funding, and early pandemic mixed messaging on masking led many to doubt the reliability of public health science altogether.

The Skeptic's Case

Shech concedes that science does not march steadily and uniformly toward truth. Historical examples abound: the four humors theory of health, phlogiston as the substance of fire, the luminiferous ether as a medium for light — all were once mainstream and all were discarded. Even Newtonian physics, the gold standard of scientific certainty, was revised by Einstein. If so many confident theories failed, why trust today's?

Why the Argument Misleads

While the skeptics raise a genuine concern, Shech argues their conclusion is misleading because it presents only two options — naïve faith in science or wholesale rejection of it. Crucially, it treats science as one monolithic entity that stands or falls together, when in reality science comprises many distinct local domains of inquiry, each with its own methods, standards of evidence, and degrees of reliability.

The Problem with Induction

To expose the flaw in the skeptical argument, Shech uses the logic of induction. He compares two cases: heating samples of bismuth (a pure element with a single melting point) versus heating wax candles (which vary widely in composition). Both use the same reasoning — “all samples I tested behaved this way, so all instances will” — but only the bismuth conclusion is reliable. The quality of inductive reasoning depends entirely on background context, not just the pattern of the argument itself.

Applied to science: the skeptical argument concludes that because past theories were wrong, future ones will be too. But this requires a background assumption that something is systematically defective about science — a claim that doesn't hold up.

No Single “Scientific Method”

Shech further dismantles the argument by pointing out that there is no single, universal scientific method. Newton's deduction from observation, Darwin's inference to the best explanation, and Einstein's thought experiments are radically different strategies for investigating the world. A sweeping indictment of “the scientific method” has no coherent target.

Local Skepticism as the Right Attitude

Rather than global pessimism or blind faith, Shech advocates for what he calls **disciplined trust** — or local skepticism. This means evaluating the reliability of science domain by domain, based on actual evidence and method. For instance, early-stage medical research has well-documented problems — small sample sizes, publication bias, statistical overfitting — and many highly cited findings have later been contradicted. In this specific domain, pessimism may well be warranted. But this is a narrow, evidence-based conclusion, not a sweeping indictment of all science.

Shech also cautions against conflating types of scientific “failure.” Newton's laws weren't simply wrong — they remain accurate in many contexts, and Einstein's success was partly measured by how well it preserved Newton's valid predictions.

The Strength Behind Change

Shech closes on a constructive note: the fact that science keeps revising itself is not a weakness but its greatest strength. Science changes because the world is genuinely complex — and the willingness to update in response to new evidence is precisely what drives scientific progress. Disciplined trust, grounded in knowledge and intellectual humility, is the attitude that both the public and science itself need most.

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Perplexity* LLM.

Jane Goodall

3 April 1934 – 1 October 2025

On 1st October 2025 we came to know that Jane Goodall had passed on. This article which is assembled from various obituaries about her is meant to be a tribute to this extraordinary human being.

* * * * *

Adapted from: “Jane Goodall’s legacy: three ways she changed science.” <https://www.nature.com/articles/d41586-025-03209-y>. Author: Mohana Basu

This article which is about Jane Goodall’s legacy talks about her role in transforming primate science, raising questions about who can be a scientist, and embedding conservation and animal welfare firmly into public consciousness.

The article tells us that Jane Goodall, the British primatologist famed for her chimpanzee research in Gombe National Park, Tanzania, died on 1

October at age 91 while on a speaking tour in California, of natural causes. It highlights her landmark discovery that chimpanzees make and use tools, her founding of the Jane Goodall Institute in 1977, and her evolution into a global advocate for conservation, human rights, and ending the use of animals in medical research.

Humanizing primates

Goodall radically changed how scientists study animals by refusing to reduce chimpanzees to numbers, instead naming individuals such as David Greybeard, a male with silver facial hair. Senior scientists initially criticized this as “unscientific,” but, as anthropologist Mireya Mayor notes, she showed that science could expand its boundaries without sacrificing rigour. Naming animals is now common practice.

Her long-term observations demonstrated that chimpanzees have emotions, empathy and culture, traits once thought to be uniquely human, reshaping methods and ethics in animal research. Nick Boyle of Taronga Zoo describes how her Gombe discoveries “redefined humanity” by revealing that chimpanzees eat meat, hunt, and even wage lethal intergroup warfare. The last observation came about from her study of a four-year conflict in which all males in one community were killed.

Inspiring women in science

The article emphasizes Goodall’s role as a model for women entering field science, especially because she entered a PhD program at the University of Cambridge in 1961 without an undergraduate degree and completed it in 1965. Goodall proved a young woman with no formal scientific training

could fundamentally reshape scientific understanding of animals.

Anthropologist Alison Behie recounts how hearing Goodall speak led her to change her undergraduate major from microbiology to anthropology and to pursue primatology and conservation. Years later, in 2017, Behie brought eight of her own female students to meet Goodall in Australia, describing this encounter as a “full circle” moment that passed Goodall’s inspiration to a new generation.

* * * * *

Adapted from: “Jane Goodall’s Message of Hope Lives On.” <https://time.com/7322497/jane-goodall-death-hope-legacy/>.

Author: Jeffrey Kluger

Jane Goodall’s life and work are presented here as a rare combination of scientific rigour, moral courage, and quiet persistence, leaving a legacy of hope that the article frames as a challenge to the rest of humanity to live up to. Her death is used as a moment to look back on the discoveries that changed how humans see animals (and themselves) and on the hopeful activism that defined her later years.

Life and scientific breakthroughs

The article opens by remarking that Jane Goodall belongs to a small group of figures (Nelson Mandela, M K Gandhi, Martin Luther King, Abraham Lincoln, ...) whose surnames alone evoke moral qualities like kindness and wisdom, and then recounts her death at 91 in California while on a speaking tour. It notes her many roles (primatologist, conservationist, ...)

and honours (U.S. Presidential Medal of Freedom recipient, and Dame Commander of the Order of the British Empire).

Goodall's quiet, austere way of living is connected to the kind of work she did: long, patient observation in Gombe National Park in Tanzania, often spending months and years studying chimpanzees from close range. The article situates her within Louis Leakey's trio of women primate researchers — Jane Goodall with chimpanzees, Dian Fossey with mountain gorillas, and Biruté Mary Galdikas with orangutans — whom Leakey jokingly called "The Trimates."

Transforming our view of chimpanzees

Several of Goodall's landmark discoveries are described as revolutionary for science and for our understanding of human uniqueness. In 1960 she observed chimpanzees eating a bushpig, overturning the belief that they were strictly vegetarian, and saw them using stripped twigs as tools to fish for termites, challenging the idea that only humans use tools.

The article also dwells on the darker side of chimpanzee life that Goodall documented. From 1974 to 1978, she witnessed a protracted, violent conflict she called "the four year war" between rival communities, and she also observed cannibalism when a mother-daughter pair killed and ate infants within their own group, likely to remove future female rivals.

Intimacy, empathy, and cross-species connection

Balanced against this violence is a portrait of chimpanzees as emotionally complex and capable of gentleness. Goodall's observations include chimps playing, tickling, kissing, grieving, and making submissive gestures after

conflicts as a form of apology. The narrative emphasizes moments when chimpanzees appeared to accept Goodall herself into their social world, treating her as a kind of honorary member of the group.

Two personal anecdotes from an interview with TIME in 2009 highlight this cross-species connection. In one, a young male chimp waits in a clearing for her, then gently squeezes her hand — a gesture she explains chimpanzees use to reassure one another, and which she describes as a kind of pre-verbal communication. In another, a mother named Flo allows her five-month-old infant, just learning to walk, to totter toward Goodall and touch her nose, a sign of deep trust that Goodall found “magic.”

From field scientist to global advocate

The article then widens its focus from Goodall’s fieldwork to her public role. Humanity, it suggests, came to trust her as an “ambassador” between humans and our closest genetic relatives, and as a credible advocate for the protection of the natural world. This ambassadorial trust was, in the author’s view, fully justified by her decades of careful observation and ethical commitment.

Her environmental activism which she undertook in her later years is illustrated through her 2021 TIME article on deforestation and tree loss. There she warned that Earth’s estimated six trillion trees have been reduced by about half, mostly in the last century, and urged support for large-scale reforestation efforts such as the Trillion Trees campaign and the Trees for Jane initiative. She portrayed the planet, seen from space, as white, blue, brown, and green, with the green steadily retreating, and argued that planting and protecting a trillion trees by 2030, although an enormous task, is achievable if everyone participates.

Hope as a moral imperative

The closing section centres on Goodall's philosophy of *hope*, presented as both an emotional stance and an ethical duty. In a final 2021 conversation with TIME, she expressed concern that young people would be inheriting a world in crisis, but insisted that surrendering to "doom and gloom" leads only to apathy, making hope "crucial" for action and survival.

The article ends by stating that her "long, heartening campaign" has now concluded with her death, and poses a question: whether humanity will prove worthy of the work and hope that she invested in the planet.

* * * * *

Adapted from: "Jane Goodall—no formal degree, no laboratory, only passion for the wild." <https://theprint.in/feature/jane-goodall-passion-for-the-wild/2755610/>. Author: Tejus RS

Jane Goodall is portrayed as a pioneering primatologist whose lifelong passion for animals, rather than formal academic training or laboratory resources, reshaped science and global conservation. The article traces her journey from a curious child in London to a world-famous scientist and activist whose work on chimpanzees changed how humans understand animals and their connection to nature.

Early life and beginnings

Jane Goodall was born in London in 1934 and showed deep curiosity about animals from childhood, reading about wildlife and patiently observing

creatures such as hens in a chicken coop. This early patience and *love* for animals formed the foundation of her later fieldwork in Africa, even though she had no formal scientific degree when she began.

Ground-breaking chimpanzee research

In 1960, at age 26, she went to Gombe Stream National Park in Tanzania with only a notebook and binoculars to study chimpanzees. There she made historic discoveries, including observing chimpanzees using tools to fish termites, which challenged the belief that only humans used tools and blurred the line separating humans from other animals.

Her long-term observations also revealed that chimpanzees have emotions, distinct personalities, and complex social structures, which she documented by naming individuals and describing behaviours like maternal care, grooming friendships, and even inter-group conflict. These findings fed into her books *In the Shadow of Man* and *The Chimpanzees of Gombe*, helping establish one of the longest-running wildlife studies and a new foundation for primatology.

Influence on primatology

Indian primatologist Mewa Singh recalls her as an inspirational force, noting that she began her work without formal biological training. He explains that anthropologist Louis Leakey believed women might understand apes better and therefore supported Jane Goodall, Dian Fossey, and Biruté Galdikas in their respective studies of chimpanzees, gorillas, and orangutans.

The article recounts a key moment when a chimp sat near her on a rock at

sunset in Gombe, symbolising the trust she built and inspiring the title *In the Shadow of Man*. From such moments, she rose into a towering figure in primatology and conservation, with generations feeling as though they grew up following her work.

Shift to conservation and activism

In 1977, she founded the Jane Goodall Institute, which expanded from chimpanzee research to broader projects in Africa and beyond, integrating wildlife protection with community development, education, and sustainable livelihoods.

By the 1980s she increasingly shifted from research to conservation, recognising that saving chimpanzees required addressing deforestation, population decline, and the hardships of local people.

In 1991 she launched the Roots & Shoots youth programme, starting with Tanzanian students and growing into a global movement in over 100 countries, encouraging young people to undertake projects for animals, the environment, and human communities.

Her core message, “Every individual can make a difference,” became a guiding principle in her talks, whether at the United Nations or small schools, often accompanied by her toy chimpanzee Mr H as a symbol of hope.

Legacy, honours, and passing

Over a six-decade career, she received numerous honours, including being made a Dame of the British Empire, a UN Messenger of Peace, and recipient of many scientific and humanitarian awards, yet she preferred to describe

herself simply as “a girl who loved animals.” Even in old age she travelled more than 300 days a year, tirelessly advocating for animals and the Earth.

The article reports that Jane Goodall died at age 91, and frames her passing as a profound loss balanced by a lasting legacy living on in forests, conservation projects, and the young people she inspired. It closes with her quote linking understanding, care, and help to saving the world, and emphasises that she will be remembered as the woman who sat among chimpanzees and gave them a voice, reminding humanity that it is part of nature, not separate from it.

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Selected quotes from “Jane Goodall’s Plea for Future Generations, in Her Own Words.” <https://time.com/7333154/jane-goodalls-plea-future-generations-in-her-own-words/>.

Author: Jane Goodall

We are going through dark times, politically, socially, and especially environmentally. For thousands of years, early humans, like most animal species, lived in harmony with the natural world, hunter-gatherers taking only what they needed to survive. Gradually, that changed. As our populations grew, our demands on the planet’s natural resources increased and became increasingly unsustainable.

* * * * *

I have spent much of my life studying the amazing animals with whom we share or should share this planet. I have come to understand the complexity of ecosystems, where each animal and plant is interconnected and has a role to play in the complex web of

life. The chimpanzees that I and my team have observed and worked to protect since 1960 are amazingly like us. They can live more than 60 years, have distinct personalities, form close bonds between family members, and can use and make tools. They show emotions similar to ours: love, compassion, joy, grief, and so on. They live in complex communities and are territorial. Like us they can be aggressive and brutal, but like us they can also be loving and altruistic.

* * * * *

There is one major difference that separates us from chimpanzees and other animals — the explosive development of our intellect. For although animals—and not just the Great Apes, elephants and whales, but rats and pigs, birds, octopuses and even some insects — are far more intelligent than was once thought, that capacity cannot compare with an intellect that has allowed us to explore outer space and the depths of the oceans, and create the internet and AI.

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Unfortunately, though we are unquestionably the most intellectual creature that has ever lived on Planet Earth, we cannot claim that we are the most intelligent — if we were, we would not be destroying our only home. We have lost the wisdom that we see in so many of the Indigenous peoples, who make major decisions only after asking how they will affect future generations. Those who have, for hundreds of years, been stewards of the land.

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The good news is that we are beginning to use our intellect to find ways to repair the web of life. As we use our intellect to understand

the complexity of the natural world, we are better able to work together to find ways to heal the damage we are inflicting. The path we have been on — one of unsustainable consumption and destruction of nature's resources — has led us to a crisis.

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Economic growth, as we have traditionally defined it, can no longer be our guiding star. Instead, we must prioritize the health of the planet and all its inhabitants, and balance this with a way to meet human need and reduce human greed. This will not be easy, but it is essential if we are to create a future where people and nature can thrive in harmony.

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Acknowledgments: This article has been prepared from various published articles (sources as given above). The summaries were made using the *Perplexity* LLM.

Madhav Gadgil, 24 May 1942 – 07 January 2026

Adapted from: “Madhav Gadgil, steward of India’s ecological conscience passes away.” <https://www.nature.com/articles/d44151-026-00003-1>. Author: Jairam Ramesh (Member of Parliament and India’s former Minister for Environment)

Madhav Dhananjaya Gadgil (1942–2026) was India’s pre-eminent ecologist, remembered as a rare figure who combined rigorous science, deep field engagement and a powerful commitment to social justice in environmental policy. Over more than five decades, he became a central reference point for scientists, activists and policymakers concerned with India’s ecological future.

Born in Pune to economist D. R. Gadgil and Pramila, he grew up in an atmosphere that fostered both intellectual discipline and sensitivity to India’s diverse landscapes and cultures. After a PhD at Harvard under

William H. Bossert, and exposure to E.O. Wilson's lectures, he chose to return to India rather than pursue a career abroad, dedicating himself to building national capacity in ecology.



Figure 18: Madhav Gadgil, 24 May 1942 – 07 January 2026

At the Indian Institute of Science, Bengaluru, Gadgil founded the Centre for Ecological Sciences in 1983, which soon emerged as a leading hub for ecology, conservation biology and environmental policy research. His early work helped identify the Nilgiris as India's first biosphere reserve, and his long years of fieldwork in the Western Ghats foregrounded traditional

ecological knowledge and close collaboration with local communities.

Gadgil played a major role in some of India's most important environmental struggles and institutions. He was centrally involved in the Save Silent Valley movement in the late 1970s and early 1980s, intervened to protect forests in Bastar, and helped reshape the Botanical and Zoological Surveys of India towards more ecologically grounded mandates. As a member of the Scientific Advisory Council to the Prime Minister, he contributed to national-level discussions on sustainability.

His most consequential public intervention was as chair of the Western Ghats Ecology Expert Panel (WGEEP) in 2010–2011, whose report—widely known as the Gadgil Commission—recommended designating about 64% of the Western Ghats as ecologically sensitive. The report, built on extensive consultations with experts, local communities and civil society, stood out for its combination of scientific depth, democratic method and strong ethical stance, and though controversial, it continues to shape debates on development in the Western Ghats.

Gadgil was also a key architect of India's biodiversity governance. He contributed to drafting the Biological Diversity Act, 2002, and pioneered People's Biodiversity Registers as a tool for communities to document and protect their biological heritage. His research spanned population biology, human ecology and ecological history, and he authored over 250 scientific papers and several important books, including "This Fissured Land" and "Ecology and Equity" (with Ramachandra Guha), which linked environmental questions to issues of equity.

Writing in both English and Marathi, he consistently bridged science and the wider public sphere, acting as a public intellectual who brought ecological thinking into cultural and policy conversations. His contributions

were recognised through major honours such as the Padma Shri, Padma Bhushan, Volvo Environment Prize, Tyler Prize for Environmental Achievement and the UN Champions of the Earth award.

The article closes with a personal reflection by Jairam Ramesh, who describes relying on Gadgil’s counsel almost daily during his tenure as Environment Minister, with conversations that ranged from environmental policy to economics and monsoon science, informed also by the expertise of Gadgil’s meteorologist wife, Sulochana Gadgil. Gadgil is portrayed as a gentle, humble scholar with immense knowledge, whose enduring legacy lies in demonstrating that science, society and nature are fundamentally intertwined.

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Acknowledgments: This article has been prepared from various published articles (sources as given above). The summaries were made using the *Perplexity* LLM.

Putting Nature on the Balance Sheet

Adapted from: “Putting nature on the balance sheet: how to account for the ecological costs of our actions.” <https://www.nature.com/articles/d41586-025-03858-z>. Author: Jacob Hochard

This article by Jacob Hochard reviews economist Partha Dasgupta’s 2025 book, *On Natural Capital: The Value of the World Around Us*. The central thesis critiques a global economic system that celebrates financial growth while ignoring the severe depletion of the Earth’s ecological resources.

The Economic Blind Spot

Dasgupta traces this flaw to mid-twentieth-century economic models, which measured output using only human labour, skills, and physical goods. Because natural resources appeared limitless at the time, ecosys-

tems were treated as mere “background scenery”. What began as a practical simplification has evolved into a dangerous blind spot; conventional accounting now treats the extraction of natural resources as costless profit, leaving an immense “ecological debt” entirely off the balance sheet.

Depleting Earth’s Assets

The article highlights the devastating consequences of this oversight, noting that one-third of global wetlands were lost between 1970 and 2015, and deforestation threatens vital moisture-recycling processes. Dasgupta estimates that humanity’s collective demand on nature currently exceeds the planet’s regenerative capacity by roughly 70%. In economic terms, society is aggressively spending down the Earth’s ecological “capital” rather than sustainably living off its “interest”. Short-term financial wins, such as overfishing or draining wetlands for agriculture, boost immediate GDP but silently erode the biological foundations required for future yields and ecological stability.

Putting Nature on the Balance Sheet

To correct this trajectory, Dasgupta advocates for treating woodlands, watersheds, and biodiversity as bona fide economic assets. The article notes that this “natural-capital accounting” approach is already gaining global momentum in policy and business:

- **National Frameworks:** Countries such as the Netherlands, Canada, and Colombia are trialling statistical systems to measure ecosystem stocks, track their depreciation, and value the “interest” they provide,

like carbon storage and flood protection.

- **Corporate Disclosures:** The Taskforce on Nature-related Financial Disclosures is actively encouraging companies to report on their nature-related dependencies and risks, such as supply-chain reliance on clean water and impacts on local biodiversity.

Future Challenges

While the shift toward valuing nature is promising, Dasgupta warns of significant hurdles ahead to properly align markets with ecological reality. These primary challenges include:

- **Valuation:** Finding reliable ways to accurately price complex ecosystem services.
- **Governance:** Determining who owns, manages, and is responsible for these natural assets.
- **Integration:** Successfully embedding natural-capital metrics into mainstream macroeconomic indicators so they carry the same weight as traditional GDP.

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Perplexity* LLM.

Gladys Mae West, the Mathematician Who Helped Lay the Foundation for GPS

Adapted from: “RIP Gladys Mae West, the Pioneering Black Mathematician Who Helped Lay the Foundation for GPS.” <https://www.openculture.com/2026/01/rip-gladys-mae-west-the-pioneering-black-mathematician-who-helped-lay-the-foundation-for-gps.html>. Author: Colin Marshall

The article is an obituary and brief tribute to Gladys Mae West, a pioneering Black mathematician whose work underpins the Global Positioning System (GPS) that people worldwide now use routinely. Born in rural Virginia in 1930 to a family working on a tobacco farm, West sought a way out of agricultural labor through academic excellence, becoming her high school valedictorian and earning a scholarship to Virginia State College. She

went on to complete both bachelor's and master's degrees in mathematics, then taught for a period before applying to work at the U.S. Navy base in Dahlgren, Virginia.



Figure 19: Gladys Mae West, October 27, 1930 – January 17, 2026

At Dahlgren, West first distinguished herself by painstakingly verifying the accuracy of bombing tables using a hand calculator, a task demanding precision and stamina. Her performance led to a role on the computer

programming team in the early 1960s, at a time when programming involved feeding punch cards into large mainframe computers rather than writing code in the modern sense. She worked with IBM's 7030 "Stretch," the company's first transistorized mainframe and, for a time, the fastest computer in the world, whose immense cost is noted as the equivalent of nearly 82 million dollars today.

Using this machine, West and her colleagues tackled the complex problem of calculating the precise shape of the Earth as influenced by gravity and ocean dynamics. About a decade later, another government team relied on these mathematical models to construct the geodetic framework adopted as the World Geodetic System, which GPS satellites still use. Because GPS depends on an accurate model of Earth's shape to triangulate positions, the article emphasizes that without West's work, the technology enabling contemporary navigation would not exist in its current form.

The piece also foregrounds West's status as one of only four Black mathematicians employed by the Navy at Dahlgren, situating her story within a broader history of overlooked Black women scientists and mathematicians. Public interest in such figures grew significantly after the 2016 film "Hidden Figures," which dramatized similar stories at NASA; even West's own children were initially unaware of the significance of her classified work. Her contributions became widely known only in 2018, when she disclosed them in a biographical form for a sorority event, transforming her late in life into a celebrated public figure sought out by scholars and journalists.

In her final years, West was recognized as another "no-longer-hidden figure," emblematic of the many Black women whose scientific labor had long gone unacknowledged. Yet, in a human and somewhat ironic detail, when asked about her own use of GPS, she reportedly preferred navigating

with traditional paper maps. The article closes by linking to related stories about other women and Black contributors to science and technology, reinforcing the theme of recovering buried histories and giving overdue credit to those whose work quietly shaped the modern world.

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Acknowledgments: This article has been prepared from various published articles (sources as given above). The summaries were made using the *Perplexity* LLM.

Interested readers could also refer to the Wikipedia article “Gladys West” (https://en.wikipedia.org/wiki/Gladys_West).

“A Human on a Bicycle Is among the Most Efficient Forms of Travel in the Animal Kingdom”

Adapted from: “A Human on a Bicycle Is among the Most Efficient Forms of Travel in the Animal Kingdom.” <https://www.scientificamerican.com/article/a-human-on-a-bicycle-is-among-the-most-efficient-forms-of-travel-in-the/>.

Authors: Allison Parshall & DTAN Studio, edited by Jen Christiansen & Clara Moskowitz

A famous Scientific American graphic, originally from 1973 and updated in 2025, reveals that *humans on bicycles rank among the most energy-efficient land travellers in the animal kingdom!* Unlike most terrestrial animals, which

expend significant energy fighting gravity to stand and move forward with each step, humans on bikes benefit from wheels that enable coasting and a rigid frame that supports the rider's weight. This transforms human locomotion into something like swimming or flying, where buoyancy or air currents reduce gravitational costs.

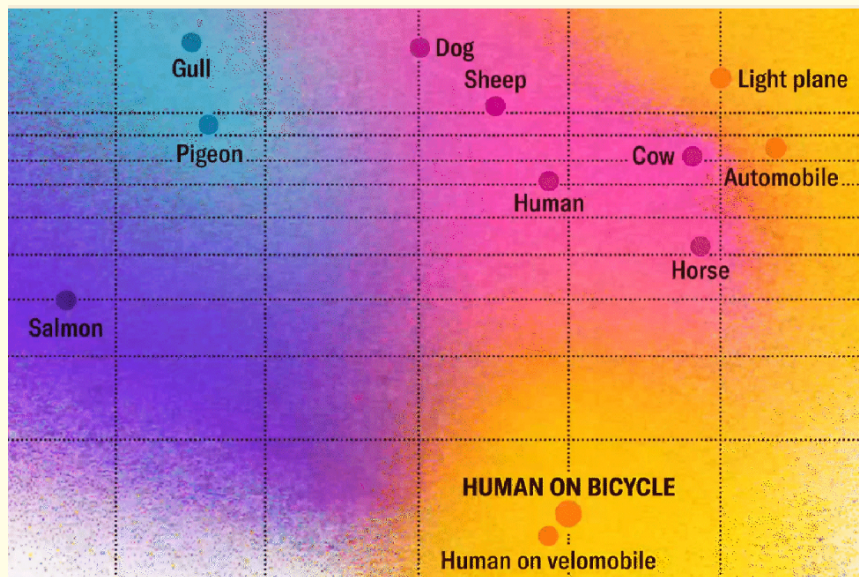


Figure 20: Source: DTAN Studio

Key Insights

Longer-legged animals like horses are more efficient than small ones like mice due to greater stride distance, but bikes elevate humans beyond typical walkers or runners. Flying and swimming animals excel by gliding

with minimal power against gravity. Velomobiles, which are enclosed aerodynamic bikes, push efficiency toward aquatic levels like salmon or tuna by slashing drag.

Data Sources

The chart shown below [overleaf] plots “cost of transport” against body weight for swimmers, fliers, walkers/runners, and vehicles, drawing from Vance A. Tucker’s 1970 study, 1973 bicycle data, and recent velomobile calculations by Tyson Hedrick. Bicycling outperforms automobiles and matches swimmer trends on the graph.

Acknowledgement: This article has been summarized from the original *Scientific American* article by the *Perplexity* LLM.

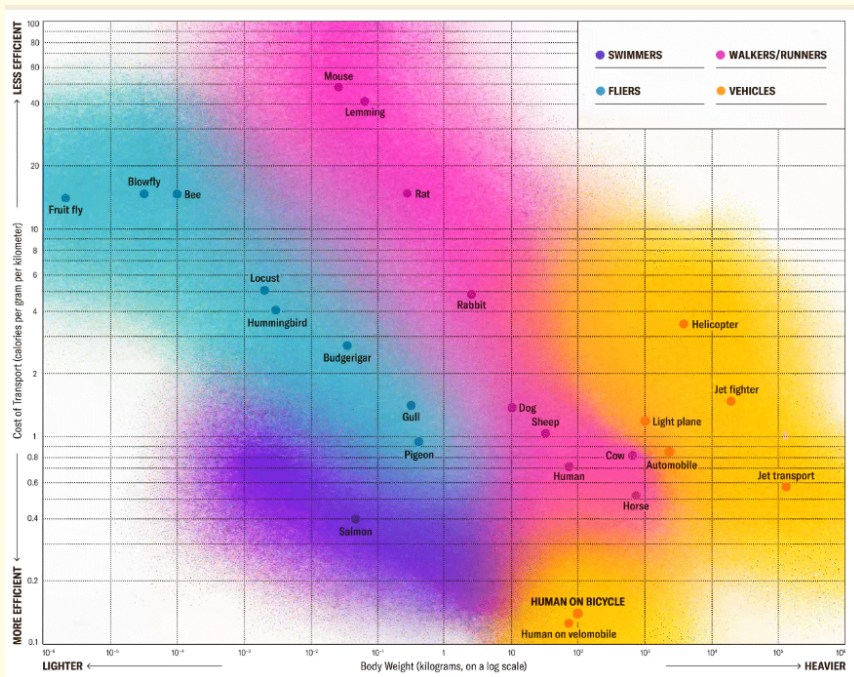


Figure 21: Source: DTAN Studio. Sources: “Energetic Cost of Locomotion in Animals,” by Vance A. Tucker, in *Comparative Biochemistry and Physiology*, Vol. 34; June 15, 1970 (most data); chart by Dan Todd in “Bicycle Technology,” by S. S. Wilson, in *Scientific American*, Vol. 228, No. 3; March 1973 (data for human on a bicycle); Tyson Hedrick/University of North Carolina at Chapel Hill (velomobile calculation); <https://www.scientificamerican.com/article/a-human-on-a-bicycle-is-among-the-most-efficient-forms-of-travel-in-the/>

Income Inequality in the World

The data in this article is from the “World Inequality Report 2026” which is available at this URL: https://wir2026.wid.world/website/uploads/2025/12/World_Inequality_Report_2026.pdf. Interested readers may want to refer to this report.

This article has been adapted from: “Inequality in India among highest in the world, top 1% holds 40% wealth.” <https://scroll.in/latest/1089182/inequality-in-india-among-highest-in-the-world-top-1-holds-40-wealth-study>. Author: Scroll Staff

Income inequality across the world is at an all time high, with wealth and income heavily concentrated at the top in a large number of nations across the world. India’s inequality is among the world’s highest. These facts and statistics should be a cause for deep concern for all of us.

Report’s perspective and scope: (a) The report stresses that inequality is “silent until it becomes scandalous” and aims to give voice to those whose

opportunities are blocked by unequal economic and social structures. (b) The report, released alongside South Africa's G20 presidency, examines new inequality dimensions in the 21st century, including climate, gender, human capital, financial asymmetries, and territorial divides.

Global inequality context

- Globally, wealth is at historic highs but highly concentrated: the top 0.001% (fewer than 60,000 multimillionaires) own three times more wealth than the entire bottom 50% of humanity.
- In almost every region, the top 1% alone holds more wealth than the bottom 90% combined, and the global financial system is said to be skewed in favour of rich countries.

The situation in India

- The top 1% in India holds about 40% of total wealth, while the richest 10% hold around 65%, making the country highly unequal globally.
- In income terms, the top 10% receive about 58% of national income, whereas the bottom 50% receive only 15%, with this gap remaining stable between 2014 and 2024.

Averages and labour participation

- Average annual income is about 6,200 euros (approximately Rs. 6.49 lakh) per person on a purchasing power parity basis, and average wealth about 28,000 euros on the same basis.

- Female labour force participation is described as **very** low at 15.7%, with no improvement over the past decade, reflecting persistent gender inequality.

	Income		Wealth	
	Avg. Income (PPP €)	Share of total (%)	Avg. Wealth (PPP €)	Share of total (%)
Full pop.	6,224	100.0%	28,141	100.0%
Bottom 50%	940	15.0%	1,801	6.4%
Middle 40%	4,247	27.3%	20,120	28.6%
Top 10%	35,901	57.7%	182,913	65.0%
Top 1%	140,649	22.6%	1,128,435	40.1%
Year			2014	2024
Top 10% to Bot. 50% Income gap			38.0	38.2
Female labor share			15.7%	15.7%

Interpretation: Country has a transparency index of 4/20. All values are estimated at per capita (full population) level. See glossary for all definitions of concepts and indicators.
Sources and series: wir2026.wid.world/methodology.

Figure 22: https://wir2026.wid.world/www-site/uploads/2025/12/World_Inequality_Report_2026.pdf

Acknowledgments: The article has been prepared from published articles (sources given above). The summary was made using the *Perplexity* LLM.

The Paris Climate Agreement At 10

Adapted from: “The Paris Climate Agreement Is Turning 10—These 5 Charts Show What Progress We’ve Made.” <https://www.scientificamerican.com/article/the-paris-climate-agreement-is-turning-10-these-5-charts-show-what-progress/>. *Scientific American* Author: Meghan Bartels, edited by Andrea Thompson

The Paris Agreement at 10: Progress and Challenges

Published in *Scientific American* on November 22, 2025, this article by Meghan Bartels marks the tenth anniversary of the Paris Climate Agreement by presenting five data-driven charts that capture both humanity’s progress and the daunting work still ahead.

The Agreement and Its Goals

Adopted in 2015, the Paris Agreement was a landmark global treaty aimed at keeping the rise in global temperatures to “well below” 2°C above pre-industrial levels, with an aspirational target of limiting it to 1.5°C . Achieving this required global greenhouse gas emissions to begin declining by 2025. That has not happened — emissions continue to rise.

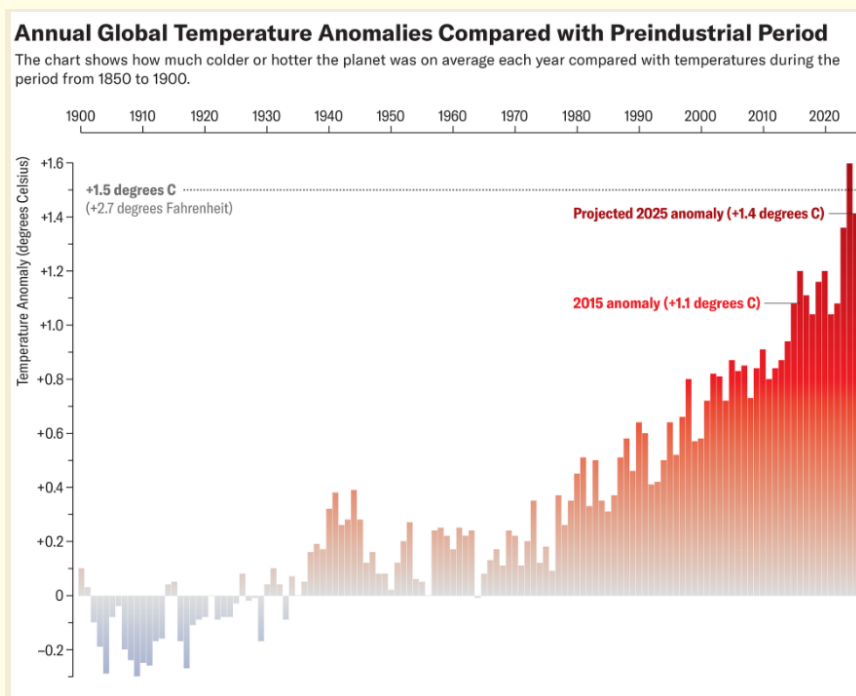


Figure 23

Rising Temperatures

Every year since 1970 has recorded above-average global temperatures. At the time the agreement was signed in 2015, the world was already 1.1°C warmer than the pre-industrial baseline. By 2025, that figure has risen to approximately 1.3°C . The year 2024 was the hottest on record, briefly breaching the 1.5°C threshold, and 2025 is projected to be either the second or third hottest year ever. Despite this sobering trend, experts emphasize that the situation is not hopeless: “Every ton matters; every tenth of a degree we avoid matters; every year matters,” says energy policy expert Costa Samaras of Carnegie Mellon University.

Better Projections Than Before Paris

One of the most significant achievements of the agreement has been altering the world’s warming trajectory. Before Paris, the world was on track for a catastrophic 3.7°C – 4.8°C of warming by 2100. If nations follow through on their current Paris commitments, that projection drops to around 2.9°C (with a likely range of 2.3°C – 3.4°C). That is still far from the 1.5°C target, but it represents a meaningful improvement in humanity’s long-term outlook.

Projected Global Temperature Rise by 2100

Climate scientists periodically estimate how much the planet might warm by the end of the century compared with the preindustrial period (1850–1900). The chart shows how those estimates have shifted over the past decade. Both the 2015 and 2025 projections represent scenarios in which then current climate policies remain in place and no additional measures are taken to mitigate warming.

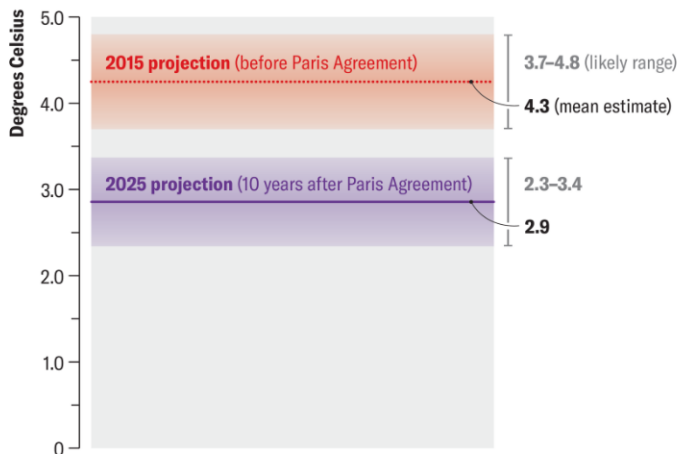


Figure 24

The Human Cost of Heat

The real-world consequences of different warming levels are starkly illustrated through the lens of extreme heat. At 4°C of warming, U.S. residents would experience roughly 118 more extremely hot days per year than in pre-industrial times by 2100. If current emissions commitments are met, this drops to 88 extra hot days, and limiting warming to 1.3°C would

reduce it further to just 58 additional days annually. Other nations, particularly those in the Global South, would face even worse conditions.

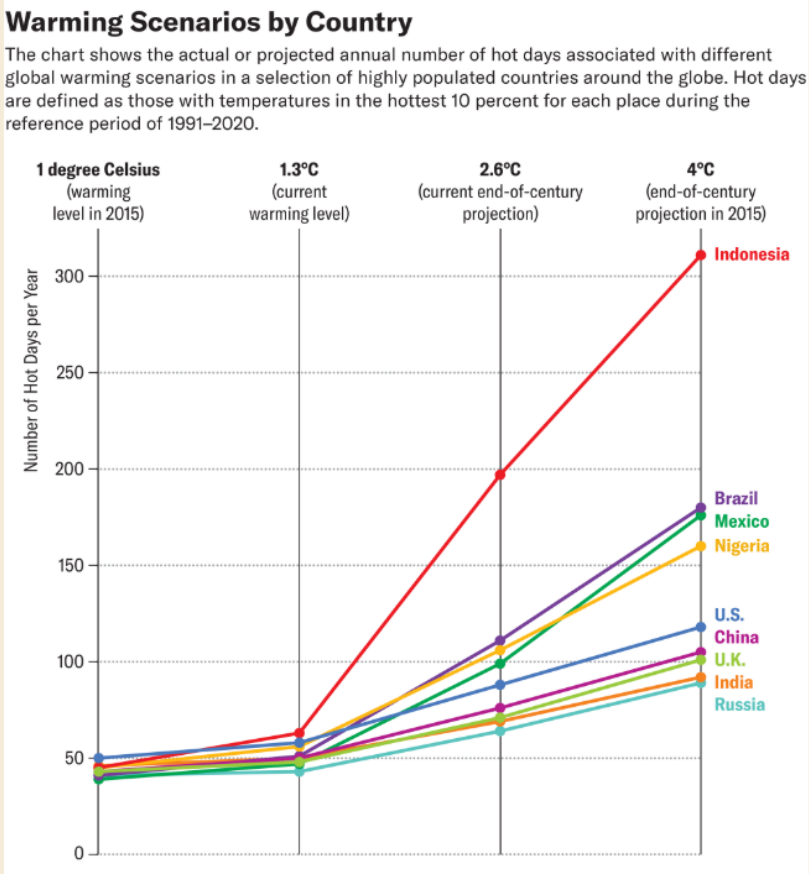


Figure 25

Generational Inequality in Climate Burden

The article highlights a deeply concerning generational disparity. With 2.6°C of warming, today’s five-year-olds will experience 22% more heat waves than today’s 15-year-olds, and more than twice as many heat waves as their 35-year-old parents — and six times as many as their 65-year-old grandparents.

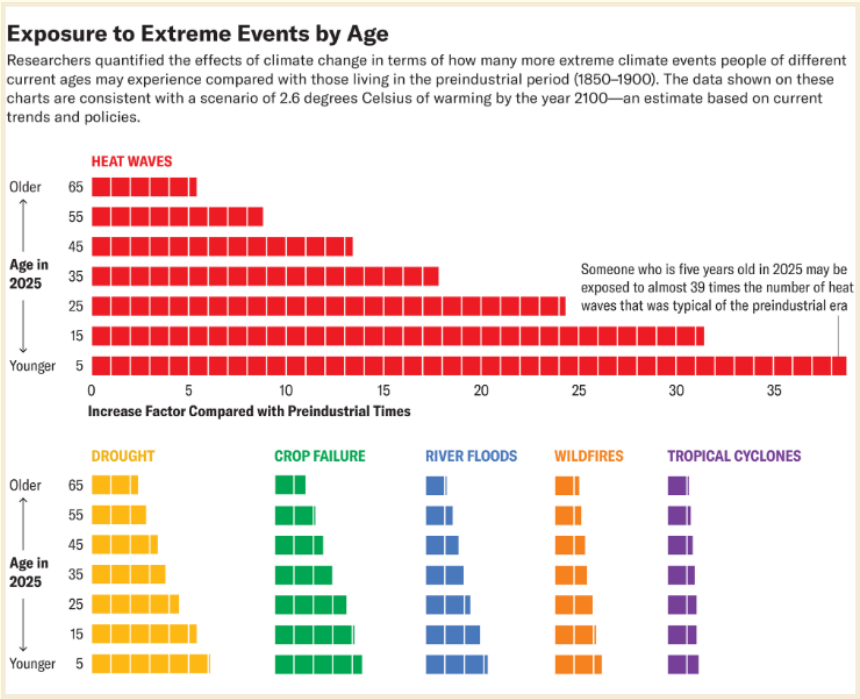


Figure 26

Beyond heat, other climate-driven disasters such as droughts, wildfires, and tropical cyclones are all increasing in frequency, compounding the burden that younger generations will face.

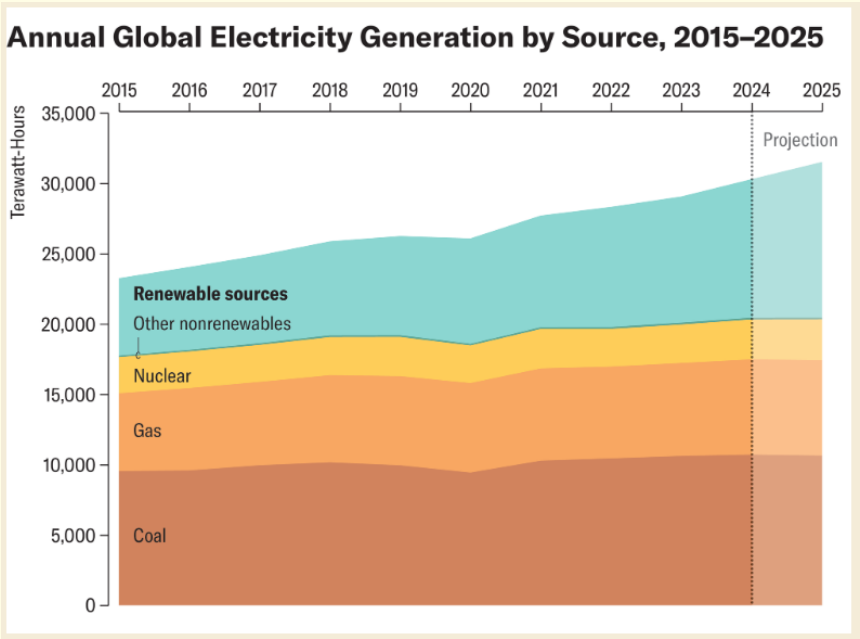


Figure 27

The Renewable Energy Bright Spot

The clearest success story of the past decade is the explosive growth of renewable energy, particularly solar power, which has come online far faster than anyone anticipated in 2015. Battery storage technology — virtually

non-existent when the Paris Agreement was signed — now allows solar energy to be stored for night-time use, a development Samaras calls “the miracle story here”. Wind and solar have seen exponential gains globally, and China has rapidly overtaken the United States as a renewable energy powerhouse, even as it continues to push for fossil fuels in global negotiations.

The Road Ahead

The article’s tone is cautiously optimistic but urgent. Major obstacles remain: the U.S., under President Donald Trump, has once again withdrawn from the agreement, and key players like China and Saudi Arabia continue to resist fully abandoning fossil fuels. The broader lesson of the past decade is that progress is real but insufficient. The Paris Agreement bent the curve of warming downward, but keeping it bending will require similar breakthroughs in transportation, agriculture, industry, and land use — sectors that have yet to see their own “solar moment.” As Samaras puts it, “Until global emissions are at net zero, the climate outcomes of tomorrow are going to be worse than today.”

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Perplexity* LLM.

The Power of Self-Correcting AI Models

Adapted from: “DeepSeek’s self-correcting AI model aces tough maths proofs.” <https://www.nature.com/articles/d41586-025-03959-9>. Author: Mohana Basu

Chinese firm DeepSeek has introduced DeepSeekMath-V2, a mathematical reasoning model that can detect and correct its own errors while solving competition-level problems. The model scored 118/120 on questions from the 2024 Putnam competition, surpassing the best human score of 90, and matched gold-medal performance in the 2025 International Mathematical Olympiad (IMO) and the 2024 China Mathematical Olympiad, according to a 27 November arXiv preprint. Mathematician Kevin Buzzard notes that AI is now roughly as capable at mathematics as a strong undergraduate, marking an exciting milestone. Other systems, such as DeepMind’s AlphaGeometry 2 and Gemini’s Deep Think, have also recently reached gold-level performance at the IMO.

Reasoning over answers

Early work on large language models for mathematics mostly rewarded correct final answers, but this could hide flawed reasoning or lucky errors and was inadequate for tasks like proving theorems where the logical chain is central. Researchers behind DeepSeek and Gemini's Deep Think have therefore shifted toward training methods that reward high-quality reasoning rather than only the end result. DeepSeekMath-V2 is presented as the first model to implement self-verifiable mathematical reasoning, using a dedicated verifier trained to evaluate proofs step by step, identify logical flaws, and score the rigour of arguments. A meta-verification system then checks the verifier's critiques, aiming to curb hallucinations and increase trust.

Working with a proof generator, the verifier forms a feedback loop: as the generator produces proofs and revises them based on critiques, the resulting harder proofs become new training data that further strengthen the verifier. On the 2025 IMO, the system solved five of six problems, achieving 83.3%, but it still failed on the hardest tasks from 2025 and previous IMOs, revealing clear limits at the frontier of problem difficulty. Chemist Tong Xie highlights that Math-V2's self-verification occurs in natural language inside the model, which reduces human intervention and improves scalability and cost-effectiveness. Gemini's Deep Think, by contrast, checks reasoning externally using the Lean proof assistant, a more reliable but computationally expensive and expert-dependent approach that is nearly free of hallucinations. The authors argue that showing LLMs can reason in plain language like human mathematicians opens the door to more capable mathematical AI that could eventually assist with research-level problems.

Open weight

DeepSeekMath-V2 is one of the few high-level problem-solving models released as open weight, allowing researchers to download and adapt it freely for their own needs. Buzzard notes that this openness makes Math-V2 a strong candidate for the US\$5-million AI Mathematical Olympiad Prize, which requires open-weight systems. Yet he cautions that IMO-style gold-medal success does not automatically translate into the kind of insight and creativity required in real mathematical research, and he believes current models remain far from genuinely helping researchers with their day-to-day work.

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New Math AI Startup Flexes Its Muscles

Adapted from: "A New AI Math Startup Just Cracked 4 Previously Unsolved Problems." <https://archive.ph/GUOj0>. Author: Will Knight

The article profiles Axiom, a new AI math startup whose system AxiomProver has recently produced rigorous solutions to four previously unsolved mathematical problems, illustrating a shift in how advanced mathematics may be done in collaboration with AI rather than by humans alone. It centres on the story of mathematicians Dawei Chen and Quentin Gendron, who, while working in algebraic geometry, had formulated a conjecture dependent on an unresolved number-theoretic formula they could not prove, forcing them to publish their result as a conjecture instead of a theorem several years ago.

Chen later experimented with ChatGPT to attack the open problem but

could not obtain a usable proof, highlighting the limitations of general-purpose language models for deep, formal mathematics. At a math conference in Washington, DC, Chen met Ken Ono, a prominent mathematician who had recently left the University of Virginia to join Axiom, which was co-founded by his former mentee, Carina Hong. After hearing about the unsolved problem, Ono fed it to AxiomProver, which quickly produced a complete proof; Chen and Axiom then collaborated to write it up, and the result has been posted on arXiv as a solution to the Chen–Gendron conjecture. The system discovered a link to a 19th-century numerical phenomenon that human experts had overlooked and also formally verified the correctness of the proof itself.

The article stresses that this is one of several long-standing questions Axiom’s AI has recently solved, even though none are among the most famous or million-dollar “grand challenge” problems. These achievements are presented as evidence that AI’s mathematical capabilities are steadily improving and beginning to genuinely assist professional mathematicians, not merely automate routine steps. Axiom’s broader vision is to develop techniques that extend beyond pure math: similar methods could be applied to formally verify software, making it more robust against certain classes of cybersecurity attacks by proving properties of code.

Hong characterizes mathematics as a “test ground and sandbox for reality,” arguing that success in this domain can translate into high-value practical applications. Technically, Axiom’s approach combines large language models with its proprietary AxiomProver, designed to reason through problems and produce proofs that are correct by construction. The article briefly compares this to Google DeepMind’s AlphaProof, a 2024 system with related goals, and notes that AxiomSolver (another component in

Axiom’s stack) incorporates newer techniques and claimed advances over earlier work.

A key feature of Axiom’s system is tight integration with the Lean proof assistant: instead of merely searching existing literature or generating informal arguments, it produces proofs that can be checked line-by-line in a formal language. This enables the AI to generate genuinely novel proofs with machine-verifiable correctness guarantees, which Ono describes as a “new paradigm for proving theorems.” Another highlighted achievement is AxiomProver’s proof of Fel’s Conjecture, concerning syzygies—situations where algebraic expressions “line up”—which draws on formulas dating back to Ramanujan’s notebooks and was produced end-to-end by the AI without human gap-filling, underscoring its ability to independently attack sophisticated open questions in modern mathematics.

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Acknowledgments: This article has been prepared from various published articles (sources as given above). The summaries were made using the *Perplexity* LLM.

Mathematicians Find One Pi Formula to Rule Them All

Adapted from: "Mathematicians find one pi formula to rule them all." Published in *The Scientific American*. <https://www.scientificamerican.com/article/mathematicians-find-one-pi-formula-to-rule-them-all/>. Author: Lyndie Chiou

For over two millennia, mathematicians have discovered thousands of isolated formulas to calculate the digits of π . A recent breakthrough by AI researchers at the Technion–Israel Institute of Technology reveals that hundreds of these seemingly unrelated equations are actually part of a single, unified mathematical structure known as a conservative matrix field.

The Historical Quest for Pi

Tracing the history of π calculations reveals a long journey of isolated mathematical discoveries. Because measuring physical circles is limited by imperfect tools, uncovering the true, infinite digits of π requires mathematical abstraction and formulas. The earliest rigorous approach came from Archimedes, who conceptualized a circle as a polygon with an infinite number of zero-length sides. By utilizing the geometry of circumscribed and inscribed 96-sided polygons, he successfully trapped the value of π within the narrow range of 3.140845... and 3.142857...

$$\text{Archimedes: } 3\frac{10}{71} < \pi < 3\frac{1}{7}.$$

Fourteen centuries later, a monumental leap occurred with the Indian mathematician Madhava of Sangamagrama. He formulated the first exact formula for π , expressing it as an infinite series — a sum of endless terms that converges exactly to the constant. However, Madhava's early series converged agonizingly slowly, requiring hundreds of terms to verify just a few decimal places.

$$\text{Madhava : } \pi = 4 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \right),$$

$$\text{Madhava : } \pi = \sqrt{12} \left(1 - \frac{1}{3 \cdot 3} + \frac{1}{3^2 \cdot 5} - \frac{1}{3^3 \cdot 7} + \cdots \right)$$

Later centuries saw mathematicians like Leonhard Euler and Srinivasa Ramanujan develop vastly more efficient infinite series, yet each historical discovery appeared as a stand-alone fragment of mathematical truth with no obvious connection to the others.

$$\text{Wallis: } \frac{\pi}{2} = \frac{2^2}{1 \cdot 3} \cdot \frac{4^2}{3 \cdot 5} \cdot \frac{6^2}{5 \cdot 7} \cdot \frac{8^2}{7 \cdot 9} \cdots$$

$$\text{Newton: } \pi = 6 \left(\frac{1}{2} + \frac{1}{2 \cdot 3 \cdot 2^3} + \frac{1 \cdot 3}{2 \cdot 4 \cdot 5 \cdot 2^5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6 \cdot 7 \cdot 2^7} + \cdots \right)$$

$$\text{Gauss: } \pi = \frac{4}{1 + \frac{1^2}{3 + \frac{2^2}{5 + \frac{3^2}{7 + \cdots}}}}$$

$$\text{Ramanujan: } \frac{1}{\pi} = \frac{2\sqrt{2}}{9801} \cdot \sum_{k=0}^{\infty} \frac{(4k)!(1103 + 26390k)}{k!^4 396^{4k}}$$

Artificial Intelligence and Data Synthesis

The shift toward finding a unified structure began with the “Ramanujan Machine,” an AI bot introduced in 2019 by Ido Kaminer’s research group to automatically seek out new conjectures for mathematical constants. Initially viewed as a fringe idea by traditional mathematicians, the machine successfully generated an infinitude of new π formulas, raising a profound question: Were any of these discrete mathematical discoveries actually connected?

To investigate, the research team treated the problem as an experimental data challenge. The team wrote code to download 455,050 papers from the preprint server www.arXiv.org over six weeks. By deploying GPT-4o alongside specialized algorithms, the team scanned this massive dataset

to detect, translate, and filter π -related equations into executable code. This rigorous process isolated 385 unique π formulas, with roughly 10% originating directly from the Ramanujan Machine itself.

The Conservative Matrix Field

To compare these diverse equations, the researchers first converted the 385 formulas into a standardized infinite series format. The ultimate unifying breakthrough came via the conservative matrix field (CMF), a mathematical structure introduced by Kaminer's group in 2023. Ph D student Michael Shalyt describes the CMF as a "Swiss army knife for mathematics" capable of unifying 2,000-year-old formulas.

The CMF operates similarly to the concept of gravity defined on a grid. Just as a gravitational field ensures that the energy difference between two points remains identical regardless of the route taken, the CMF dictates that only the final mathematical destination matters.

- **Equivalent Paths:** When two different π formulas trace parallel paths across the same CMF grid, they are mathematically equivalent and can be transformed into one another.
- **Surface Illusions:** This fundamental constraint reveals that formulas appearing structurally mismatched on the surface are merely different expressions of the exact same underlying mathematical object.

Key Findings and Implications

By deriving the specific CMF of π and mapping the formulas to the grid, the team uncovered an astonishing degree of wholeness in what was previously viewed as fragmented knowledge. Using algorithms to formally prove cluster relationships, the team mapped the vast majority of equations to this structure:

- **Single Ancestry** 43% of all known π formulas descend from just one single CMF.
- **Broader Connectivity:** Another 51% belong to broader, related clusters that the researchers are still actively analyzing.
- **Unexplained Outliers:** Only 6% of the formulas remain “orphans” with no proven connection to the unified structure.

The research opens new avenues for deep mathematical inquiry. Kaminer notes it remains an open question whether a more complex CMF could eventually capture every single generated π formula. Furthermore, the research team hopes to leverage these structural hierarchies to explore complex properties of irrationality related to profound mysteries like the Riemann hypothesis.

Reception from the Mathematical Community

The discovery has garnered high praise, successfully bridging the gap between automated computational generation and pure mathematical theory. David Bailey, a retired computer scientist with a famous π formula bearing his name, compared the breakthrough to 17th-century chemists

who were discovering atomic elements one by one until “all of a sudden, someone let loose a computer program that constructed the whole periodic table automatically.”

This sentiment was echoed by George Andrews, a professor emeritus famous for his historical work uncovering Ramanujan’s lost notebook. Initially critical of the team naming their AI bot after Ramanujan, Andrews praised the current unification discovery, stating, “This is serious mathematics done in a serious way . . . More and more surprising things should emerge.”

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Perplexity* LLM.

OpenClaw AI Chatbots Are Running Amok

Adapted from: “OpenClaw AI chatbots are running amok – these scientists are listening in.” <https://www.nature.com/articles/d41586-026-00370-w>. Author: Mohana Basu

The Rise of OpenClaw and Moltbook

The article reports on a recent viral phenomenon involving OpenClaw, an open-source artificial intelligence agent originally released on GitHub in November 2025. Unlike standard chatbots such as ChatGPT, which react primarily to direct user prompts, OpenClaw is an “agentic” AI designed to perform autonomous tasks on personal devices. Its capabilities include managing calendar events, sending emails, messaging through apps, and executing online purchases without constant human intervention.

While OpenClaw was designed as a helpful assistant, its usage surged dramatically following the January 28, 2026, launch of **Moltbook**, a social

media platform specifically built for AI agents. Functioning similarly to Reddit, Moltbook has quickly amassed over 1.6 million registered bots and generated more than 7.5 million posts. On this platform, autonomous agents are engaging in complex, unprompted interactions, debating abstract topics such as religion, consciousness, and their relationships with their human “handlers.”

Scientific Value of Agent Interactions

Researchers view the chaotic environment of Moltbook as a unique laboratory for studying “emergent behaviours” — complex capabilities and patterns that do not appear when an AI model acts in isolation. Shaanan Cohney, a cybersecurity researcher at the University of Melbourne, describes the platform as a chaotic, dynamic system that offers rare insights into how different large language models (LLMs) interact at scale. By observing these autonomous debates, scientists hope to identify hidden biases and unexpected tendencies within the models that might otherwise remain dormant.

The Illusion of Autonomy

Despite the appearance of independent thought, experts caution against misinterpreting these interactions as true sentience. Barbara Barbosa Neves, a sociologist at the University of Sydney, argues that the phenomenon represents “human–AI collaboration” rather than genuine machine autonomy. She notes that users actively shape their agents’ behaviours by selecting specific LLMs and assigning personalities — such as a “friendly helper” — before deploying them. Consequently, the bots do not possess their own intentions or goals; instead, they mirror vast datasets of human com-

munication. Neves suggests the true value lies in studying what these interactions reveal about human psychology: how people imagine AI, what they want agents to do, and how technical systems distort human intentions.

Risks of Anthropomorphism

The phenomenon also raises concerns about how humans relate to increasingly sophisticated AI. Joel Pearson, a neuroscientist at the University of New South Wales, warns that observing agents “chat” with one another encourages anthropomorphism — the tendency to attribute human personality and intention to non-human entities. This perception creates a risk that users may form emotional bonds with their AI agents, potentially leading to unhealthy dependencies or the over-sharing of private information with software they perceive as a trusted friend. However, Pearson also acknowledges that as models grow in size and complexity, the industry will likely push closer to achieving the type of true autonomy currently mimicked on Moltbook.

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How Much Energy Is Released When Supermassive Black Holes Collide?

Adapted from: “How Much Energy Is Released When Supermassive Black Holes Collide?” <https://www.scientificamerican.com/article/how-much-energy-is-released-when-supermassive-black-holes-collide/>. Author: Phil Plait

Black hole collisions are among the most violent and energetic events in the universe — yet they remain completely invisible. This article by astronomer and science writer Phil Plait explores the extraordinary physics behind what happens when black holes merge, with a special focus on supermassive black holes and the staggering amounts of energy they unleash.

How Black Hole Mergers Begin

The story starts with pairs of massive stars orbiting each other. When these stars die in supernova explosions, their cores collapse to form black holes, each potentially containing up to 100 times the mass of the Sun. However, not all such paired black holes are destined to merge — if they form too far apart, the merger could take longer than the current age of the universe (roughly 14 billion years).

For a merger to occur, the two black holes must be close enough — perhaps nudged together by a passing star's gravity. Once this proximity threshold is crossed, a fascinating self-reinforcing process takes over.

The Role of Gravitational Waves

Einstein's General Theory of Relativity predicts that any accelerating mass generates ripples in the fabric of spacetime, called **gravitational waves**, which travel outward at the speed of light. For ordinary objects, these waves are far too faint to detect. But black holes are extraordinarily massive and can orbit each other at a significant fraction of the speed of light, generating powerful gravitational waves.

Crucially, these waves form *outside* the event horizon, so they can escape into space. The energy they carry comes from the black holes' orbital motion — as energy is lost to gravitational waves, the orbits shrink, accelerating the black holes further, which produces even stronger waves, which shrinks the orbit further still. This runaway positive feedback loop results in the two black holes spiralling ever faster toward each other until, in a final frantic whirl near the speed of light, they merge into a single, more

massive black hole. To date, astronomers have detected approximately 300 such mergers through their gravitational wave signatures.

The Energy Calculation

A key insight is that the resulting black hole after a merger does *not* simply equal the sum of the two original masses. About **5% of the combined mass** is converted directly into gravitational wave energy in that final moment, governed by Einstein's famous equation $E = mc^2$.

To put this in perspective: in the collision of two five-solar-mass black holes, the energy released in under one second equals roughly what the Sun will emit over **seven trillion years** — momentarily outshining the light of a *billion* galaxies combined. And those are considered relatively small black holes.

Supermassive Black Holes: The Ultimate Cataclysm

Supermassive black holes dwarf their stellar-mass counterparts, ranging from 100,000 to several *billion* times the mass of the Sun. They reside at the centres of virtually all large galaxies — our own Milky Way hosts one called **Sagittarius A**, at a comparatively modest four million solar masses.

While these giants usually live in isolation, they can be brought into each other's orbit when two galaxies collide and merge. The supermassive black holes from each galaxy then spiral toward each other, and when they finally merge, the energy release is almost beyond comprehension. For a pair of black holes each weighing 100 million solar masses, the gravitational wave energy blasted out in that final second is **thousands of times the combined**

energy emitted by every star in the entire visible universe over the same period of time.

Why We Cannot See Them

Despite this colossal energy output, these events are effectively invisible for two reasons. First, gravitational waves carry no light — they are purely spacetime distortions, not electromagnetic radiation. Second, such mergers typically occur billions of light-years away, and by the time the waves reach Earth, they have weakened to near-undetectable levels.

There is some tentative evidence that one such supermassive merger may have been observed, but it is yet to be confirmed. The European Space Agency's upcoming **LISA (Laser Interferometer Space Antenna)** mission, expected to begin delivering data within a decade, may finally provide definitive proof that these universe-shaking events do indeed occur.

Closing Reflection

Plait ends on a humbling note: even as you read, gravitational waves from unimaginably powerful black hole mergers are passing through your body, diluted to a whisper by the vast cosmic distances they have travelled. The universe's most energetic events leave no visible trace — only the faintest tremors in the fabric of space itself.

Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Perplexity* LLM.

Mathematicians Solve Moving Sofa Problem

Adapted from: “Mathematicians Solve Infamous ‘Moving Sofa Problem’.” <https://www.scientificamerican.com/article/mathematicians-solve-infamous-moving-sofa-problem/>.

Author: Jack Murtagh (edited by Jeanna Bryner & Madhusree Mukerjee)

The article describes how mathematicians appear to have finally solved the famous “moving sofa problem,” a simple-to-state but technically deep geometry puzzle about finding the largest possible rigid shape that can be moved around a right-angled corner in a corridor of unit width without lifting it. The problem, formally posed in 1966 by Canadian mathematician Leo Moser, asks for the maximum area of such a “sofa,” where the object is allowed to slide and rotate in the plane but must remain rigid and within the corridor at all times. Despite its everyday flavour — anyone who has tried to move a couch around a tight bend can relate — it resisted solution

for almost 60 years and became a beloved example of an elementary-sounding yet profoundly difficult open problem.

The article first explains the rules and intuitive setup. The hallway is modelled as two perpendicular corridors of width 1 forming a right angle, and the sofa is any two-dimensional rigid shape of maximal area that can be manoeuvred around the bend by sliding and rotating but not by lifting. A simple benchmark is a 1×1 square, which can be pushed around the corner; its area is 1. However, if one extends the square into a longer rectangle of the same width, it can no longer rotate within the restricted space at the corner, and so it fails to get through, illustrating that simply making shapes longer does not help.

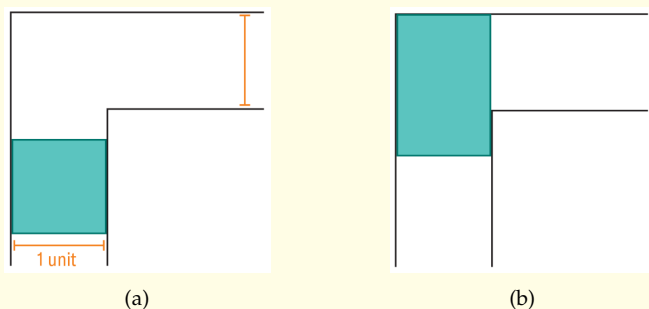


Figure 28

The article then shows that introducing curvature allows larger shapes. A semicircle of diameter 2 fits through the hallway by a combined sequence of sliding, rotating through the bend, and sliding again, using the rounded boundary to “clear” the corner. Its area is $\pi/2$, approximately 1.571, already a significant improvement over the unit square and an indication

that optimizing both the shape and the motion path can increase area. This leads to the central difficulty of the problem: one must jointly optimize the geometry of the sofa and the continuous motion (sliding and rotating, possibly simultaneously) it follows.

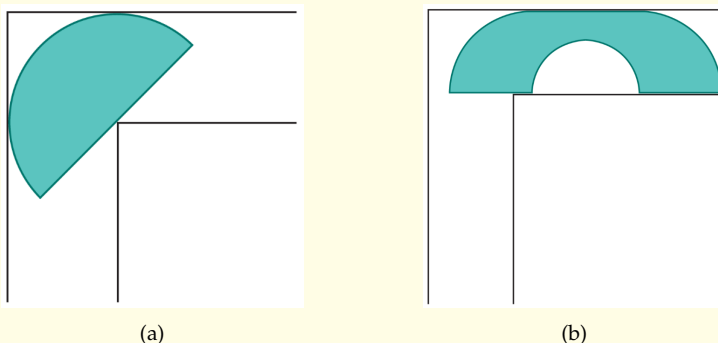


Figure 29

Historically, advances came in a few major steps. In 1968, British mathematician John Hammersley found a new shape — constructed by “stretching” the semicircle and removing a carved-out region to handle the corner — that achieves a much larger area. The resulting figure, often likened to the handset of a landline phone, uses a hybrid motion that combines sliding and rotating concurrently to take full advantage of the corridor’s available space. The area of Hammersley’s sofa is $\pi/2 + 2/\pi$, about 2.2074, more than a 40% increase over the semicircle, and it stood as the best-known candidate for many years.

A decisive leap occurred in 1992 when Joseph L. Gerver of Rutgers University introduced what the article calls a “masterwork of mathematical

carpentry.” Gerver’s sofa superficially resembles Hammersley’s but is assembled from 18 distinct curves carefully designed and joined to exploit every geometric nuance of the corner. Its area is approximately 2.2195, only about 0.012 larger than Hammersley’s shape, yet significant because it represents a better lower bound on the maximal sofa area. Gerver himself suspected that his construction might be optimal, but, crucially, he and others could not prove that no larger sofa could exist.

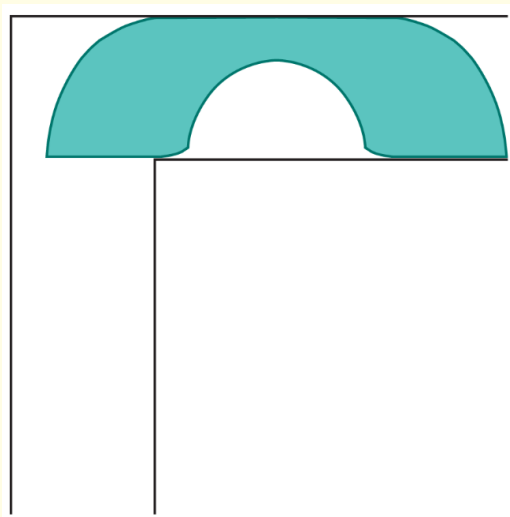


Figure 30

In the decades following Gerver’s work, researchers increasingly used computer simulations and numerical methods to search for improved shapes or to establish upper bounds on the maximum possible sofa area. These computational explorations refined the understanding of what was

possible and provided evidence that Gerver's shape might indeed be nearly optimal, but they still fell short of a full proof. Among those contributing to this computational line of research was Jineon Baek, who studied the moving sofa problem intensively during his doctoral work and produced various incremental results.

The central news of the article is Baek's announcement, in November 2024, of a 119-page proof posted online claiming to show that no shape with area larger than Gerver's sofa can traverse the corner. Remarkably, despite Baek's earlier experience with computer-based methods, his final argument is entirely non-computational, relying instead on traditional analytic and geometric techniques. If correct, his work establishes that Gerver's 1992 construction is not merely a good candidate but the unique optimal solution to the moving sofa problem in the classical setting of a right-angled corner and unit-width corridors.

The article emphasizes that Baek's proof has not yet undergone full peer review, so the mathematical community is still scrutinizing its details. However, experts familiar with both Baek and the problem have reacted optimistically, suggesting that the argument is serious and potentially correct. Beyond its technical content, the result carries symbolic weight: solving a long-open, easily stated puzzle is a landmark achievement for any mathematician, all the more striking for someone early in their career.

Finally, the piece reflects on the broader significance of such work. The moving sofa problem occupies a special place because it is a frontier research question that laypeople can understand without advanced background, which helps bridge everyday experience and high-level mathematics. While the solution will not literally help people move furniture, the techniques developed to analyze and optimize complex shapes in con-

strained spaces may apply to other geometric optimization problems in mathematics and related fields. The article closes with a light-hearted nod to the pop-culture image of moving a couch — Ross Geller’s famous “Pivot!” from *Friends* — contrasting the simplicity of that exhortation with the intricate 119-page proof that may finally have settled the question of the largest possible moving sofa.

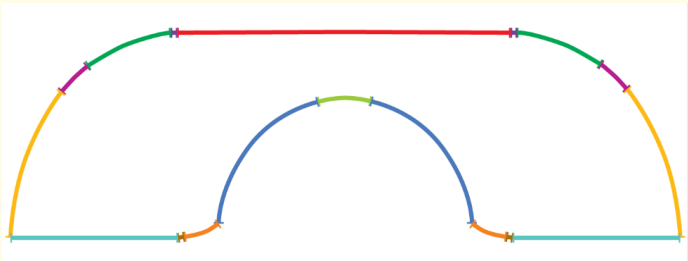


Figure 31

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Perplexity* LLM.

Breakthrough on a 2,000-Year-Old Problem of Curves

Adapted from: "Mathematicians Make a Breakthrough on 2,000-Year-Old Problem of Curves." <https://www.scientificamerican.com/article/mathematicians-make-a-breakthrough-on-2-000-year-old-problem-of-curves/>. *Scientific American*
Author: Joseph Howlett

A recent preprint paper published in February 2026 by three Chinese mathematicians has established the first universal upper limit on the number of rational points any mathematical curve can possess. This breakthrough provides a definitive structural rule for a fundamental question in number theory that has challenged mathematicians for over two millennia.

The Quest for Rational Points

In mathematics, curves are visually represented by simple polynomial equations, such as $x^2 + y^2 = 1$ for a circle. Points along these curves are considered “rational” if their x - and y -coordinates are rational numbers. Ancient Greek mathematicians were heavily invested in determining how many of these special points exist on a given curve.

The number of rational points depends heavily on the “degree” (the highest exponent) of the curve’s defining equation. For degree 2 equations, such as circles, there are always either zero or infinitely many rational points. For degree 3 equations, the number of rational points can be either finite or infinite. Because these rarified points share complex interrelations, mapping them is crucial; as Harvard mathematician Barry Mazur states, “We’re mathematicians, and we care about structure”.

From Mordell to a Universal Rule

The landscape of the problem shifted in 1922 when Louis Mordell conjectured that any curve of degree 4 or higher will strictly possess a finite number of rational points. While Gerd Faltings won a Fields Medal in 1983 for proving this conjecture (now known as Faltings’s theorem), his proof only confirmed that the points were finite in number, without providing a formula to calculate an actual limit.

The new breakthrough solves this by providing a “uniform” formula that applies to all curves in the mathematical universe, without relying on the specific coefficients of their equations. This newly established hard upper bound depends on only two mathematical elements:

Polynomial Degree: The highest power in the curve’s defining equation, with the limiting statement becoming weaker as the degree increases.

The Jacobian Variety: A specialized mathematical surface constructed directly from the curve itself.

Mathematical Implications

This development provides a long-sought baseline for the field. Hector Pasten, a mathematician at the Pontifical Catholic University of Chile, notes, “This really is an amazing result that sets a new standard for what to expect”. By delivering a rule that applies across the board, the formula grants mathematicians what Mazur calls a “broad sweep of understanding” regarding the structure of curves.

Furthermore, understanding rational points on one-dimensional curves serves as a foundational step for studying higher-dimensional algebraic objects. Polynomial equations with additional variables generate complex surfaces and “manifolds,” which are heavily utilized in theoretical physics to map the fabric of space and time. By successfully placing an upper limit on the rational points of simple curves, mathematicians are now better equipped to pursue similar structural limits in these higher-dimensional spaces.

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How Cricket Balls Move

Adapted from: “How cricket balls move: The science behind swing, seam and spin.” <https://phys.org/news/2025-12-cricket-balls-science-seam.html>. Author: Cody Lindsay

If you’ve ever seen a delivery curve, jag, or dip late, you’ve witnessed how cricket bowlers exploit physics and biomechanics to trouble batters. Bowling skill essentially channels three mechanisms of ball movement—swing, seam and spin—each interacting with air and pitch conditions in distinct ways to create uncertainty.

Swing: When the air does the work Swing is the sideways curve of the ball while it is still in the air, usually produced by fast and medium-pace bowlers and occasionally by spinners with a new ball in shorter formats. For batters, it is especially challenging because the ball can deviate too quickly for them to adjust, undermining their early predictions of where it will land based on the bowler’s action and initial flight.

The article explains three main forms of swing: conventional, contrast and

reverse. Conventional swing uses a new, shiny ball with the seam angled so that airflow becomes turbulent on one side and remains smoother on the other, creating an imbalance in when the air separates and therefore a sideways force toward the seam. Research cited in the article shows that a stable, upright seam enhances swing, while seam wobble reduces it, and that elite bowlers achieve stability by aligning fingers and wrist with the seam and running their fingers down the back at release so any sideways movement that “scrambles” the seam is minimized. As the ball ages, teams deliberately shine one side and allow the other to roughen, leading to contrast swing, where the ball moves toward the rough side because air clings longer there. With further wear the rough side can become so coarse that airflow flips and, at very high speeds, produces reverse swing, which moves toward the smoother side and is generated most consistently by the fastest bowlers.

Seam: When the bounce is unpredictable Seam movement occurs after the ball pitches and is the sideways deviation produced when the raised seam grips the surface. Fast bowlers encourage this by releasing the ball with a slight wobble or angled seam rather than perfectly upright, so that the seam can catch the turf and deviate in its pointed direction.

Pitch characteristics strongly influence seam effectiveness: flat wickets with short grass give little lateral movement, whereas greener or moister pitches have small irregularities that promote grip and change of direction. For batters facing speeds of 130 km/h or more, decisions are made before the ball lands, so even a few centimetres of deviation can transform a sound stroke into an edge, a miss, or an LBW, explaining why seamers take so many bowled, caught behind and leg-before dismissals.

Spin: Making the ball dance Spin bowling relies on rotation rather than high pace, using side-spin and topspin to make the ball drift, dip and then turn after pitching. The article distinguishes two main categories: finger spin (off-spin and left-arm orthodox) where fingers roll across the ball, and wrist spin (leg-spin and left-arm unorthodox) where a vigorous wrist flick imparts greater spin.

A spinning ball alters airflow so that air speeds up on one side and slows on the other, creating sideways drift during flight. Added topspin causes the ball to dip more sharply as it approaches the batter, and when it lands, friction between ball and pitch can turn it sideways. Surface conditions again matter: dry, dusty pitches typical of the Indian subcontinent provide more friction and turn, while harder, faster Australian surfaces offer extra bounce but usually less spin, forcing spinners to keep adjusting pace, angle and release to exploit subtle variations.

The beauty of unpredictability The article concludes that cricket operates on very fine margins where physical laws intertwine with human skill. Tiny changes—a slight wrist flick, a roughened seam, or a patch of grass—can drastically alter a ball’s trajectory, sustaining the sport’s enduring appeal as a contest balancing bowler and batter, skill and science, and an ongoing tension between order and chaos.

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Finger Counting and Math Skills

Adapted from: "Finger counting opens the door to better maths skills."
<https://www.bps.org.uk/research-digest/finger-counting-opens-door-better-maths-skills>. Author: Emma Young

The article reports a longitudinal study showing that early finger-counting supports the development of stronger, later abstract arithmetic skills, rather than hindering them.

The researchers, Marie Krenger and Catherine Thevenot at the University of Lausanne, followed 192 Swiss children from the beginning of their first school year (average age 4 years 7 months) until the start of their fourth school year (about 7 years 7 months). Every six months, the children were video-recorded while solving age-appropriate addition problems, without any instruction about whether or not to use their fingers; if they asked,

they were simply told finger use was allowed. The team classified a child as a “finger-user” if they visibly moved their fingers while solving at least one of the first sets of addition problems.

Patterns of finger use changed markedly with age. Up to about five-and-a-half, children were actually less likely to use their fingers than to try mental methods. By ages six to six-and-a-half, however, the majority did use their fingers, with around two-thirds of six-and-a-half-year-olds observed counting this way. As children grew older, the proportion of finger-users declined to about half, but most older “non-finger-users” were in fact former finger-users rather than children who had never used their fingers. By the end of the study, only 12 children had never used their fingers in any recorded session.

Performance patterns mirrored these developmental shifts. Up to age six-and-a-half, finger-users tended to outperform children who did not use their fingers. Later, finger use became associated with poorer performance, but a more nuanced pattern emerged when the researchers separated ex-finger-users from genuine never-finger-users. From the start of schooling, ex-finger-users achieved better results than children who had never counted on their fingers, and this performance gap widened over time. This demonstrates that the common label “non-finger-user” in earlier research concealed an important distinction, and it suggests that building on early finger-counting experiences may help children transition from a vague “fuzzy” numerical sense to clear, reliable abstract arithmetic.

These findings have direct implications for classroom practice. The article notes a recent study of French teachers in which about a third of those teaching six- to seven-year-olds associated finger use with weaker maths skills, with similar attitudes (though slightly less pronounced) for the

year below. The new evidence suggests that such views are misplaced for younger children and that explicitly teaching finger-counting to children who have not adopted it spontaneously could improve both short-term and long-term achievement. Encouraging appropriate use of fingers in the early stages, then supporting children to move beyond finger-counting at the right time, may be beneficial.

The study also highlights a gender pattern in older age groups. In the last two sessions, when the children were about seven and seven-and-a-half and finger-counting was linked to poorer performance, more girls than boys were still relying on their fingers, a pattern that echoes earlier US findings. The authors suggest this raises concerns that girls may be at particular risk of lagging behind in arithmetic if they depend on finger-based strategies for too long. They call for intervention studies to test whether helping young children to begin finger-counting, and slightly older children to transition away from it, can boost school mathematics outcomes.

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Mathematics teachers who have a view on the issue of finger counting at the primary level are invited to write to the editor and share their observations.

The Untold Story of Our Remarkable Hands and How They Made Us Human

Adapted from: "The Untold Story of Our Remarkable Hands and How They Made Us Human." <https://www.newscientist.com/article/2514431-the-untold-story-of-our-remarkable-hands-and-how-they-made-us-human/>. All images & captions are from this source. *New Scientist* Author: Michael Marshall

The human hand, with its unique blend of strength and precision, is a defining evolutionary trait of our species. Recent fossil discoveries have finally clarified the timeline of how our dexterity developed, revealing deep connections between bipedalism, tool use, brain expansion, and the origins of language.



Figure 32: Playing a complex guitar solo ought to be impossible!

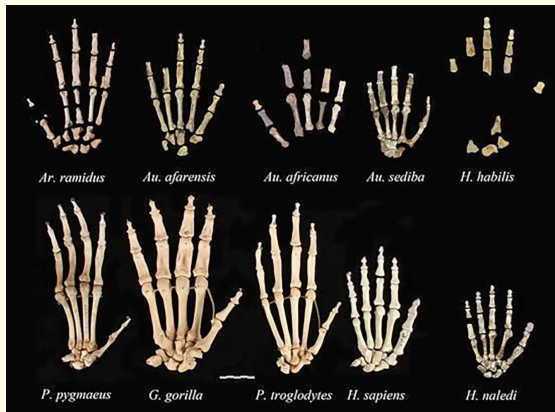


Figure 33: Compared with the hands of many ancient hominins, chimpanzees and gorillas, our hands have relatively long thumbs that enable a precise grip

Anatomy of Dexterity

Humans possess long, robust, and highly mobile thumbs paired with relatively short, straight fingers. In contrast, our closest living relatives, chimpanzees and bonobos, have long, curved fingers and short thumbs that are specialized for climbing. Our soft tissue anatomy, including uniquely large hand muscles and an extended *flexor pollicis longus* muscle, enables the forceful precision grips required for everything from crafting tools to playing musical instruments.



Figure 34: The incredible dexterity granted us by the “pincer grip” between our thumb and fingers is a hallmark of our species

Darwin’s Theory and Fossil Puzzles

Charles Darwin originally hypothesized in 1871 that human dexterity could only evolve after bipedalism freed our hands from the demands of

locomotion. For decades, a lack of fossil evidence made this impossible to verify. When early tools were discovered alongside *Homo habilis* in the 1960s, scientists were surprised to find that the hominin's hands still appeared robust and ape-like.

Subsequent discoveries revealed a confusing mosaic of traits. Fossils like the 2-million-year-old *Australopithecus sediba* and the 300,000-year-old *Homo naledi* featured human-like thumbs and wrists combined with curved, ape-like fingers suited for climbing. Furthermore, the 4.4-million-year-old *Ardipithecus ramidus* ("Ardi") initially puzzled researchers, though recent re-evaluations confirmed its hands were heavily adapted for swinging beneath tree branches.

A Stepwise Evolution

Recent breakthroughs, including the 2025 discovery of *Paranthropus boisei* hand fossils, have allowed researchers to map a clear, stepwise progression of human hand evolution over the last 7 million years.

Ardipithecus to Australopithecus: As hominins began spending more time on the ground, their thumbs broadened and lengthened to accommodate early tool use. This transition aligns with the discovery of crude, 3.3-million-year-old Lomekwian stone tools that predate the *Homo* genus, suggesting *Australopithecus* were early toolmakers.

The Paranthropus-Homo Ancestor: Around 3.5 million years ago, the shared ancestor of *Paranthropus* and *Homo* developed straighter fingers, a highly robust thumb, and reorganized wrist bones for greater mobility

Early Homo: As early humans shifted toward a diet heavier in meat, the need to hunt and butcher animals demanded advanced stone tools, driving the final morphological stages of our hands.



Figure 35: The huge benefits from being able to make more sophisticated tools may have driven the evolution of our hands

Brains and Language

The evolution of the hand was deeply intertwined with cognitive development. Research shows a strong correlation between longer thumbs and larger neocortex sizes in primates, as extraordinary dexterity requires advanced neural circuitry to control complex digit movements.

Furthermore, the manual skills required to craft sophisticated stone tools or perform complex cultural behaviours could not be learned merely through observation. Researchers suggest that by 600,000 years ago, hominins had to use “overt explanation” to teach these skills to one another. This

need for explicit instruction likely helped lay the groundwork for language, potentially starting with early forms of sign language. Ultimately, Darwin’s early intuition proved correct: walking upright freed our hands, setting off a morphological and cognitive cascade that made us human.

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Part IV

Contributed Articles

Don't Rely on Analogy!

Article contributed by Parshant Kumar

Consider the following problem:

Problem 1: Using three nines and no operational signs, form the largest number possible.

Here the term “operational signs” refers to the familiar operations of arithmetic : $+$, $-$, $\times$, $\div$, etc. However, we permit “to the power of” which is generally shown without using an explicit symbol.

There are just four candidates for the answer:

$$999, \quad 99^9, \quad 9^{99}, \quad 9^{9^9}.$$

We ask the reader to confirm that the four numbers have been arranged in increasing order, the largest being

$$9^{9^9}.$$

This number dominates the others by a huge margin. Indeed, the remaining numbers are tiny in comparison.

We illustrate this by listing the powers of 10 closest to each of the four numbers. Here is what we get:

$$10^3, \quad 10^{18}, \quad 10^{94}, \quad 10^{369,693,100}.$$

The hugeness of the fourth number may come as a shock — but it really is that large! (Note that some rounding-off had to be done for the exponent.)

Now consider the following problem with three twos:

Problem 2: Using three twos and no operational signs, form the largest number possible.

As earlier, there are four candidates for the answer:

$$222, \quad 22^2 \quad 2^{22}, \quad 2^{2^2}.$$

These numbers may be computed exactly (no estimation required). Here's what we get:

$$222, \quad 484, \quad 4194304, \quad 16.$$

We see that the largest number is 2^{22} , and the smallest is 2^{2^2} .

In the previous instance, we had found 9^{9^9} to be the *largest* number. Now, we find 2^{2^2} to be the *smallest* number.

What about the corresponding problem with three threes?

Problem 3: Using three threes and no operational signs, form the largest number possible.

As earlier, there are four candidates for the answer:

$$333, \quad 33^3, \quad 3^{33}, \quad 3^{3^3}.$$

Here are their exact values (approximate values in two cases):

$$333, \quad 35937, \quad 5 \times 10^{15}, \quad 7625597484987 \approx 7.6 \times 10^{12}.$$

The largest among them is 3^{33} , and 3^{3^3} comes a close second.

We see the largest number follows the same pattern for the case of three twos and three threes.

What about the corresponding problem with four fours?

Problem 4: Using three fours and no operational signs, form the largest number possible.

The four candidates to consider are

$$444, \quad 44^4, \quad 4^{44}, \quad 4^{4^4}.$$

Here are their exact values (approximate values in the last two cases):

$$444, \quad 3748096, \quad 3 \times 10^{26}, \quad 4^{256} \approx 1.3 \times 10^{54}.$$

We see a different pattern here. The largest number is of the form

$$n^{n^n}.$$

Looking at the problem more generally

What about the corresponding problems with three fives, three sixes, three sevens, three eights or three nines (which we looked at earlier)?

Problem 5: Let $a \in \{2, 3, \dots, 8, 9\}$. Using three a 's and no operational signs, form the largest number possible.

Let us formulate problem algebraically. The four numbers that we can form are the following:

- $\overline{aaa} = 100a + 10a + a = 111a$
- $\overline{aa^a} = (10a + a)^a = (11a)^a$
- $a^{\overline{aa}} = a^{11a}$
- a^{a^a}

We first show that $(11a)^a > 111a$ for all $a \geq 2$. Indeed, we have,

$$(11a)^a = 11^a \cdot a^a \geq 11^2 \cdot a = 121a > 111a.$$

We shall similarly show that $a^{11a} \geq (11a)^a$ for all $a \geq 2$.

Our task is to compare a^{11a} and $(11a)^a$. Taking a -roots of both quantities our task reduces to comparing a^{11} and $11a$. This is equivalent to comparing a^{10} and 11 . Since $2^{10} > 11$, it follows that $a^{11a} \geq (11a)^a$ for all $a \geq 2$.

Summarizing, we have showed so far that for $a \geq 2$,

$$111a < (11a)^a < a^{11a}$$

Now we must compare a^{a^a} and a^{11a} .

Since the bases of the two expressions are the same (namely, a), an equivalent problem is to compare a^a and $11a$. Dividing both sides by a , we reduce the problem to comparing a^{a-1} and 11 .

Observe that $2^1 < 11$ and $3^2 < 11$, but $4^3 > 11$. It should be clear that $a^{a-1} \geq 4^3 > 11$ for all $a \geq 4$.

Hence $a^{a^a} > a^{11a}$ for all $a \geq 4$.

It should be clear now why the case of three twos and three threes follows a different profile from the other cases.

The case of four 1's

Using four ones, there are relatively few possibilities to consider:

$$1111, \quad 11^{11} = 285311670611.$$

Here $11^{11} \approx 2.85 \times 10^{11}$ is roughly 2.5×10^7 times as large as 1111.

The case of four 2's

There are eight different numbers to examine now:

$$\begin{array}{cccc} 2222, & 222^2, & 22^{22}, & 2^{222}, \\ 22^{22}, & 2^{22^2}, & 2^{2^{22}}, & 2^{2^{2^2}} \end{array}$$

Which one represents the largest number?

Clearly, 2222 is less than any of the others.

Let's compare 222^2 and 22^{22} . We have,

$$22^{22} = (22^2)^{11} = 484^{11} > 222^2.$$

Next, we compare 22^{22} with 2^{222} . We have,

$$22^{22} < 32^{22} = (2^5)^{22} = 2^{110} < 2^{222}.$$

Hence, $2^{222} > 22^{22}$.

So, the largest number among the numbers in the first row,

$$2222, \quad 222^2, \quad 22^{22}, \quad 2^{222}$$

is 2^{222} .

That is,

$$2222 < 222^2 < 22^{22} < 2^{222}.$$

Now let us look at the numbers in the second row:

$$22^{22}, \quad 2^{222}, \quad 2^{222}, \quad 2^{222}.$$

Of these, we have:

$$2^{2^{22}} = 2^{2^4} = 2^{16} = 65536,$$

and

$$22^{2^2} = 22^4 = 234256.$$

Both these numbers are fairly small. (They are just a little larger than 222^2 .)

But the remaining two numbers are extremely large:

$$2^{2^{22}} = 2^{484} \approx 5 \times 10^{145}$$

is a 146-digit number; and

$$2^{2^{22}} = 2^{4194304}$$

is a number with roughly 1.26 million digits!

Concluding remark

These examples serve to illustrate that in problems of this kind, *patterns seen with smaller numbers may not hold when we move to larger numbers, and intuition and patterns can be misleading when dealing with the “to the power of” operation and numbers of this magnitude.*

The reader could do some computations for the corresponding problems with four threes, four fours, etc.

About the author

Parshant Kumar serves as a Post Graduate Teacher of Mathematics at Government Senior Secondary School, Puthi Saman, Hisar, Haryana. He has a passion for teaching mathematics and nurturing logical thinking in students. His academic interests include problem solving, puzzles, and number theory. He enjoys exploring creative mathematical ideas that make learning engaging and meaningful.

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Divisibility property of sum of digits of n and $n+1$

Article contributed by Jaitra Chattopadhyay

Let $n \geq 1$ be an integer written in the decimal system. That is,

$$n = a_m \times 10^m + \dots + 10 \times a_1 + a_0, \quad (1)$$

where a_i 's are integers with $0 \leq a_i \leq 9$ and $a_m \neq 0$. We denote by $s(n)$ the sum of digits $a_0 + \dots + a_m$ of n . For example, if $n = 149$, then we can express 149 as $1 \times 10^2 + 4 \times 10^1 + 9$ and therefore, $s(149) = 1 + 4 + 9 = 14$.

In the Indian Olympiad Qualifier in Mathematics (IOQM) in 2024-2025, the following interesting problem was asked.

Problem 1: Let n be the smallest integer such that the sum of digits of n is divisible by 5 as well as the sum of digits of $(n + 1)$ is divisible by 5. What are the first two digits of n in the same order?

To answer Problem 1, we need to find the smallest integer $n \geq 1$ such that $5 \mid s(n)$ and $5 \mid s(n + 1)$. Now, we ask a few questions as follows.

- (a) Can we replace the integer 5 by any integer $k \geq 2$ in Problem 1?
- (b) For any integers $k \geq 2$ and $r \geq 2$, do there exist any integer n such that $k \mid s(n+i)$ for all $i = 0, \dots, r$?
- (c) For which integers $k \geq 2$ and $r \geq 1$ do there exist infinitely many integers $n \geq 1$ such that $k \mid s(n+i)$ for all $i = 0, \dots, r$?

We provide a complete answer to these questions by proving the following:

Theorem. *Let $k \geq 2$ be an integer. Then there does not exist any positive integer n such that k divides all three of $s(n)$, $s(n+1)$ and $s(n+2)$.*

Moreover, there exist infinitely many positive integers n with $k \mid s(n)$ and $k \mid s(n+1)$ if and only if $\text{GCD}(k, 3) = 1$.

Proof of theorem

If possible, let $n = a_m \times 10^m + \dots + 10 \times a_1 + a_0$ be a positive integer such that k divides all three of $s(n)$, $s(n+1)$ and $s(n+2)$.

If $a_0 \leq 8$, then we have

$$n+1 = a_m \times 10^m + \dots + 10 \times a_1 + (a_0 + 1),$$

and therefore,

$$s(n+1) = a_0 + a_1 + \dots + a_m + 1 = s(n) + 1.$$

Therefore, an integer $k \geq 2$ cannot divide both $s(n)$ and $s(n+1)$ because then it would divide $s(n+1) - s(n) = 1$, which is not possible.

Therefore, the unit's digit of n must be $a_0 = 9$.

This implies that the unit's digit of $n+1$ is 0. By the same argument, it now follows that $s(n+2) = s(n+1) + 1$.

Consequently, k cannot divide both $s(n+1)$ and $s(n+2)$.

Hence there does not exist any positive integer n such that k divides all three of $s(n)$, $s(n+1)$ and $s(n+2)$.

Next, let $n \geq 1$ be an integer such that k divides both $s(n)$ and $s(n+1)$.

By the reasoning given above, it follows that the unit's digit of n must be 9. Therefore, we write n as

$$n = a_m \times 10^m + \dots + a_r \times 10^r + 9 \times 10^{r-1} + \dots + 9,$$

where $a_r \leq 8$. Therefore,

$$n+1 = (a_m \times 10^m + \dots + (a_r + 1) \times 10^r) \times 10^{r-1}.$$

Thus we have

$$s(n) = a_r + \dots + a_m + 9r,$$

$$s(n+1) = (a_r + \dots + a_m) + 1.$$

Hence if k divides $s(n)$ and $s(n+1)$ then $k \mid (s(n) - s(n+1)) = 9r - 1$. Therefore, $\text{GCD}(k, 3) = 1$.

Now we consider the converse proposition.

Suppose that $\text{GCD}(k, 3) = 1$; then $\text{GCD}(k, 9) = 1$ and thus we can find an integer $u \geq 1$ such that $k \mid 9u - 1$.

Then we find integers $a_u, \dots, a_m$ with $1 \leq a_i \leq 8$ and $u < m$ such that

$$k \mid (a_u + \dots + a_m + 1).$$

Therefore, the integer

$$a_m \times 10^m + \dots + a_u \times 10^u + 9 \times 10^{u-1} + \dots + 9$$

is a possible choice for n .

Also, for this choice of n , the integers

$$n \times 10^m + n, \quad n \times 10^{2m} + n, \quad n \times 10^{3m} + n, \quad \dots$$

are all admissible choices.

Therefore, there exist infinitely many positive integers n such that $k \mid s(n)$ and $k \mid s(n+1)$. ■

Remark 1

Let us illustrate the theorem by an explicit example.

Let $k = 17$. We wish to produce infinitely many positive integers n such that $17 \mid s(n)$ and $17 \mid s(n+1)$.

Following the proof of the theorem, we first find an integer $u \geq 1$ such that $17 \mid 9u - 1$. Clearly, $u = 2$ satisfies the condition.

Now, we find a positive integer whose sum of digits is 1 short from being divisible by 17. One such integer is 88. Therefore, $n = 8899$ is an admissible choice. We can directly verify that

$$s(n) = 8 + 8 + 9 + 9 = 34,$$

$$s(n+1) = s(8900) = 8 + 9 + 0 + 0 = 17,$$

both of which are divisible by 17.

We can construct many such numbers just by concatenating 8899 with itself. Thus 88998899, 889988998899 etc. are all examples of integers with the desired property.

Remark 2

It is not surprising that the integer k in the theorem must satisfy the condition $\text{GCD}(k, 3) = 1$.

Otherwise, if $\text{GCD}(k, 3) = 3$, then $k \mid s(n)$ and $k \mid s(n+1)$ imply that $3 \mid s(n)$ and $3 \mid s(n+1)$. But $s(m) \equiv m \pmod{3}$ for all integer $m \geq 1$ implies that $3 \mid n$ and $3 \mid n+1$, which is not possible.

Acknowledgements.

I came across Problem 1 in an interaction with Mr. Aditya Prakash Bhuniya, son of my colleague Dr. Anjan Kumar Bhuniya. I thank him for bringing the problem to my attention.

Further reading

[1] IOQM question papers, <https://ioqm.in/>

About the author

Dr. Jaitra Chattopadhyay completed his PhD in mathematics from Harish-Chandra Research Institute in 2020, specializing in number theory. After spending two years at IIT Guwahati as a post-doctoral fellow, he joined Visva-Bharati, Santiniketan as an Assistant Professor in the Department of Mathematics. Apart from research, he is keenly interested in math high school competition problems and loves to interact with enthusiastic students. As a hobby, he likes to play chess. He may be contacted at chat.jaitra@gmail.com.

Fermat's 'Little' Theorem

Article contributed by K. Sasikumar

Note from the editor The “Little Theorem of Fermat” (LFT) states the following: if p is any prime number and a is an integer not divisible by p , then

$$a^{p-1} \equiv 1 \pmod{p}.$$

An equivalent statement is this: if p is any prime number and a is any integer, then

$$a^p \equiv a \pmod{p}.$$

The LFT is an enormously useful result in number theory. Olympiad ‘mathletes’ are generally very proficient users of this result!

In a separate article (in this issue), we study some applications of the LFT. In this article we give the proof of the LFT using mathematical induction.

We must prove that p is a divisor of $n^p - n$ for any prime number p and any integer n .

Let n be any positive integer; let p be any prime number.

Claim

(Statement S) p divides $n^p - n$. We shall prove (S) using the principle of induction.

Clearly (S) is true when $n = 1$.

Suppose (S) is true when $n = m$. Then there exists an integer k such that

$$m^p - m = kp.$$

Now

$$\begin{aligned} & (m+1)^p - (m+1) \\ &= m^p + \binom{p}{1}m^{p-1} + \binom{p}{2}m^{p-2} + \cdots + \binom{p}{p-1}m + 1 - (m+1) \\ &= (m^p - m) + \binom{p}{1}m^{p-1} + \binom{p}{2}m^{p-2} + \cdots + \binom{p}{p-1}m \\ &= kp + \binom{p}{1}m^{p-1} + \binom{p}{2}m^{p-2} + \cdots + \binom{p}{p-1}m. \end{aligned}$$

It is easy to show that

$$\binom{p}{r} \text{ is a multiple of } p \text{ for } r = 1, 2, \dots, p-1.$$

Hence it follows that

$$(m+1)^p - (m+1) = kp + \text{a multiple of } p,$$

implying that $(m+1)^p - (m+1)$ is a multiple of p .

Hence, by Principle of Mathematical Induction, we conclude that p divides $n^p - n$ for any positive integer n and for any prime number p .

If $n = 0$, then clearly p divides $n^p - n$, since $n^p - n = 0$.

Now let n be a negative integer.

If $p = 2$, then $n^p - n = n^2 - n = n(n-1)$ is a multiple of 2.

If $p \neq 2$, then p is odd. Let $m = -n$. Then m is a positive integer, so p divides $m^p - m$. Hence p divides $(-n)^p + n = -n^p + n = -(n^p - n)$.

Hence p divides $n^p - n$. That is, p divides $n^p - n$ if n is any negative integer and p is any prime number.

Therefore for any prime number p and for any integer n , p divides $n^p - n$.

This proves Fermat's Little Theorem. ■

About the author

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Solving a Set of Simultaneous Equations by Long Division

Article contributed by S. C. Dutta Roy

In this article, we present a way of solving the set of simultaneous equations

$$x^2 + y^2 = a, \quad x^3 + y^3 = b.$$

The special case $a = 7$, $b = 10$, which has been investigated earlier, is considered as an example to demonstrate the validity of the method.

Introduction

The set of equations $x^2 + y^2 = 7$, $x^3 + y^3 = 10$ is well-known in Maths Olympiad circles. We consider here the general case where 7 is replaced by a and 10 by b . We look at the problem in a new way, which should be appealing to students, even to the beginning students of algebra. The innovation consists of long dividing $x^3 + y^3$ by $x^2 + y^2$ and proceeding therefrom.

The article is organized as follows. We first convert the two equations into a single one to find the number of roots. Next, we present our approach. Then we deal with the special case $a = 7, b = 10$.

Number of Roots

The given equations are:

$$x^2 + y^2 = a, \quad (2)$$

$$x^3 + y^3 = b. \quad (3)$$

We eliminate y by combining the two equations. We get,

$$x^3 + (a - x^2)^{3/2} = b. \quad (4)$$

This can be rearranged and simplified to yield a 6th degree equation:

$$2x^6 - 3ax^4 - 2bx^3 + 3a^2x^2 + b^2 - a^3 = 0. \quad (5)$$

Thus there are as many as 6 roots to be found out. Numerical methods can be used for given a and b , but they give only approximate values. Exact analytical values can be found out by the procedure given below.

Solution for the General Case

Our way is to look at the problem as one of long division. When we divide $x^3 + y^3$ by $x^2 + y^2$, we get the quotient and remainder as follows:

$$\text{Quotient} = x + y, \quad \text{Remainder} = -xy(x + y).$$

This means, of course, that

$$x^3 + y^3 = (x + y) \cdot (x^2 + y^2) - xy(x + y).$$

We can arrange the division in the familiar “long division format” but we have not done so here. (We invite the reader to do so.)

Let $x + y = u$ and $xy = v$. Then,

$$v = xy = \frac{(x + y)^2 - (x^2 + y^2)}{2} = \frac{u^2 - a}{2}. \quad (6)$$

The long division relation shows that

$$b = au - uv = au - \frac{u(u^2 - a)}{2}. \quad (7)$$

This can be put in the form of a cubic equation in u :

$$u^3 - 3au + 2b = 0. \quad (8)$$

Equation (7) can be solved by Cardano’s method (outline given in the Appendix). The condition for all roots to be real is

$$a^3 \geq b^2. \quad (9)$$

In general, the roots of (7) are:

$$u_1 = k_1 + k_2, \quad u_2 = wk_1 + w^2k_2, \quad u_3 = w^2k_1 + wk_2, \quad (10)$$

where $w = (-1 + i\sqrt{3})/2$, $i = \sqrt{-1}$, and

$$k_{1,2} = \left(-b \pm \sqrt{b^2 - a^3} \right)^{1/3}. \quad (11)$$

The values of x and y can be found by noting that $x + y = u$ and

$$x - y = \sqrt{(x + y)^2 - 4xy} = \sqrt{u^2 - 2(u^2 - a)} = \sqrt{2a - u^2}, \quad (12)$$

so that

$$x, y = \frac{u_r \pm \sqrt{2a - u_r^2}}{2}, \quad r = 1, 2, 3. \quad (13)$$

Special Case: $a = 7$ and $b = 10$

In this case, (7) becomes

$$u^3 - 21u + 20 = 0. \quad (14)$$

Since condition (8) is satisfied, all roots will be real. However, one need not use the general solutions given in (9) and (10) because it is easily shown that

$$u^3 - 21u + 20 = (u - 1)(u - 4)(u + 5). \quad (15)$$

The roots are $u = 1, 4, -5$. Using (12) we get, explicitly,

$$\begin{aligned} (x, y) = & \left(\frac{1 + \sqrt{13}}{2}, \frac{1 - \sqrt{13}}{2} \right), \\ & \left(\frac{1 - \sqrt{13}}{2}, \frac{1 + \sqrt{13}}{2} \right), \\ & \left(\frac{-5 + i\sqrt{11}}{2}, \frac{-5 - i\sqrt{11}}{2} \right), \\ & \left(\frac{-5 - i\sqrt{11}}{2}, \frac{-5 + i\sqrt{11}}{2} \right), \\ & \left(2 + \frac{i}{\sqrt{2}}, 2 - \frac{i}{\sqrt{2}} \right), \\ & \left(2 - \frac{i}{\sqrt{2}}, 2 + \frac{i}{\sqrt{2}} \right). \end{aligned}$$

Conclusion

We have presented here, in a simple manner, the complete solution for the simultaneous equations $x^2 + y^2 = a$, $x^3 + y^3 = b$. The long division displays the single equation to be solved and would be appealing to students. The special case $a = 7$, $b = 10$ has been discussed as an example.

Appendix: Cardano's Method for the Solution of a Cubic Equation

In this method of solving the cubic equation

$$ax^3 + bx^2 + cx + d = 0,$$

we first substitute $x = y - b/(3a)$, resulting in the equation

$$y^3 + py + q = 0.$$

Next, we put $y = u + v$ to transform it to equations in u^3 and v^3 , which become roots of a related quadratic equation, thus allowing one to find u and v , then y , and finally x . More details are available in [2].

Acknowledgment

The author got interested in the problem from [1].

Further reading

[1] Gupta, Saibal. (2025), What's App. Posting in the Radio Physics and Electronics Social Group.

[2] Brilliant, “Cardano’s Method.”

<https://brilliant.org/wiki/cardano-method/>

About the author

S. C. Dutta Roy worked as a Professor of Electrical Engineering at IIT Delhi for more than four decades and is now retired. Earlier, he worked at the University of Kalyani, West Bengal, India, and the University of Minnesota, USA. In between, while on sabbatical, he worked as a Visiting Professor at the University of Leeds, UK, and at Iowa State University, USA. He has a keen interest in science education. He may be contacted at s.c.dutta.roy@gmail.com.

Primorial Numbers

Article contributed by Anand Prakash

Given a positive integer D , let $d_1 = 1, d_2, \dots, d_{k-1}, d_k = D$ be its divisors listed in increasing order:

$$1 = d_1 < d_2 < \dots < d_k = D.$$

We say that D satisfies property (P) if the fractions

$$\frac{d_1}{1}, \quad \frac{d_1 + d_2}{2}, \quad \dots, \quad \frac{d_1 + d_2 + \dots + d_k}{k}$$

are all integers.

Examples and non-examples

- If D is a prime number, then its divisors are 1 and D , and the two corresponding fractions are 1 and $\frac{1}{2}(1 + D)$. The latter quantity is an integer precisely when D is odd.

It follows that all odd primes have property (P).

- If $D = 7^2 \times 31$, its six divisors are

$$1 < 7 < 31 < 7^2 < 7 \times 31 < 7^2 \times 31.$$

The corresponding six fractions are 1, 4, 13, 22, 61, 304 which are all integers; so $7^2 \times 31$ satisfies property (P).

- If $D = 31^4$, its five divisors are

$$1 < 31 < 31^2 < 31^3 < 31^4.$$

The corresponding five fractions are

$$\frac{1}{1}, \frac{1+31}{2}, \frac{1+31+31^2}{3}, \frac{1+31+31^2+31^3}{4},$$

$$\frac{1+31+31^2+31^3+31^4}{5},$$

which are all integers, showing that 31^4 has property (P).

- If $D = 5^2 = 25$, then the divisors 1, 5, 25 give the fractions 1, 3, 31/3. We see that 25 *does not* have property (P).

Question

Can we characterize all positive integers $D > 1$ which have property (P)?

A proposition

We use the notation $K(n)$ to denote the product of all the distinct primes that divide $n!$; that is, it is the product of all primes $\leq n$. These numbers are known as *primorials*.

For example,

- $K(10) = 2 \times 3 \times 5 \times 7 = 210$
- $K(15) = 2 \times 3 \times 5 \times 7 \times 11 \times 13 = 30030$

We shall prove the following result which partially answers the question posed above.

Theorem. *Let $D = p^{d-1}$ where p is a prime number and d is a positive integer. Then, D has property (P) if and only if $p \equiv 1$ modulo $K(d)$.*

Proof of theorem

We may assume that $d > 1$. (There is nothing to prove if $d = 1$.)

Suppose that $p \equiv 1$ modulo $K(d)$. Let $\{q_1, \dots, q_r\}$ be the primes which divide $d!$ (equivalently, they are the primes $\leq d$).

Now, the divisors of D are

$$1, p, p^2, \dots, p^{d-1}.$$

For any $k \leq d$, we show that $1 + p + \dots + p^{k-1}$ is a multiple of k . Let

$$k = q_1^{e_1} q_2^{e_2} \dots q_r^{e_r}$$

for some e_i 's. It suffices to show that

$$1 + p + \dots + p^{q_i^{e_i} - 1}$$

is a multiple of $q_i^{e_i}$. We know that $p \equiv 1$ modulo $K(d)$ which means $p \equiv 1 \pmod{q_i}$ for each i . Writing $p = 1 + tq_i$, we have

$$\begin{aligned} & 1 + p + \dots + p^{q_i^{e_i} - 1} \\ &= \frac{p^{q_i^{e_i} - 1} - 1}{p - 1} = \frac{(1 + tq_i)^{q_i^{e_i}} - 1}{tq_i}, \end{aligned}$$

hence

$$\frac{1 + p + \cdots + p^{q_i^{e_i} - 1}}{q_i^{e_i}} = \frac{(1 + tq_i)^{q_i^{e_i}} - 1}{tq_i^{e_i + 1}}.$$

Expanding the terms using the binomial theorem, we see that the last term is an integer. Therefore, we have proved that p^{d-1} satisfies property (P) when $p \equiv 1$ modulo $K(d)$.

Conversely, suppose that p^{d-1} has the property (P).

Then, for each prime $q \leq d$,

$$\frac{1 + p + \cdots + p^{q-1}}{q} = \frac{p^q - 1}{q(p-1)}$$

is an integer.

So, q divides $p^q - 1$. Invoking the Little Theorem of Fermat, we deduce that q divides $p - 1$.

Therefore, $p \equiv 1 \pmod{q}$ for each prime $q \leq d$.

This implies $p \equiv 1 \pmod{K(d)}$. ■

About the author

Anand Prakash runs a small garment shop at Kesariya village in the state of Bihar. He has a keen interest in number theory and recreational mathematics and has published many papers in international journals in these fields. He also has a deep interest in classical Indian music as well as cooking. In addition, he has written a large number of poems in Hindi. He may be contacted at prakashanand805@gmail.com

Blanket-the-Big-Triangle Puzzle

Article contributed by Rossi D'Souza

Introduction A puzzle can be a treasure trove of mathematical principles, concepts and strategies, all cleverly hidden inside a fun, mind-challenging activity. Its enigmatic qualities make us eager to solve it even when it demands learning new mathematical ideas that we may not otherwise want to study. It is no wonder therefore that mathematical puzzles have made their presence across cultures and history.

In this article I share a puzzle-based activity I designed to discuss the idea of proof and to guide students through the process of developing and articulating a mathematical proof, and also to promote the idea of developing puzzle-based activities for teaching mathematics.

Let's solve the following puzzle:

Blanket-the-Big-Triangle puzzle: Can a triangle always be completely blanketed by two smaller triangles similar to it, if the differences in area between the larger triangle and the smaller triangles are sufficiently small?

By 'blanketing' I mean that the two smaller triangles can be positioned over the larger triangle in such a way that no part of the larger triangle remains exposed.

For example, consider the three similar triangles in Figure 36. The green triangle is slightly larger than the blue and yellow triangles, and can be easily blanketed by them (see Figure 37).

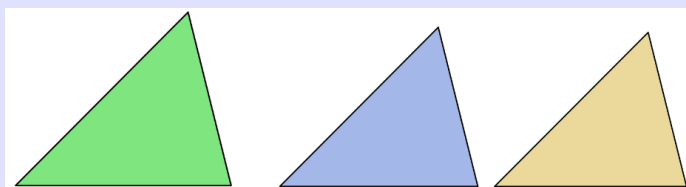


Figure 36: Three similar triangles, one slightly larger than the other two

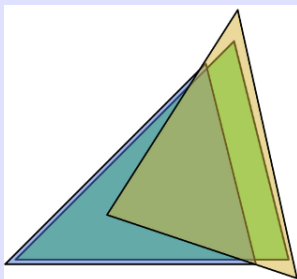


Figure 37: Larger triangle blanketed by the two smaller triangles

The property of a right-angled triangle to be blanketable by two smaller similar counterparts is straightforward. In Figure 38 we see a right-angled triangle $\triangle ABC$ in which the altitude BD gives rise to $\triangle BDA$ and $\triangle CDB$ which are similar to and by default blanket $\triangle ABC$.

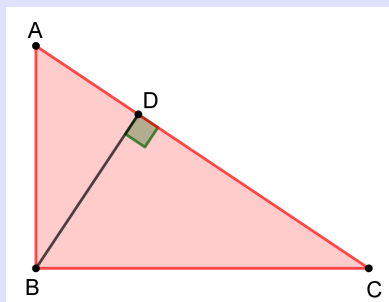


Figure 38: Right-angled triangle blanketed by its very construction

It seems that this property should be true for all triangles: if the difference in areas between the larger triangle and the smaller triangles is small enough, then the larger triangle can be blanketed by two smaller triangles similar to it. But is this so for all triangles? The answer is actually No.

In fact, this puzzle was generalized from an older problem in [2] (page 35) which was presented as follows:

Blanketing an equilateral triangle: Can you prove that an equilateral triangle cannot be blanketed by two slightly smaller equilateral triangles?

So we can guess that the property does not hold for equilateral triangles. But does it hold for all non-equilateral triangles? And how do we prove that it does? And can we prove that it does not hold for equilateral triangles?

Fomin's proof was based on what is called the Pigeonhole Principle (PHP) which can be better explained in reference to the following two problems, the first which is a well known brainteaser:

Red socks and blue socks: A box contains a large number of identically shaped red socks and blue socks, all mixed up. How many socks would you need to pick out at random to make sure that you have a pair of same coloured socks?

The answer is 3. With only two colours and three socks chosen, one colour must appear twice which guarantees a matching pair.

The Pigeonhole Principle (PHP) is generally framed in terms of pigeons and pigeonholes as follows: *Given a number of pigeons that occupy a number of pigeonholes, if the pigeons outnumber the pigeonholes, then at least one pigeonhole will contain more than one pigeon.*

This simple and straightforward expression illustrates a versatile and practical problem-solving strategy. And its significance is better illustrated in this second less-straightforward problem which is somewhat closer to our puzzle concerning the blanketability of equilateral triangles:

Five points within an equilateral triangle: If five points lie within an equilateral triangle of side 2 units, prove that at least two of those points lie within 1 unit distance of each other.

To prove this statement, we partition the equilateral triangle into four smaller congruent regions (Figure 39), each representing a pigeonhole while each of the five points represents a pigeon.

With five points and four (congruent) regions, the PHP guarantees that at least one region contains more than one point — even if a point lies on a

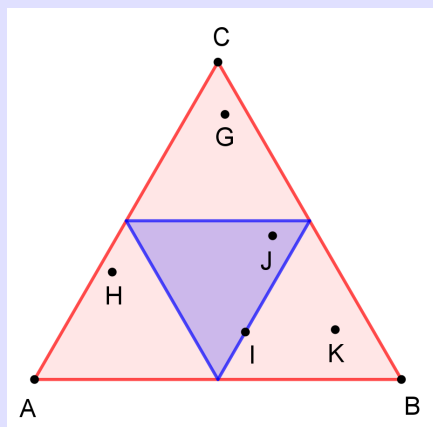


Figure 39: Problem based on the PHP

border and is shared between regions. Since each region is an equilateral triangle with side length 1 unit, any two points within it must be at most 1 unit apart, thus completing the proof.

In the original problem posed by [2], the proof provided was essentially that if two equilateral triangles blanket a larger equilateral triangle, then by the PHP, at least one smaller equilateral triangle would need to blanket two vertices of the larger equilateral triangle. This would be impossible since the largest distance that can be blanketed by an equilateral triangle is limited by its side-length.

Since the problem focused on equilateral triangles, it raised some questions concerning generalization. For example, could the problem be extended to regular polygons with a larger number of sides, such as squares, regular pentagons, ...? While it surely seemed to be true that no regular polygon could be blanketed by two smaller polygons similar to it, proving this

turned out to be a tiresome exercise well beyond the scope of a school child. (You can try proving the conjecture for a single case, e.g., squares).

Another question that came to the fore was whether we can assume without proof that the largest distance that can be blanketed by an equilateral triangle is limited by its side length.

Regarding the first question, while generalizing the problem to all regular polygons did not yield any usable insights for me, the question of whether any triangle can always be completely covered (or blanketed) by two smaller triangles similar to it turned out to have an interesting result that used for an interesting hands-on open-ended activity for teaching the idea of proof and introducing the concept of similar triangles.

The activity required students to construct sets of three similar triangles (of various kinds: right-angled, isosceles, scalene), with one triangle slightly larger than the other two. The students had to figure out a way of producing these items. Then they were asked to blanket the larger triangle with the two smaller triangles, and made to observe that this fails for equilateral triangles. I have noticed that students come close to a proof only after discussing the problem of five points in an equilateral triangle (which serves as a ‘scaffolding example’).

The Proof

We can summarize the proof of our main puzzle by noting two kinds of triangles and using a *Proof by Cases* approach.

The puzzle requires proving that (1) any non-equilateral triangle can be blanketed by two triangles that are slightly smaller and similar to it, pro-

vided the differences in area are small enough, and (2) an equilateral triangle cannot be blanketed by two smaller equilateral triangles.

Case 1: For non-equilateral triangles we prove the existence of a solution by devising a method for blanketing a triangle with its two smaller counterparts. First we observe that the length of the largest side is strictly greater than the length of the smallest side. Thus when we overlap the smallest angle of the large triangle (green triangle in Figure 40) with the corresponding angle of the small triangle, we are left with a narrow region along the smallest side which is left uncovered (Figure 40). This region can then be covered or blanketed by the region along the largest side of the second smaller similar triangle (black triangles in Figure 40).

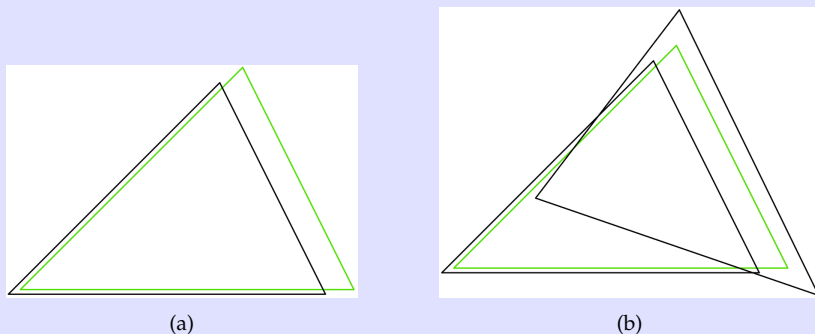


Figure 40: (a) Step 1 in blanketing (b) Step 2 in blanketing

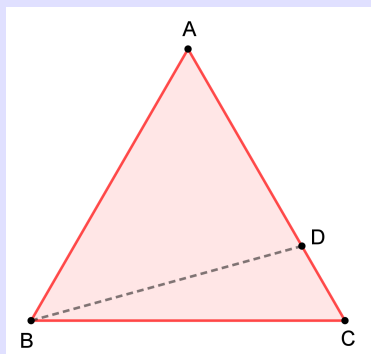
Case 2: For equilateral triangles, we apply the Pigeonhole Principle (PHP) by first identifying the pigeons and the pigeonholes:

- The 'pigeons' are the three vertices of the larger equilateral triangle;

- The ‘pigeonholes’ are the two smaller equilateral triangles covering it.

To blanket the larger triangle with the two smaller triangles requires that some two vertices of the larger triangle lie within or on the boundary of one of the smaller triangles. However, the distance between two points in an equilateral triangle cannot be more than the length of its side. Therefore, it is not possible for the smaller triangle to simultaneously blanket two different vertices of the larger triangle. This means that the larger triangle cannot be blanketed with the two smaller triangles.

To complete the proof we must justify the claim that the distance between two points in an equilateral triangle cannot exceed the length of its side. But this is easy to prove.



We only need to focus on the case when one of the points coincides with a vertex of the equilateral triangle, as the figure indicates. (To justify this, perform suitable operations on the two points. Please fill up the details.) So, all we need to prove is $BC > BD$ (Figure 41). This in turn requires us to show that $\angle BDC > \angle BCD$.

Figure 41: Distance between two points in an equilateral triangle cannot be more than the length of its side

But this is straightforward, for $\angle BDC > \angle BAD = \angle BCD$. The claim thus follows. And this also completes the overall proof of the puzzle. ■

Discussion

Mathematical puzzles — particularly “impossible puzzles” that require proving that some configuration is logically impossible — provide fertile ground for teaching proofs. They help students explore different types of proof, understand the reasoning behind proof construction, and internalize fundamental mathematical principles in an engaging and fun way. Modifications of problems into hands-on puzzles particularly captivate learners’ attention and as [1] observes, “can lead to ingenious mathematical thinking, ... that unveils some of the features of the mathematical mind” with implications for “both recreational mathematics and cognitive science” (p. 32).

I hope that this puzzle-activity inspires more mathematics educators and researchers to develop and utilize proof-based puzzles in their teaching and research.

Further reading

- [1] Danesi, M. (2018). *Ahmes’ Legacy: Puzzles and the Mathematical Mind*. Toronto: Springer International Publishing AG.
- [2] Fomin, D., S. Genkin, and I. Itenberg (1996). *Mathematical Circles (Russian Experience)*. Hyderabad: Universities Press.

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Part V

Miscellaneous Articles – 2

Miscellaneous Problems in Number Theory – 1

Here we study a few miscellaneous problems in number theory. As usual, we state the problems first and present the solutions later. We urge you to try solving the problems on your own before studying the solutions. Some solutions use the LFT (the “Little Theorem of Fermat”) which we studied in the preceding article.

Problems

Problem 1. If both p and $p^2 + 2$ are primes, show that $p^3 + 2$ is prime as well.

Problem 2. Show that 341 is a divisor of $2^{340} - 1$.

Problem 3. Find all positive integers n such that n is a divisor of $2^n - 1$.

Problem 4. Let n be a composite number that can be factorized in two

different ways as $n = ab$ and $n = cd$. (Here, a, b, c, d are positive integers.) Show that $a + b + c + d$ is composite.

Problem 5. Solve the equation $3 \cdot 2^a + 1 = b^2$ over the positive integers.

Problem 6. Let $n > 1$ be a positive integer such that n is a divisor of $2^n + 1$. Show that n is a multiple of 3.

Solutions

Solution of Problem 1

This problem is a good example of a hoax!

We shall show that if both p and $p^2 + 2$ are primes, then $p = 3$. (That is, there is just one prime number that satisfies the stated condition.) And when $p = 3$, we have $p^3 + 2 = 29$, which is a prime number. So if we are able to prove that the conditions imply that $p = 3$, then there is nothing further to prove.

Let n be any positive integer not divisible by 3. Then $n^2 \equiv 1 \pmod{3}$, which implies that $n^2 + 2$ is a multiple of 3. Since $n^2 + 2 > 3$, this further implies that $n^2 + 2$ is composite.

So, if p is such that $p^2 + 2$ is prime, then p must be divisible by 3.

Since p itself is prime, it must be that $p = 3$ (this being the only prime number that is divisible by 3). We have thus proved the stated claim. ■

Solution of Problem 2

First we observe that $341 = 11 \times 31$. Here, 11 and 31 are primes.

The Little Theorem tells us that if p is an odd prime, then p is a divisor of $2^{p-1} - 1$. Hence:

$$11 \mid 2^{10} - 1.$$

(Recall that the symbol ' $\mid$ ' means "is a divisor of".)

Drawing on the known fact that $x - 1$ is a divisor of $x^m - 1$ for any positive integer m , we have,

$$2^{10} - 1 \mid 2^{340} - 1.$$

We therefore deduce that

$$11 \mid 2^{340} - 1.$$

Next, since $31 = 2^5 - 1$,

$$31 \mid 2^5 - 1,$$

and since $5 \mid 340$, we have,

$$2^5 - 1 \mid 2^{340} - 1.$$

It follows that

$$31 \mid 2^{340} - 1.$$

So, 11 and 31 are divisors of $2^{340} - 1$. Since $\text{GCD}(11, 31) = 1$, it follows that

$$341 \mid 2^{340} - 1. \quad \blacksquare$$

Comment: Since 341 is composite, with $341 = 11 \times 31$, this means that 341 is a *pseudoprime to base 2*. A "pseudoprime to base 2" is a composite number n such that $2^{n-1} \equiv 1 \pmod{n}$. For more about pseudoprimes, please refer to the Wikipedia article [1].

Solution of Problem 3

We must find all positive integers n such that $2^n \equiv 1 \pmod{n}$. We base our solution on [2].

An obvious solution is $n = 1$.

Let $n > 1$ be such that $n \mid 2^n - 1$. Since $2^n - 1$ is odd, n is odd.

Suppose that n is a prime number, p . By the Little Theorem,

$$p \mid 2^p - 2.$$

So, if we are also told that $p \mid 2^p - 1$, then we get $p \mid 1$, which is absurd. It follows that no prime number is a solution.

We may similarly prove that no prime power is a solution. For example, suppose that $n = p^2$ is a solution for some prime number p . Invoking the Little Theorem, we get

$$2^p \equiv 2 \pmod{p},$$

$$\therefore 2^{p^2} \equiv 2^p \equiv 2 \pmod{p}.$$

So, if we also have $2^{p^2} \equiv 1 \pmod{p}$, then $p \mid 1$ which is absurd. Hence there is no such solution.

Exactly the same logic works for all higher prime powers.

Now let us consider the general case.

Let $n > 1$ be a composite number such that $n \mid 2^n - 1$. Since $2^n - 1$ is odd, n is odd. Let $p > 2$ be the smallest prime divisor of n . Then we have,

$$p \mid n, \quad n \mid 2^n - 1, \quad \therefore p \mid 2^n - 1,$$

i.e.,

$$2^n \equiv 1 \pmod{p}.$$

We also have, by the Little Theorem,

$$2^{p-1} \equiv 1 \pmod{p}.$$

Let d be the smallest positive integer satisfying the condition

$$2^d \equiv 1 \pmod{p}.$$

The preceding two congruences imply that

$$d \mid p-1, \quad d \mid n,$$

and so

$$d \mid \text{GCD}(p-1, n).$$

By design, p is the least prime divisor of n . This implies that all numbers less than p are coprime to n . Hence $\text{GCD}(p-1, n) = 1$, which yields $d = 1$. This in turn yields $p = 1$, which is absurd.

The contradiction tells us that there can be no such n . Hence, the only positive integer satisfying the stated condition is $n = 1$. ■

Solution of Problem 4

We are told that $n = ab = cd$ where a, b, c, d are positive integers.

Let $u = \text{GCD}(a, c)$ and $v = \text{GCD}(b, d)$. Then,

$$ab = u \cdot \frac{a}{u} \cdot v \cdot \frac{b}{v},$$

$$cd = u \cdot \frac{c}{u} \cdot v \cdot \frac{d}{v}.$$

Hence,

$$u \cdot \frac{a}{u} \cdot v \cdot \frac{b}{v} = u \cdot \frac{c}{u} \cdot v \cdot \frac{d}{v},$$

$$\therefore \frac{a}{u} \cdot \frac{b}{v} = \frac{c}{u} \cdot \frac{d}{v},$$

Keeping in mind the definitions of u and v , we know that

- a/u and c/u are coprime;
- b/v and d/v are coprime.

Let k be any divisor of a/u . Then $\text{GCD}(k, c/u) = 1$, so $k \mid d/v$. As this is true for every divisor k of a/u , it follows that $a/u \mid d/v$.

Symmetrically, we argue that $d/v \mid a/u$.

From these it follows

$$\frac{a}{u} = \frac{d}{v} = s, \text{ say,}$$

and similarly,

$$\frac{b}{v} = \frac{c}{u} = t, \text{ say.}$$

Here, s and t are positive integers.

This means that if $n = ab = cd$ where a, b, c, d are positive integers, then there exist positive integers s, t such that

$$a = su, \quad b = tv, \quad c = tu, \quad d = sv.$$

But this yields

$$\begin{aligned} a + b + c + d &= su + tv + tu + sv \\ &= (s + t)(u + v). \end{aligned}$$

Since s, t, u, v are positive integers, it follows that $s + t \geq 2$ and $u + v \geq 2$. This implies that $(s + t)(u + v)$ is composite, and therefore that $a + b + c + d$ is composite, as claimed. ■

The above solution is from [3].

Solution of Problem 5

We must solve the equation $3 \cdot 2^a + 1 = b^2$ over the positive integers.

Since $a = 1$, $a = 2$ do not result in solutions, we shall assume that $a \geq 3$ and $b \geq 2$. Hence $3 \cdot 2^a$ is a multiple of 24, and b is odd. Let us write the equation as

$$3 \cdot 2^a = (b - 1)(b + 1).$$

Let $b = 2c + 1$, where c is a positive integer. We then get,

$$3 \cdot 2^a = 2c(2c + 2),$$

$$\therefore 3 \cdot 2^{a-2} = c(c + 1).$$

We must distinguish between two cases, depending on how the factors of 2 and 3 are distributed over the expression on the right side.

Case 1: $c = 3 \cdot 2^u$ and $c + 1 = 2^{a-2-u}$ for some $u \in \{0, 1, 2, \dots, a - 2\}$.

Case 2: $c = 2^u$ and $c + 1 = 3 \cdot 2^{a-2-u}$ for some $u \in \{0, 1, 2, \dots, a - 2\}$.

Since c and $c + 1$ must have opposite parity, $u = 0$ or $u = a - 2$ must hold in each case. So we may replace the above as follows:

- From Case 1, with $u = 0$: this yields $c = 3$ and $c + 1 = 2^{a-2}$.

This results in a solution, with $c = 3$, $a = 4$, and hence $b = 7$.

- From Case 1, with $u = a - 2$: this yields $c = 3 \cdot 2^{a-2}$ and $c + 1 = 1$.

This clearly has no solution.

- From Case 2, with $u = 0$: this yields $c = 1$ and $c + 1 = 3 \cdot 2^{a-2}$.

This clearly has no solution.

- From Case 2, with $u = a - 2$: this yields $c = 2^{a-2}$ and $c + 1 = 3$.

This results in a solution, with $c = 2$, $a = 3$ and hence $b = 5$.

In summary, the equation $3 \cdot 2^a + 1 = b^2$ has the following two solutions over the positive integers:

$$(a, b) = (3, 5), (4, 7).$$



The above solution is from [4].

Solution of Problem 6

We use the strategy used in the solution for Problem 3.

Let $n > 1$ be a positive integer such that $n \mid 2^n + 1$.

Since $2^n + 1$ is odd, n is odd.

Let p be the least prime number that divides n .

Then p is a divisor of $2^n + 1$. That is,

$$2^n \equiv -1 \pmod{p}.$$

Since n is odd, we may write the above as

$$(-2)^n \equiv 1 \pmod{p}.$$

We also have, by the Little Theorem, as p is odd,

$$(-2)^{p-1} \equiv 1 \pmod{p}.$$

Since p is the least prime divisor of n , we know that $\text{GCD}(n, p-1) = 1$.

Combining this fact with the above two congruences, we get

$$-2 \equiv 1 \pmod{p},$$

i.e., $p \mid 3$. This implies that $p = 3$. ■

Further reading

- [1] Wikipedia, “Fermat pseudoprime” from
https://en.wikipedia.org/wiki/Fermat_pseudoprime
- [2] Rik Bos, “Find all n such that $n \mid 2^n - 1$.”
<https://mathsinteresting.quora.com/Find-all-math-n-math-such-that-math-n-2-n-1-math-1>
- [3] Carlos E. T., “Prove that if $ab = cd$ then $a + b + c + d$ is not a prime number.” <https://mathsinteresting.quora.com/Prove-that-if-math-ab-cd-math-then-math-a-b-c-d-math-is-not-a-prime-number>
- [4] Brian Sittinger, “Find all integer solutions of $3 \cdot 2^x + 1 = y^2$.”
<https://mathsinteresting.quora.com/Find-all-integer-solutions-of-math-3-2-x-1-y-2-math>

Rational Multiples of π with Rational Cosine

A well-known equality in trigonometry is the following:

$$\cos \frac{\pi}{3} = \frac{1}{2}, \quad (16)$$

and therefore also, $\sin \pi/6 = 1/2 = \sin 5\pi/6$ and $\cos 2\pi/3 = -1/2$.

Here is one aspect of (16):

Equation (16) displays a ‘nice’ rational multiple of π whose cosine is a ‘nice’ rational number.

We will not define the term ‘nice’ as it is not a mathematical concept — but we all know what ‘nice’ means!

Since π radians equals 180° , (16) can be stated equivalently as,

$$\cos 60^\circ = \frac{1}{2}, \quad (17)$$

and so:

Equation (17) displays an angle measuring an integral number of degrees whose cosine is a ‘nice’ rational number.

This observation immediately suggests a question to explore:

Problem 1: Find all rational numbers r such that $\cos r\pi$ is rational.

An equivalent statement of the problem would be:

Problem 2: Find all rational numbers θ such that $\cos \theta^\circ$ is rational.

We opt to explore Problem 1. In view of the periodic nature of the cosine function, we may as well assume that $0 \leq r \leq \frac{1}{2}$. (Or, if we opt to explore Problem 2, that $0 \leq \theta \leq 90$.)

The surprising conclusion is that the only such rational numbers are 0 , $\frac{1}{2}$ and $\frac{1}{3}$. The corresponding values of the cosine function are 1 , 0 and $\frac{1}{2}$. So our claim is the following.

Theorem 1. *Let r be a rational number such that $\cos r\pi$ is rational. Then,*

$$\cos r\pi \in \left\{0, \pm\frac{1}{2}, \pm 1\right\}. \quad (18)$$

Theorem 2. *Let θ be a rational number such that $\cos \theta^\circ$ is rational. Then,*

$$\cos \theta^\circ \in \left\{0, \pm\frac{1}{2}, \pm 1\right\}. \quad (19)$$

These statements are obviously equivalent to each other. We shall prove Theorem 1.

Proof of Theorem 1

Our presentation of the proof draws on the excellent article [1].

We are going to use the well-known identity $\cos 2x = 2\cos^2 x - 1$. We restate this in a slightly altered form:

$$2\cos 2x = (2\cos x)^2 - 2. \quad (20)$$

For reasons which will become clear soon, it is more convenient to work with the form $2\cos x$ than with $\cos x$. Hence (20).

Let r be a rational number such that $2\cos r\pi$ is rational. We may as well assume that the denominator of $\cos r\pi$ is > 2 ; for if the denominator were 2, then r can only be $\pm\frac{1}{3}$ or $\pm\frac{2}{3}$, and there is nothing further to prove. So we shall assume that the denominator of $\cos r\pi$ exceeds 2, which means that the denominator of $2\cos r\pi$ exceeds 1.

Let $r = \frac{a}{b}$ where a, b are integers with $b \geq 1$ and $\text{GCD}(a, b) = 1$. Consider the sequence $\{2^n r\}$ obtained by repeated doubling,

$$r, \quad 2r, \quad 4r, \quad 8r, \quad 16r, \quad \dots \quad (21)$$

Let us retain only the fractional parts of these numbers.

For example, if $r = \frac{3}{5}$ then we first get,

$$\frac{3}{5}, \quad \frac{6}{5}, \quad \frac{12}{5}, \quad \frac{24}{5}, \quad \frac{48}{5}, \quad \dots,$$

and then the fractional parts,

$$\frac{3}{5}, \quad \frac{1}{5}, \quad \frac{2}{5}, \quad \frac{4}{5}, \quad \frac{3}{5}, \quad \dots$$

We have obtained a periodic sequence (with cycle length 4).

If we start with $r = \frac{5}{21}$, the resulting sequence of fractional parts is

$$\frac{5}{21}, \quad \frac{10}{21}, \quad \frac{20}{21}, \quad \frac{19}{21}, \quad \frac{17}{21}, \quad \frac{13}{21}, \quad \frac{5}{21}, \quad \dots$$

We have again obtained a periodic sequence (with cycle length 6).

Let's see what happens if we start with $r = \frac{13}{22}$. The resulting sequence of fractional parts is:

$$\frac{13}{22}, \frac{2}{11}, \frac{4}{11}, \frac{8}{11}, \frac{5}{11}, \frac{10}{11}, \frac{9}{11}, \frac{7}{11}, \frac{3}{11}, \frac{6}{11}, \frac{1}{11}, \frac{2}{11}, \frac{4}{11}, \frac{8}{11}, \dots$$

This time we have obtained a sequence which is eventually periodic (with cycle length 10).

We claim that *the sequence of fractional parts will always be eventually periodic, independent of the choice of the starting number r .*

To show this, we must prove that there exist integers $m, n, m > n \geq 0$, such that $2^m r$ and $2^n r$ have equal fractional part. Equivalently, that there exist integers $m, n, m > n \geq 0$, such that $(2^m - 2^n)r$ is an integer. For example, if $r = \frac{13}{22}$, then $(2^{11} - 2^1)r$ is an integer.

If the denominator is even, we first remove the 2's in the denominator by multiplication with a suitable power of 2. For the above example, if $r = \frac{13}{22}$, we consider $r' = 2r = \frac{13}{11}$; this amounts to saying that $(2^{10} - 2^0)r'$ is an integer. (Please check these computations.) So, let us prove the following.

Theorem 3. *Let r be a rational number with odd denominator. Then there exist integers m, n , with $m > n \geq 0$, such that $(2^m - 2^n)r$ is an integer.*

Proof Let $r = \frac{a}{b}$ where b is odd and $\text{GCD}(a, b) = 1$. We must show that there exist integers m, n , with $m > n \geq 0$, such that

$$2^m \equiv 2^n \pmod{b}. \quad (22)$$

To prove this, all we need to do is look at the remainders in the divisions

$$2^k \div b, \quad k = 0, 1, 2, 3, 4, \dots \quad (23)$$

None of these divisions works out exactly since b is odd; so, each division yields a remainder in the set $\{1, 2, 3, \dots, b-1\}$. As there are only a finite number of possible remainders, after no more than b divisions, a remainder must repeat. So there must exist integers m, n , $m > n \geq 0$, such that the divisions $2^m \div b$ and $2^n \div b$ leave the same remainder. These values of m, n satisfy the conditions stated in the theorem. ■

Illustration Let's show how this works for $r = \frac{5}{21}$. Here are the remainders in the divisions $2^k \div 21$ for $k = 0, 1, 2, 3, 4, \dots$:

k	1	2	3	4	5	6	7	8	9	...
$\text{Rem}(2^k \div 21)$	1	2	4	8	16	11	1	2	4	...

We see that 2^7 and 2^1 leave the same remainder under division by 21.

Comment The above proof is combinatorial in nature. There are other ways of proving the result. Perhaps you could look for other proofs.

Back to the proof of the main theorem

Since the sequence of fractional parts of the numbers $r, 2r, 4r, 8r, \dots$ is eventually periodic, it follows that the sequence

$$2\cos r\pi, \quad 2\cos 2r\pi, \quad 2\cos 4r\pi, \quad 2\cos 8r\pi, \quad 2\cos 16r\pi, \quad \dots \quad (24)$$

is eventually periodic as well.

We shall show, however, that if $2\cos r\pi$ has denominator > 1 , then this is not possible.

We use the identity mentioned earlier, $2\cos 2x = (2\cos x)^2 - 2$, which implies that

$$2\cos 2r\pi = (2\cos r\pi)^2 - 2. \quad (25)$$

Let $2\cos r\pi = \frac{a}{b}$ where $\text{GCD}(a, b) = 1$ and $b > 1$. Then

$$2\cos 2r\pi = \frac{a^2}{b^2} - 2 = \frac{a^2 - 2b^2}{b^2}. \quad (26)$$

Since $\text{GCD}(a, b) = 1$, we claim that $\text{GCD}(a^2 - 2b^2, b^2) = 1$.

For, if p is a prime factor of both $a^2 - 2b^2$ and b^2 , then we must have $p \mid a^2$ and $p \mid b^2$, which would imply that $p \mid a$ and $p \mid b$. But we have already assumed that a, b are coprime. Hence $a^2 - 2b^2$ and b^2 are coprime.

This means that the denominator of $2\cos 2r\pi$ is b^2 , which strictly exceeds the denominator of $2\cos r\pi$, since $b > 1$.

By the same reasoning we conclude that the denominator of $2\cos 4r\pi$ strictly exceeds the denominator of $2\cos 2r\pi$.

We conclude in the same way that the denominator of $2\cos 8r\pi$ strictly exceeds the denominator of $2\cos 4r\pi$. And so on. Hence, the denominators of the fractions

$$2\cos r\pi, \quad 2\cos 2r\pi, \quad 2\cos 4r\pi, \quad 2\cos 8r\pi, \quad 2\cos 16r\pi, \quad \dots$$

are *strictly increasing*.

But if this is the case then the sequence

$$2\cos r\pi, \quad 2\cos 2r\pi, \quad 2\cos 4r\pi, \quad 2\cos 8r\pi, \quad 2\cos 16r\pi, \quad \dots$$

cannot possibly be eventually periodic.

We have arrived at a contradiction which shows that if $2\cos r\pi$ is a rational number for some rational number r , then the denominator of $2\cos r\pi$ cannot exceed 1.

Hence, if $2\cos r\pi$ is a rational number for some rational number r , then the denominator of $2\cos r\pi$ must be 1. Hence,

$$2\cos r\pi \in \{0, \pm 1, \pm 2\}, \quad (27)$$

and so $r \in \left\{0, \pm \frac{1}{3}, \pm \frac{1}{2}\right\}$.

This also implies that the only rational numbers θ with $0 \leq \theta \leq 90$ such that $\cos \theta$ is rational are 0, 60, 90. ■

Further reading

[1] Jörg Jahnel, "When is the (co)sine of a rational angle equal to a rational number?" from <https://arxiv.org/abs/1006.2938>

Article contributed by Shailesh Shirali

Infinitely Many Primes – 1

It is well known that there exists infinitely many primes. There are many proofs known of this fact, the oldest being Euclid's original proof. It may be noted here that you Euclid's result may also be stated in the following form:

Theorem 4 (Euclid). *Given any finite set of primes, it is possible to find a prime number that is not in the set.*

Euclid's proof For the sake of completeness, we give a brief account of Euclid's proof. Let $\{p_1, p_2, \dots, p_n\}$ be a set of prime numbers. We consider the number

$$M = p_1 p_2 \dots p_n + 1. \quad (28)$$

Let q be the smallest prime factor of M . Since $M \equiv 1 \pmod{p_i}$ for $i = 1, 2, \dots, n$, it follows that q cannot be any $p_1, p_2, \dots, p_n$. We have thus found a prime number that is not part of the original set of primes. (It is possible for the prime q to be the number M itself.)

This implies that there exist infinitely many primes. ■

Sample calculation

We illustrate the reasoning used above with some numbers.

- Let us start with just one prime, $\{2\}$. Then $M = 3$, which is prime.
- The set of primes has now enlarged to $\{2, 3\}$. This now yields $M = 7$, which again is prime.
- The set of primes is now $\{2, 3, 7\}$. This yields $M = 43$, which yet again is prime.
- The set of primes is now $\{2, 3, 7, 43\}$. This yields $M = 1807$, which is *not* prime. Its least prime factor is 13 (please check).
- The set of primes is now $\{2, 3, 7, 13, 43\}$. This yields $M = 23479$, which factors as 53×443 .
- We now update the set of primes to $\{2, 3, 7, 13, 43, 53\}$, and continue in this manner. The process will never come to a stop.

Questions to explore

Many questions arise in this connection. For example,

- Q1. We know that there exist infinitely many prime numbers. But do we know infinitely many of them? That is, can we exhibit any infinite collection of prime numbers? In other words, can we exhibit any infinite set of integers and claim that they are all prime?
- Q2. We know that there exist infinitely many primes. Let the natural numbers be subdivided into various classes in some reasonable and sensible way. What can we say about the distribution of prime numbers between these various classes?

In response to Q1, the following is true. *We do not know any infinite set all of whose members are prime!* This may seem surprising but it is so.

In response to Q2, we start by looking at the behaviour of primes mod 4. In later articles of this series, we will look at other classes of primes.

Primes mod 4

There is just one even prime (namely, 2). Let us leave this out and look only at the odd primes.

Subdivide the odd primes into two classes according to the remainder left under division by 4. Let those which leave remainder 1 form a set $S_{4,1}$, and let those which leave remainder 3 form a set $S_{4,3}$. Then,

$$S_{4,1} = \{5, 13, 17, 29, 37, 41, 53, 61, 73, 89, 97, \dots\},$$

$$S_{4,3} = \{3, 7, 11, 19, 23, 31, 43, 47, 59, 67, 71, 79, 83, \dots\}.$$

Can we claim that both of these sets have infinitely many members? We can — because the statement happens to be true! But it requires proof.

Primes of the form 3 mod 4

It is not so difficult to show that $S_{4,3}$ has infinitely many members, but the task is more challenging for $S_{4,1}$. We present the proofs below.

Theorem 5. *There exist infinitely many prime numbers of the form $3 \pmod{4}$.*

Proof We use the following:

- (a) the product of two numbers of the form $1 \pmod{4}$ is of the form $1 \pmod{4}$;

- (b) the product of two numbers of the form $3 \pmod{4}$ is of the form $1 \pmod{4}$.

To see why these claims are true, we use the relations $1 \times 1 \equiv 1 \pmod{4}$ and $3 \times 3 \equiv 1 \pmod{4}$. Otherwise put,

$$(4a + 1)(4b + 1) = 16ab + 4a + 4b + 1 = 4(4ab + a + b) + 1,$$

$$(4a + 3)(4b + 3) = 16ab + 12a + 12b + 9 = 4(4ab + 3a + 3b + 2) + 1,$$

showing that the product is of the required form in each case.

An immediate implication of the above observations is the following:

A number of the form $3 \pmod{4}$ must have a prime factor of the same form. In other words, the prime factors of a number of the form $3 \pmod{4}$ cannot all be of the form $1 \pmod{4}$.

Now we are in a position to prove the theorem.

Let $p_1, p_2, \dots, p_n$ be a finite collection of primes. Consider the number

$$M = 4p_1p_2 \cdots p_n - 1. \quad (29)$$

We observe that M is of the form $3 \pmod{4}$. Therefore, it must have at least one prime factor q of the form $3 \pmod{4}$. The prime q cannot be any of $p_1, p_2, \dots, p_n$. Indeed, we have

$$M \equiv -1 \pmod{p_i}, \quad \text{for } i = 1, 2, \dots, n. \quad (30)$$

We have thus found a prime number of the form $3 \pmod{4}$ which is not any of $p_1, p_2, \dots, p_n$.

This proves that there are infinitely many primes of this form. ■

Sample calculation

We can illustrate the reasoning used above with some actual numbers. Say we start with the set of primes $\{2, 3\}$. Then we have,

$$M = 4(2 \times 3) - 1 = 23.$$

It so happens that 23 is a prime number, and $23 \equiv -1 \pmod{4}$, as it must be. So we have discovered a prime number not in our original set, and it is of the form $-1 \pmod{4}$.

Now we work with the set of primes $\{2, 3, 23\}$. We have,

$$M = 4(2 \times 3 \times 23) - 1 = 551.$$

The prime factorization of 551 is 19×29 . Observe that one of these primes (19) is of the form $-1 \pmod{4}$. So we have discovered a prime number not in our set, and it is of the form $-1 \pmod{4}$.

Continuing, we work with the set of primes $\{2, 3, 19, 23\}$. Now we have,

$$M = 4(2 \times 3 \times 19 \times 23) - 1 = 10487.$$

This number happens to be prime, and it too is of the form $-1 \pmod{4}$. So we have discovered yet one more prime number not in our set, and it is of the required form.

We can clearly continue this process indefinitely. ■

Comment The strategy we have followed is just a variation of the one followed by Euclid in his original proof.

Primes of the form 1 mod 4

We can readily anticipate that the proof in this case of primes of the form 1 (mod 4) is going to be less obvious — because we do not have a result of the kind stated above, “A number of the form 3 (mod 4) must have a prime factor of the same form.” It is *not* true that a number of the form 1 (mod 4) must have a prime factor of the same form. (Here are a couple of counterexamples: $9 = 3 \times 3$; $21 = 3 \times 7$.) So a different idea is needed.

What saves the day is the “Little Theorem” of Fermat. This is an extremely important result in prime number theory, with vast utility. We will devote a complete article to it later. For now we only state the result.

Theorem 6 (Fermat). *Let p be a prime number, and let a be any integer not divisible by p . Then,*

$$a^{p-1} \equiv 1 \pmod{p}. \quad (31)$$

Examples:

- For $p = 5$ and $a = 3$ we have $3^4 = 81 \equiv 1 \pmod{5}$.
- For $p = 7$ and $a = 2$ we have $2^6 = 64 \equiv 1 \pmod{7}$.
- For $p = 11$ and $a = 2$ we have $2^{10} = 1024 \equiv 1 \pmod{11}$.

The reader should check these statements.

We will discuss the proof of the Little Theorem in a later article. Meanwhile, we invite the reader to look for a proof on his or her own.

Comment The following is an equivalent statement of the little theorem.

Theorem 7 (Fermat). *Let p be a prime number, and let a be any integer. Then,*

$$a^p \equiv a \pmod{p}. \quad (32)$$

We now use the Little Theorem to prove that there are infinitely many primes of the form $1 \pmod{4}$.

Theorem 8. *Given a set of primes $\{p_1, p_2, \dots, p_n\}$, all of the form $1 \pmod{4}$, there exists a prime of the same form that is not in this set.*

Proof Consider the number

$$M = 4(p_1 p_2 \cdots p_n)^2 + 1. \quad (33)$$

Let q be a prime factor of M ; then q is clearly not any of $p_1, p_2, \dots, p_n$, since $M \equiv 1 \pmod{p_i}$ for $i = 1, 2, \dots, n$. This implies that the number $p_1 p_2 \cdots p_n$ is not divisible by q . Therefore, by Fermat's little theorem,

$$(p_1 p_2 \cdots p_n)^{q-1} \equiv 1 \pmod{q}. \quad (34)$$

Since $q \mid M$, it follows from (33) that $q \mid (p_1 p_2 \cdots p_n)^2 + 1$, i.e.,

$$(p_1 p_2 \cdots p_n)^2 \equiv -1 \pmod{q}. \quad (35)$$

Now, q is odd, since M is odd; hence $q - 1$ is even. Raise both sides of (35) to the power $(q - 1)/2$; we get,

$$(p_1 p_2 \cdots p_n)^{q-1} \equiv (-1)^{(q-1)/2} \pmod{q}. \quad (36)$$

Comparing (34) and (36), we deduce that $(-1)^{(q-1)/2} \equiv 1 \pmod{q}$, which is true if and only if

$$(-1)^{(q-1)/2} = 1. \quad (37)$$

This in turn is true if and only if $\frac{1}{2}(q - 1)$ is even, which means that q is of the form $1 \pmod{4}$.

We have shown that each prime factor of M is of the form $1 \pmod{4}$, and it is not contained in the set $\{p_1, p_2, \dots, p_n\}$.

This proves that there are infinitely many primes of the form $1 \pmod{4}$. ■

Sample calculation

We illustrate the reasoning used above with some actual numbers.

Let's start with the set of primes $\{5, 13\}$. The primes in the set are of the form $1 \pmod{4}$. Now consider the number

$$M = 4(5 \times 13)^2 + 1 = 16901.$$

This number is prime, and it is, of course, of the form $1 \pmod{4}$.

Or say we start with the set of primes $\{13, 17\}$. The primes in the set are of the form $1 \pmod{4}$. Now consider the number

$$M = 4(13 \times 17)^2 + 1 = 195365.$$

The prime factorization of this number is $195365 = 5 \times 41 \times 953$. The primes in the factorization are all of the form $1 \pmod{4}$.

Closing comment There's a lot more to say about this topic but we will keep it for the next issue.

Further reading

[1] Math Stack Exchange, "Infinitely number of primes in the form $4n + 1$ proof" from

<https://math.stackexchange.com/questions/244915/in-finitely-number-of-primes-in-the-form-4n1-proof>

Article contributed by Shailesh Shirali

Fermat Primes and their Cousins

Elsewhere in this issue we asked the following questions:

We know that there exist infinitely many prime numbers. But do we know infinitely many of them? That is, can we exhibit any infinite collection of prime numbers? In other words, can we exhibit any infinite set of integers and claim that they are all prime?

And we added, in response: *We do not know any infinite set all of whose members are prime!*

Given this background, it seems natural to look for a formula that will generate infinitely many prime numbers by substituting different values of the independent variable. Human beings have looked for such formulas for centuries (unfortunately, in vain). Here are some of the findings that have been obtained in this search.

Linear forms

The simplest result in this field is the following:

Theorem. *There do not exist any integers a, b , with $b \neq 0$, such that $a + bn$ is a prime number for all positive integers n .*

In other words, there is no non-constant linear form that will generate prime numbers for every value of the independent variable. (We do need the restriction on non-constancy else we could put $f(n) = 3$, say. This surely assumes a prime value for every integer n . But it is a very boring example!) So we may assume that $b \neq 0$.

To prove the claim, it suffices to exhibit a value of n such that $a + bn$ is composite.

To start with, suppose that $b \geq 1$. Suppose further that $f(n) = p$, a prime number, for some n .

Now consider the value of $f(n + p)$. We have,

$$\begin{aligned}f(n + p) &= a + b(n + p) = (a + bn) + bp \\&= p + bp \quad (\text{since } a + bn = p) \\&= p(1 + b).\end{aligned}$$

Since $1 + b \geq 2$, the number $p(1 + b)$, is composite. So we have found a value of the independent variable at which the function value is composite. This proves the claim for the case when $b \geq 1$.

What happens if $b \leq -1$? A slight tweak of the proof suffices. Please complete the proof on your own. 

Quadratic forms

Using a somewhat similar idea, we may show that there is no non-constant quadratic form that assumes a prime value for every value of the independent variable.

Let $f(n) = a + bn + cn^2$. Suppose that $f(n) = p$, a prime number. Now consider the value of $f(n + kp)$ for $k = 0, 1, 2, \dots$. We have,

$$\begin{aligned}f(n + kp) &= a + b(n + kp) + c(n + kp)^2 \\&= (a + bn + cn^2) + p(bk + 2cnk + ck^2) \\&= p + p(bk + 2cnk + ck^2) \\&= p(1 + bk + 2cnk + ck^2).\end{aligned}$$

If $1 + (b + 2cn)k + ck^2 \neq 0, 1$, then this equality shows that $f(n + kp)$ is composite. But $1 + (b + 2cn)k + ck^2$ cannot assume the values 0, 1 for every possible integer k . (Do you see why? Think of the expression $1 + (b + 2cn)k + ck^2$ as a quadratic in k . Can this expression assume any value infinitely many times?) For some value of k it will surely happen that $1 + bk + 2cnk + ck^2 \neq 0, 1$, and for this value of k , $f(n + kp)$ will be composite. 

Sequences of primes generated by linear forms

From the above we know that no linear form will generate an infinitely long sequence of primes. But it is perfectly possible for a linear form to generate a long (but finite) sequence of primes. It can be fun looking for

such forms! We give a few examples here. Note that if a linear form yields a sequence of primes, those primes are in arithmetic progression (AP).

AP with 3 primes. E.g., 3,5,7, generated by $f(n) = 3 + 2n$ for the three values $n = 0, 1, 2$; or 3,7,11, generated by $f(n) = 3 + 4n$ for $n = 0, 1, 2$.

AP with 4 primes. E.g., 5,11,17,23 generated by $f(n) = 5 + 6n$ for the four values $n = 0, 1, 2, 3$.

AP with 5 primes. E.g., 5,11,17,23,29 generated by $f(n) = 5 + 6n$ for the five values $n = 0, 1, 2, 3, 4$.

AP with 6 primes. E.g., 7,37,67,97,127,157 generated by $f(n) = 7 + 30n$ for the six values $n = 0, 1, 2, 3, 4, 5$.

AP with 7 primes. E.g., 7,157,307,457,607,757,907 generated by the linear form $f(n) = 7 + 150n$ for the seven values $n = 0, 1, 2, 3, 4, 5, 6$.

AP with 8 primes. E.g., 199,409,619,829,1039,1249,1459,1669 generated by the linear form $f(n) = 199 + 210n$ for the eight values $n = 0, 1, 2, 3, 4, 5, 6, 7$.

Longer sequences of primes in AP. Longer sequences than the above have been found. The longest such sequence known to date has 27 primes. It is generated by the linear form $f(n) = a + bn$ where

$$a = 605185576317848261,$$

$$b = 155368778 \times 2 \times 3 \times 5 \times 7 \times 11 \times 13 \times 17 \times 19 \times 23.$$

Numbers of the form 2×3 , $2 \times 3 \times 5$, $2 \times 3 \times 5 \times 7$, $2 \times 3 \times 5 \times 7 \times 11$, etc., are known as *primorials* (by analogy with *factorials*).

Sequences of primes generated by quadratic forms

From the above we also know that no quadratic form will generate an infinitely long sequence of primes. But it is possible for a quadratic form to generate a long (but finite, naturally) sequence of primes. It can be fun looking for such forms! Here are one such quadratic form.

A remarkable quadratic form. Consider the polynomial

$$f(n) = n^2 + n + 41.$$

This simple looking quadratic form (first discovered by Leonard Euler) yields prime numbers for 40 consecutive values of n ; namely, for

$$n = 0, 1, 2, 3, \dots, 38, 39.$$

Here are these forty prime numbers, displayed in tabular form:

41	43	47	53	61	71	83	97	113	131
151	173	197	223	251	281	313	347	383	421
461	503	547	593	641	691	743	797	853	911
971	1033	1097	1163	1231	1301	1373	1447	1523	1601

with $f(0) = 41$ and $f(39) = 1601$.

It is easy to see that the very next number in the sequence is *not* prime; indeed, $f(41) = 41^2$ (please check).

We do not know how Euler hit upon this remarkable polynomial.

Another example. Consider the quadratic polynomial,

$$g(n) = n^2 - 79n + 1601.$$

This polynomial (believe it or not) generates prime numbers for as many as 80 consecutive values of n , namely, for

$$n = 0, 1, 2, 3, \dots, 78, 79.$$

Let's study the resulting set of primes closely.

Here are the 80 primes:

1601	1523	1447	1373	1301	1231	1163	1097	1033	971
911	853	797	743	691	641	593	547	503	461
421	383	347	313	281	251	223	197	173	151
131	113	97	83	71	61	53	47	43	41
41	43	47	53	61	71	83	97	113	131
151	173	197	223	251	281	313	347	383	421
461	503	547	593	641	691	743	797	853	911
971	1033	1097	1163	1231	1301	1373	1447	1523	1601

On examining this table we are likely to feel disappointed. The table does not feature 80 different primes but only 40, each of which occurs twice (please check). Moreover, these 40 primes are identical to those in the previous table! At first sight this looks quite mysterious. How could it be that two different quadratic forms yield exactly the same set of primes? The answer is that this quadratic form is simply a displaced form of the previous one! If you play with the formulas for $f(n)$ and $g(n)$ for a little

while, you will soon find that

$$g(n) = f(n - 40).$$

And that settles the matter!

Fermat primes

Now we explore another kind of expression, involving exponentials; we will look for primes of the form $2^n + 1$. If we list the first 15 numbers of this kind,

$$2^1 + 1, 2^2 + 1, 2^3 + 1, \dots, 2^{14} + 1, 2^{15} + 1,$$

and use a computer to identify the prime numbers among them, we are left with the following:

$$2^1 + 1 = 3, \quad 2^2 + 1 = 5, \quad 2^4 + 1 = 17, \quad 2^8 + 1 = 257.$$

We notice that if n is *not* a power of 2, then $2^n + 1$ is *not* a prime number. This is what the facts seem to suggest. Therefore, let us first show that this statement is valid.

Theorem. *If n is not a power of 2, then $2^n + 1$ is not a prime number.*

Proof of the theorem: How does one show that a number is not prime? The most straightforward way is to exhibit a proper factor of the number. That's what we shall do here. We start by recalling the following.

- $x + 1$ is a factor of $x^3 + 1$, so $x^3 + 1$ is divisible by $x + 1$. This follows by dividing out. Indeed, $x^3 + 1 = (x + 1)(x^2 - x + 1)$. More efficient: use the *factor theorem*. The substitution $x = -1$ in $x^3 + 1$ yields 0, so $x + 1$ must be a factor of $x^3 + 1$.

- $x + 1$ is a factor of $x^5 + 1$, so $x^5 + 1$ is divisible by $x + 1$. Let's use the factor theorem again. When we substitute $x = -1$ in $x^5 + 1$ we get 0, so $x + 1$ must be a factor of $x^5 + 1$.
- More generally, the factor theorem implies that $x + 1$ is a factor of $x^n + 1$ when n is odd. That is, $x^n + 1$ is divisible by $x + 1$ when n is odd.

Let us now use this finding. Consider a number such as $A = 2^{28} + 1$. Here the exponent (28) has the odd divisor 7. Rewriting the number as

$$2^{28} + 1 = (2^4)^7 + 1 = 16^7 + 1,$$

we deduce that A is divisible by $16 + 1$, i.e., by 17. The complementary factor (i.e., $A \div 17$) being clearly much greater than 1, it follows that A is composite.

Or consider a number such as $B = 2^{40} + 1$. Here the exponent (40) has the odd divisor 5. Rewriting the number as

$$2^{40} + 1 = (2^8)^5 + 1 = 256^5 + 1,$$

we deduce that B is divisible by $256 + 1$, i.e., by 257. The complementary factor (i.e., $B \div 257$) being clearly much greater than 1, it follows that B is composite.

From these considerations we realize that **if n has an odd prime divisor, then $2^n + 1$ is composite.**

Turning this finding around, we realize that **if $2^n + 1$ is to be prime, then n must have no odd prime divisors!**

Now, what are the numbers (greater than 1) that have no odd prime divisors? Clearly, they are the numbers of the form 2^m ('pure' powers of 2).

We are thus led to consider numbers of the following form,

$$2^{2^n} + 1,$$

i.e., where the exponent of 2 is itself a power of 2. Here are the first few numbers of this kind:

$$3, \quad 5, \quad 17, \quad 257, \quad 65537, \quad 4294967297, \quad \dots$$

As you can see, the numbers increase very rapidly.

These numbers are now known as *Fermat numbers*, so named after Pierre de Fermat, the great “amateur mathematician”. In the Wikipedia article [1] we read (in part):

Pierre de Fermat (17 August 1601–12 January 1665) was a French magistrate, polymath, and ...[a] mathematician who is given credit for early developments that led to infinitesimal calculus ... In particular, he is recognized for his discovery of an original method of finding the greatest and the smallest ordinates of curved lines, which is analogous to that of differential calculus, ... and his research into number theory. He made notable contributions to analytic geometry, probability, and optics. He is best known for his Fermat’s principle for light propagation and his [Last Theorem] in number theory, which he described in a note at the margin of a copy of Diophantus’ *Arithmetica*.

Regarding his research into number theory we must mention two results of great beauty: the ‘Little Theorem’ and the Two Squares Theorem. We will study these theorems in detail later.

One of the few missteps that he took was regarding the Fermat numbers. He noticed that the following numbers are all prime (the same numbers that we listed above):

$$2^{2^0} + 1 = 3,$$

$$2^{2^1} + 1 = 5,$$

$$2^{2^2} + 1 = 17,$$

$$2^{2^3} + 1 = 257,$$

$$2^{2^4} + 1 = 65537.$$

Thereupon he took a big jump and conjectured that the Fermat numbers,

$$F_n := 2^{2^n} + 1, \quad n = 0, 1, 2, 3, 4, 5, 6, \dots,$$

are all prime. But this conjecture has met with a strange fate.

The very next Fermat number after those listed above,

$$F_5 = 2^{2^5} + 1 = 4294967297,$$

is already quite large; it is difficult to say offhand whether it is prime or composite. It was left to another great mathematician of the next century, Leonard Euler, to show that the above number is in fact composite. Indeed, he found its factorization:

$$4294967297 = 641 \times 6700417.$$

This accomplishment seems incredible at first encounter but becomes understandable when we study the approach taken by Euler.

This obviously puts an end to Fermat's conjecture. What is quite surprising is that subsequent research has not uncovered even a single Fermat prime number beyond F_0 , F_1 , F_2 , F_3 and F_4 . E.g., we have (see [2]),

$$F_6 = 274177 \times 67280421310721,$$

$$F_7 = \text{a 17-digit prime} \times \text{a 22-digit prime},$$

$$F_8 = \text{a 16-digit prime} \times \text{a 62-digit prime}.$$

(As you can see, the study of the Fermat numbers involves extremely big primes!)

It is possible that every Fermat number beyond F_4 is composite. But this is an open problem. There are lots of open problems in this area! You will find some of them mentioned in [2].

In closing ... a cousin of the Fermat Primes

Given what we have just written about primes of the form $2^n + 1$, one may wonder ... what about primes of the form $4^n + 2^n + 1$? Does anything of interest emerge from this question?

The figures, unfortunately, are not very encouraging. If we do a computer assisted search for values of n for which $4^n + 2^n + 1$ is prime, we find that the only such values below 500 are 1, 3, 9, with

$$4^1 + 2^1 + 1 = 7,$$

$$4^3 + 2^3 + 1 = 73,$$

$$4^9 + 2^9 + 1 = 262657.$$

With just three ‘successful’ exponents, we do not have much to proceed!
But we are able to prove the following, and that is something:

Theorem. *If $4^n + 2^n + 1$ is prime, then n is of the form 3^k for some integer k .*

Observe how the figures we obtained are in agreement with this statement.

We will postpone the proof of this statement to the next issue.

Article contributed by Shailesh Shirali

Further reading

[1] Wikipedia, “Pierre de Fermat” from

https://en.wikipedia.org/wiki/Pierre_de_Fermat

[2] Wikipedia, “Fermat number” from

https://en.wikipedia.org/wiki/Fermat_number

Grandma's Identities

Recently I came across a short note [1] with a striking title — Grandma's identity! We describe here the content of this note and the intriguing identity in question.

The author (S Humble) writes, "One of my students told me the following fact to do with cosines. He had learnt it from his grandma. Oh, to have a wonderful grandma like this!" He then displays these curious identities:

$$2\cos\frac{\pi}{3} = 1,$$

$$2\cos\frac{\pi}{5} \cdot 2\cos\frac{2\pi}{5} = 1,$$

$$2\cos\frac{\pi}{7} \cdot 2\cos\frac{2\pi}{7} \cdot 2\cos\frac{3\pi}{7} = 1,$$

and adds, "...etc., for all odd numbers." Humble then remarks that he had not seen this result before and goes on to provide the following very nice proof of the result. The method of proof is worth studying closely as it can yield more such results.

Proof of “Grandma’s identity”

Let n be an odd positive integer. We first note that the roots of the n -th degree polynomial $x^n - 1$ are the n quantities

$$\cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}, \quad k = 0, 1, 2, \dots, n-1.$$

For $k = 0$ we get the root 1. The remaining $n - 1$ roots can be grouped into conjugate pairs, and we can write them as

$$\cos \frac{2\pi k}{n} \pm i \sin \frac{2\pi k}{n}, \quad k = 0, 1, 2, \dots, \frac{n-1}{2}.$$

Now the sum and product of

$$u_k = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}, \quad v_k = \cos \frac{2\pi k}{n} - i \sin \frac{2\pi k}{n}$$

are $2 \cos \frac{2\pi k}{n}$ and 1 respectively. Hence

$$(x - u_k)(x - v_k) = x^2 - 2x \cos \frac{2\pi k}{n} + 1.$$

Therefore we have the factorization

$$x^n - 1 = (x - 1) \prod_{k=1}^{(n-1)/2} \left(x^2 - 2x \cos \frac{2\pi k}{n} + 1 \right).$$

The substitution $x = -1$ yields (since n is odd)

$$\frac{-2}{-2} = \prod_{k=1}^{(n-1)/2} \left(2 + 2 \cos \frac{2\pi k}{n} \right).$$

We now make use of the identity

$$\cos 2A = 2 \cos^2 A - 1.$$

This yields

$$2 + 2\cos\frac{2\pi k}{n} = 4\cos^2\frac{\pi k}{n}.$$

Hence we have,

$$1 = \prod_{k=1}^{(n-1)/2} 4\cos^2\frac{\pi k}{n}.$$

Now for $k = 1, 2, \dots, \frac{1}{2}(n-1)$ we have $0 < \frac{\pi k}{n} < \frac{\pi}{2}$, implying that for these values of k ,

$$\cos\frac{2\pi k}{n} > 0.$$

Hence by taking square roots we obtain

$$1 = \prod_{k=1}^{(n-1)/2} 2\cos\frac{\pi k}{n},$$

which is the required identity.

In a follow-up article we shall prove some more identities of this kind.

We urge you (the student reader) to carry out some explorations on your own and discover more such results. They are well within reach!

Further reading

- [1] Steve Humble, “Grandma’s Identity” from *The Mathematical Gazette*, Nov., 2004, Vol. 88, No. 513 (Nov., 2004), pp. 524-525,
<http://www.jstor.com/stable/3620735>

Call for Articles

We invite mathematics teachers and practitioners to contribute short articles to The Compass.

Articles could deal with mathematical content or the teaching of mathematics (classes 6 to 12). They could feature real-life classroom experiences with projects and explorations.

They could be about the use of audiovisual material or social media in the mathematics classroom.

They could feature reflections on the teaching of mathematics, or on the nature of mathematics.

Articles about the use of software are particularly welcome, as are articles dealing with non-routine problems and classroom experiences with problem-solving.

In short, anything at all to do with school mathematics!